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Research

Rules of Engagement

Mapping the Next Phase of China-US Ties

September 2026

Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Tactical détente: The near-term outlook



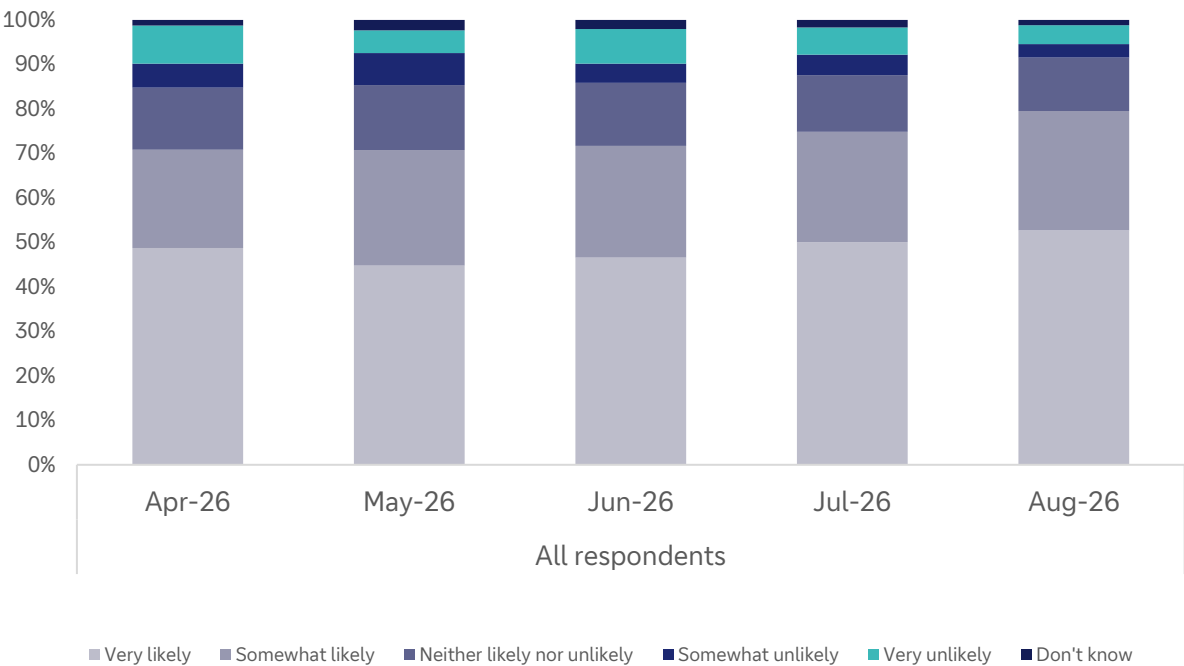
Political commitment and shifting domestic priorities create space for stability into 2027

The Trump-Xi meeting in Washington met our key expectation: It delivered a subdued-yet-constructive summit resulting in a modest number of [concrete deliverables](#) that will minimise the near-term risks of abrupt decoupling, a security crisis, or a disruptive trade war. **This was the best plausible outcome for markets, given the structural nature of China-US competition.** The brief two-month extension of the Busan agreement until 10 Jan 2027 will further delay new tariffs and export controls on both sides. It will also increase pressure to deliver a broader trade and investment package, and further trade-truce extensions, by the end of the year. We anticipate more talks between US Treasury Secretary Scott Bessent and Chinese Vice Premier He Lifeng in Q4 as both sides seek [a larger deal](#).

For China, stability with Washington will likely remain a priority as it manages **rising trade tensions with the EU** and prepares for top leadership reshuffles at its 2027 Party Congress. Combined with shifting US public perceptions of China, these factors **could create near-term space for more dealmaking** on the sidelines of the upcoming APEC (18-19 Nov) and G20 (14-15 Dec) summits. **As our surveys below show, US voters are now more open to increased trade and investment with China, even at the cost of geopolitical concessions**, while **concerns over the cost of living** have intensified. Both trends are now at their highest levels since April.

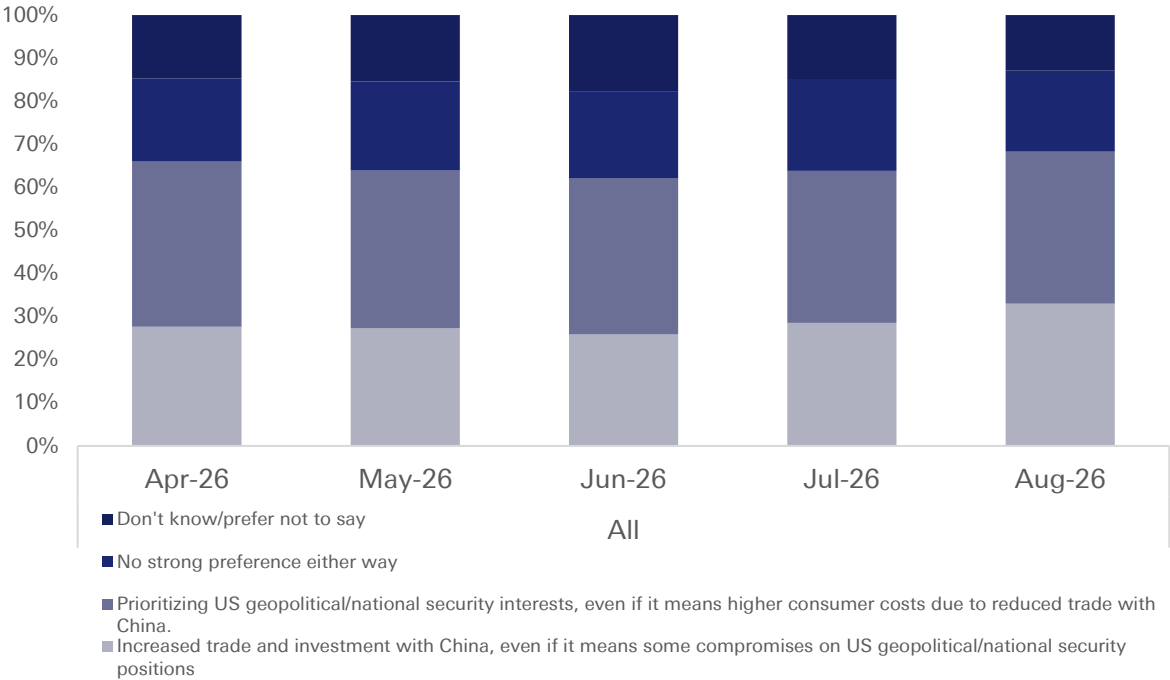
Nevertheless, political guardrails are not a panacea for a relationship that remains characterized by mistrust and competing national security priorities. **As the next slide will show, the outlook for China-US relations becomes more complex beyond this near-term horizon....**

US participants were asked: How likely or unlikely is it that current cost-of-living concern will influence your vote in the midterms?



Source: dbDataInsights. Responses collected April-August 2026 from an Online Nationally Representative sample of ~600 US respondents.

Which of the following outcomes for US relations with China do you most prefer?



Source: dbDataInsights. Responses collected April-August 2026 from an Online Nationally Representative sample of ~600 US respondents.

The policy horizon: Key events to watch (2026–2029)

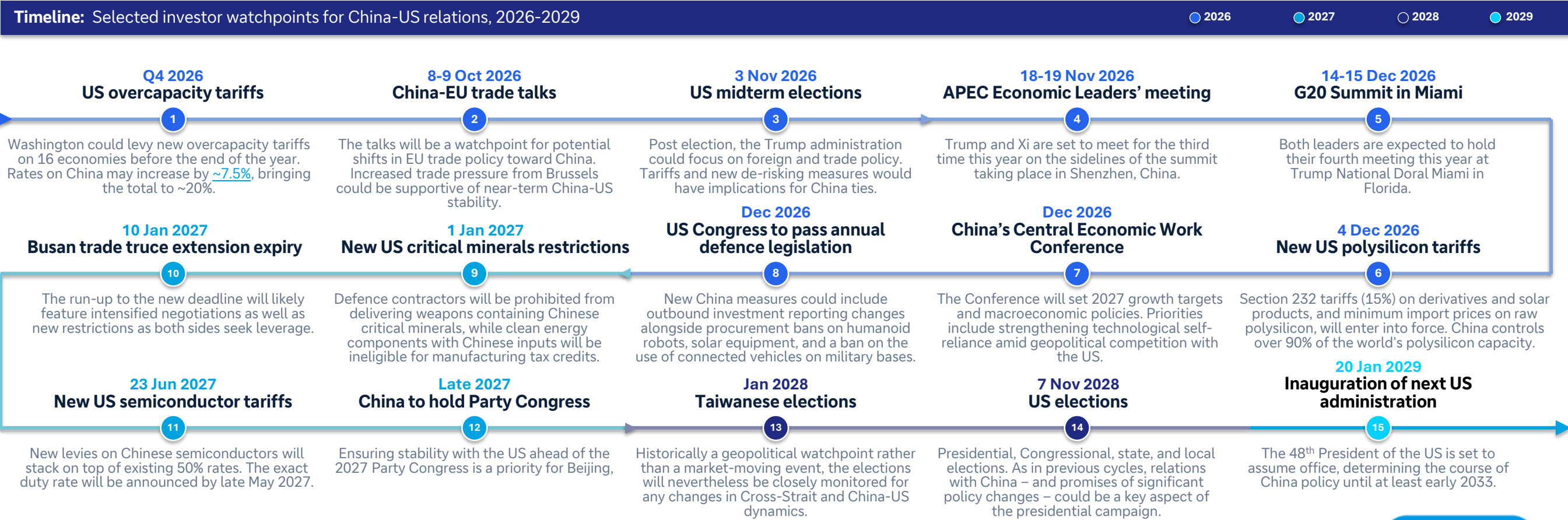
The next several years will see more summits, tariffs, major political events, and key elections



An analysis of the geopolitical calendar reveals a complex slate of tariffs, import curbs, export controls, and sensitive political events. Even as leader-level diplomacy has reduced the risk of a near-term breakdown, it has done little to stem the tide of new bilateral economic restrictions. These have proliferated during 2026 to include new export controls, sanctions, and tariffs (see [list](#)), with significant implications for high-growth industries including drones, AI, electric vehicles, and robotics.

Competition is unlikely to slow, and events could repeatedly test the fragile trade truce. Q4 2026 will see further China-US dialogue and China-EU trade talks. Should talks with Brussels fall short of averting increased European trade pressure, Beijing may further prioritise near-term US stability. But the coming months will also bring challenges in the form of new **US overcapacity tariffs and polysilicon levies**, while next year will usher in **curbs on critical minerals and additional semiconductor tariffs**. The run-up to the **10 Jan Busan truce** expiry could also bring new economic restrictions as both sides seek negotiating leverage. Both countries will also hold major political events (**China's Party Congress in 2027** and **US elections in 2028**), with potential implications for bilateral ties, while **Taiwan's Jan 2028 election** will be another key watchpoint.

But these are just the events whose outcomes and impacts we can estimate in advance. **Understanding probable China-US developments that are not on the calendar is equally important for investors.** We should assume that competitive dynamics across myriad policy issues will drive other developments and policy changes, often announced with little or no advance notice. **We examine some of the key issues in the next slide.**



Structural strains: Testing the guardrails

Navigating the trade, technology, and regional pressure points

Unless consistently managed, structural competition could move markets and test stability through the end of the decade and beyond. We highlight five watchpoints for markets:

- 1. The AI race:** Dialogue is set to strengthen AI safety guardrails, but national and economic security imperatives mean it is unlikely to significantly slow the China-US AI race, much less meaningfully reverse existing restrictions. Any attempts to pace AI development would touch on questions of tech restrictions, including Beijing's perceptions of US technological containment, which could increase amid potential restrictions on [remote access to AI chips](#) and Chinese [data centre components](#).
- 2. Tariff pressures:** Trade tensions have subsided given a lower US trade deficit and the new Board of Trade, which is set to reduce tariffs on US\$60 billion of non-sensitive goods. But significant [differences](#) persist: Washington plans to unveil new **polysilicon and semiconductor tariffs**, as well as **overcapacity levies** on 16 economies (including China). The US is also seeking to restrict Chinese inputs in North American supply chains and target alleged **transshipment**, while Beijing shores up its supply chain dominance in areas of US import dependence, such as **advanced drone components** and processed **critical minerals** – which continue to be a source of [friction](#).
- 3. Regulatory toolkits:** The wider tech sector remains central to expanding export controls and restrictions. China is strengthening its toolkit. Its 2026 codification of **anti-derisking regulations** allows Beijing to penalize companies that comply with foreign sanctions or export controls. Meanwhile, its **outbound investment regulations**, effective 1 Jul, cover technology, data, talent and capital. In the US, Congress is drafting legislation to change outbound investment reporting requirements for **biotech and biopharma**, restrict government procurement of **humanoid robots and solar equipment**, and ban the use of **China-linked connected vehicles** on military installations.
- 4. Geopolitical stress-tests:** The **war in Iran** is unlikely to derail relations in the near term, but the conflict has [strained](#) bilateral ties. Any potential broadening of the conflict in scale and intensity, or a further expansion of US [sanctions](#) that impact Chinese interests, could test guardrails. Additionally, while the probability of conflict in the Taiwan Strait remains low, differences over **Taiwan** are set to remain unresolved across successive administrations in both Washington and Beijing into the 2030s and beyond.
- 5. Political anchors: Two domestic events will shape relations into the 2030s.** Beijing's priority is stability ahead of the late **2027 Party Congress**. For China, dialogue can help reduce the risk of costly escalation, potentially slow the pace of new US economic restrictions, and buy time for efforts to strengthen self-reliance in strategic sectors such as semiconductors (where China is aiming for **80% self-sufficiency by 2030**).

In the US, a durable bipartisan consensus on China could make visible concessions to Beijing increasingly difficult ahead of the **2028 presidential election**. Recent bipartisan criticism of the Trump administration's decision to allow limited Nvidia H200 chip sales to China reflects the political constraints facing a transactional approach to China, raising the prospect that guardrails will strain closer to election day.



Source: istockphoto

Appendix 1



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The end of the safety net: Life without the macroeconomic stabilisers

28 September 2026

For several decades, financial markets and the wider economy benefited from an implicit safety net. If a slowdown or a downturn arrived, there was a monetary put from central banks, and a fiscal put from governments. The monetary put would see interest rates slashed and liquidity injected via QE. The fiscal put would see governments run larger deficits to cushion any downturn. These policy stabilisers smoothed out the business cycle, and supported asset valuations too.

These policy interventions were increasingly normalised from the late-1980s, and became more aggressive with each passing cycle. This reached its peak during the Covid-19 pandemic, when governments ran fiscal deficits unprecedented in peacetime, whilst many central banks expanded their balance sheets with huge asset purchases.

But after an era of ever-bigger interventions, these old stabilisers now face major obstacles. On the fiscal side, policy is constrained today by sovereign yields at multi-year highs, alongside the highest debt-to-GDP ratios in several decades. Then on the monetary side, policy is constrained by persistent inflation that's increasingly supply-driven.

This represents a big shift from the last few decades, but it does echo previous regimes where the macroeconomic stabilisers were either blunted or left unused. A good example is the 1970s experience, when stagflation and supply shocks posed a dilemma to traditional policy tools. Even further back, the pre-WWI environment of the classical gold standard and balanced budget orthodoxy saw governments play a very different and more limited role in macroeconomic stabilisation.

Looking forward, this environment means policymakers will have a much more difficult time stabilising the business cycle in the coming years. So absent a sharp fall in yields or a collapse in inflation, higher macro volatility will be here to stay.

Authors

Henry Allen

Macro Strategist



1. Background: How the stabilisers used to work.

Economic policymakers have long used interventions to stabilise output. This became increasingly fashionable after WWII, when Keynesian economics established a more active use for fiscal policy to stimulate the economy in a downturn. This meant budget deficits became a more regular feature of life, rather than used solely for wartime emergencies.

It was from the late 1980s, however, that the extent of policy interventions became increasingly sizeable. Indeed, it was here that the origins of a central bank “put” for financial markets began, in the crisis of Black Monday in 1987. That was a huge stock market crash, where the S&P 500 fell over 20% in a single day, marking the biggest daily decline in its entire history. The size of the falls even went beyond the Great Depression, raising fears of wider contagion, particularly if it were to spark a liquidity crisis that caused financial institutions to fail. However, the Federal Reserve responded by cutting rates and providing liquidity, offering a macroeconomic stabiliser against the prospect of wider economic stress. This offered support against a wider downturn, and the US economic expansion continued until 1990.

Whilst the genesis of a central bank “put” came in 1987, the idea gained further traction in the late-1990s, with the Asian financial crisis in 1997, followed by the Russian financial crisis in 1998. Fears of an economic contraction mounted, particularly after issues at Long-Term Capital Management (LTCM) raised concerns that the US wasn’t immune from these global pressures. So once again, the Fed delivered three 25bp rate cuts pre-emptively, seeking to support the economy.

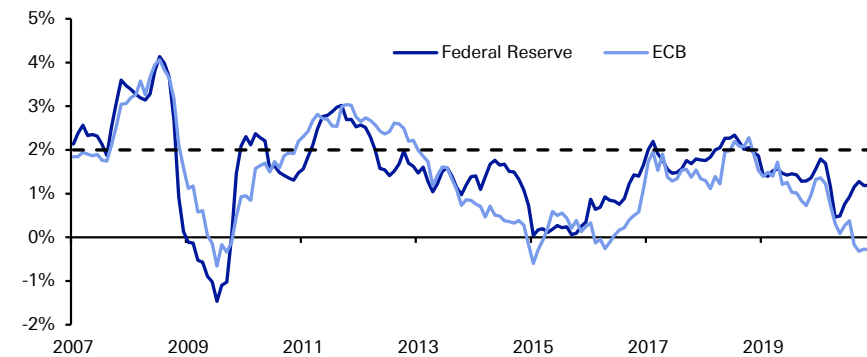
A few years later, the bursting of the dot com bubble in early 2000 set the stage for yet further stimulus. As equity markets declined, the Fed began to cut rates from early 2001, ultimately delivering 475bps of cuts that year that took rates from 6.5% to just 1.75%. That was followed up by further rate cuts in 2002 and 2003, leaving rates at 1% by mid-2003 into 2004. Moreover, the Bush tax cuts were also delivered in 2001 and 2003, acting as a fiscal stimulus as the wider US economy entered recession.

In hindsight, these moves seem modest compared to what then happened in 2008, when the Global Financial Crisis led to even bigger policy interventions. Suddenly, the world faced the biggest economic contraction since the Great Depression, with genuine fears that a repeat might happen. To avoid this, policymakers launched into the biggest economic stimulus ever seen in a peacetime environment. That included massive fiscal stimulus, with the US federal deficit at 9.8% of GDP in 2008, the highest since 1945. Moreover, central banks slashed rates to the zero lower bound, and many like the Federal Reserve launched into Quantitative Easing programmes, purchasing assets to ease financial conditions even more. Over the coming years, many began pledging to keep rates low well into the future, pioneering the concept of forward guidance.

These levels of policy intervention weren’t confined to the immediate crisis however. In the subsequent years, active attempts were made to prolong and extend the business cycle, particularly as the 2010s became an era of low inflation and low interest rates. In this environment, borrowing was historically cheap, and low inflation meant central banks could cut rates without seeing a big overshoot of their targets. Indeed, both the Fed and the ECB undershot their 2% inflation target on average in the 2010s, and the ECB took their own deposit rate into negative territory in 2014.



Figure 1: The Federal Reserve and the ECB frequently missed their 2% inflation targets on the downside in the 2010s (PCE inflation for the US, HICP inflation for the Euro Area)

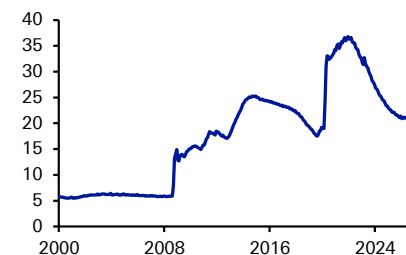


Source : Haver Analytics, Deutsche Bank

By 2019, the year before the pandemic, the intellectual climate had continued to move in the direction of more policy intervention. Government bond yields hit new lows, with Germany's 10-year bund yield at -0.7% in summer 2019. This environment of negative yields meant borrowing was basically a "free lunch", and that summer also saw the Fed begin a 75bp easing cycle, as inflation was below their target and they wanted to sustain the expansion.

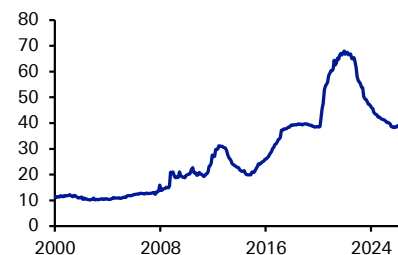
The peak of this policy interventionism was seen with the Covid-19 pandemic in 2020. Just as in 2008, the world was facing a huge and sudden economic contraction. Moreover, the consensus among policymakers was that they could have done more to speed up the recovery after 2008, potentially avoiding years of very high unemployment. As such, 2020 saw the most aggressive economic stimulus packages ever seen in a peacetime environment, well above those in 2008. Central banks immediately slashed rates back to the zero lower bound, pledged to keep rates low for years to come, and launched into major QE programmes again. Then on the fiscal side, record peacetime deficits were run. Indeed, the US federal deficit of 14.5% of GDP in 2020 was the highest since 1945.

Figure 2: Federal Reserve Balance Sheet (% of GDP)



Source : Bloomberg Finance LP, Deutsche Bank

Figure 3: ECB Balance Sheet (% of GDP)



Source : Bloomberg Finance LP, Deutsche Bank



During the pandemic period, many initially expected these interventions to remain for years to come. After all, in mid-2020, financial markets continued to price a continuation of the low inflation and low interest rates of the 2010s. This belief was clear on several fronts, and the 10-year Treasury yield closed at just 0.51% in August 2020. Meanwhile, the Federal Reserve even adjusted its own framework to allow for average inflation targeting, i.e. permitting an overshoot of the inflation target after periods of an undershoot. This reflected the consensus view that inflation was unlikely to be a serious threat.

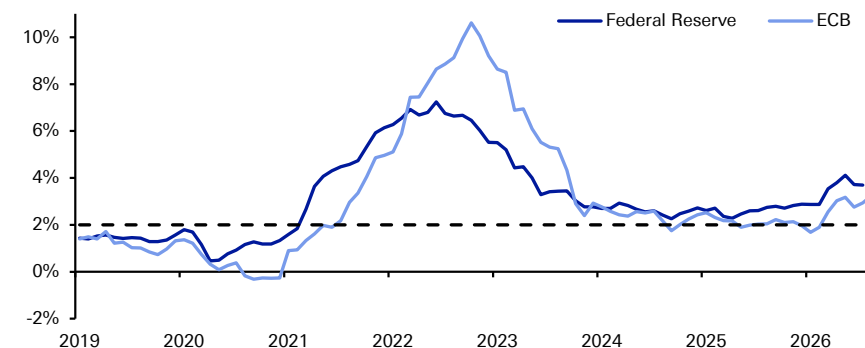
But as we now know, inflation returned in a serious way after the pandemic, forcing central banks into rate hikes. Meanwhile, public debt-to-GDP ratios rose significantly thanks to the pandemic stimulus, helping to push up sovereign bond yields too. In other words, many of the usual policy tools of the previous few decades were either blunted or no longer available.

2. The Monetary Constraint: High inflation means the central bank put isn't available

In the pre-2020 world described above, central banks regularly grappled with how to generate enough inflation. In part, this was thanks to several structural forces that put downward pressure on global prices. For instance, deepening globalisation acted as a positive supply shock that kept costs down. Meanwhile, the expansion of the global labour force in these years also boosted the supply side. In other words, inflation was low and central banks generally had the space to cut interest rates when they needed to.

But since the pandemic, we have entered a very different environment of above-target inflation across many of the world's biggest economies. Initially, that was caused by a huge wave of pent-up demand after the pandemic, alongside disruption to supply chains and labour markets. But the subsequent years also brought further shocks. Those included Russia's invasion of Ukraine in early 2022, driving a big spike in European energy prices, and pushing Euro Area inflation above 10% briefly. In 2026, the closure of the Strait of Hormuz then acted as a fresh energy shock, pushing inflation further away from target and contributing to a fresh tightening cycle from the major central banks.

Figure 4: In the 2020s, inflation has generally been above central bank targets (PCE inflation for the US, HICP inflation for the Euro Area)



Source : Haver Analytics, Deutsche Bank



Alongside those shocks, the structural forces keeping inflation down have gone into reverse. For instance, demographic trends have seen working-age populations shrink in many places. Meanwhile, the trend towards deeper globalisation has stalled, and the focus has turned to near-shoring and supply-chain resilience, rather than locating production in the cheapest place. Simultaneously, other trends like the push for higher defence spending are also adding to demand-side pressures.

Because of all this, the 2020s so far have brought persistently above-target inflation, and central bankers have found it much more difficult to ease policy in response to growth wobbles or financial stability fears.

This initially became apparent in 2022, amidst the biggest inflation wave for several decades. Central banks had to keep hiking rates, even in the face of slowing growth and financial market stress. Indeed, despite the S&P 500 being in bear market territory by mid-2022, the Federal Reserve still launched into a series of 75bp hikes, the biggest hikes since the 1990s. Then in 2023, despite the collapse of Silicon Valley Bank and fears of wider financial contagion, the Fed and ECB still continued with their rate hikes to combat inflation.

That tightening cycle in 2022-23 was a significant moment, because high inflation meant monetary policy could not simply act as a stabiliser against growth or financial stress. After all, policymakers realised that if they didn't act to tackle high inflation, they risked a repeat of the 1970s and the potential for even harsher monetary tightening to get inflation under control again.

Today in 2026, this is a familiar problem again. We are in a globally synchronised hiking cycle, with the Fed, ECB and Bank of Japan all having delivered rate hikes. So far at least, global growth has been robust and financial conditions remain accommodative by historical standards. But because inflation remains clearly above target, central banks don't have the space they did in the pre-pandemic world to simply cut rates at the first sign of difficulty. There is still an inflation mandate to be achieved, and given we have now seen multiple years of elevated inflation, there is a much higher risk of inflation expectations becoming unanchored in that context, particularly if another shock occurred.

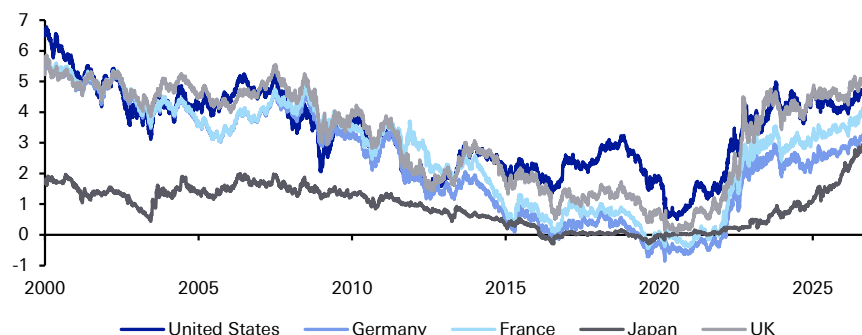
3. The Fiscal Constraint: High yields and high public debt mean fiscal stimulus isn't easily available either

Just as monetary policy is constrained by high inflation, fiscal policy today is also constrained by higher yields and higher public debt. This is a big transition from the world of 1980-2020, when yields were generally falling throughout. Indeed, despite all the fiscal stimulus around the GFC, the continuous decline in yields over the 2010s meant that debt dynamics remained manageable, because lower yields helped to offset the impact of higher debt.

Since the pandemic however, we've been in a very different world, as the dual forces of higher debt and higher yields have made debt dynamics far more unfavourable around the world. And this isn't just a nominal story, as real yields have also risen to multi-year highs as well.



Figure 5: 10yr Government Bond Yields (%) have risen to multi-year highs across several countries

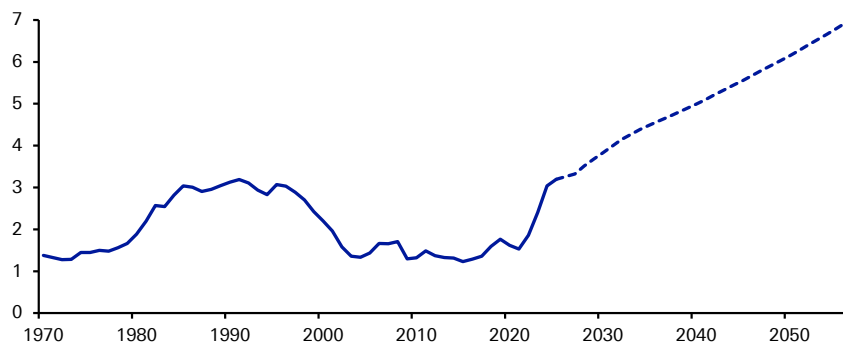


Source : Bloomberg Finance LP, Deutsche Bank

In essence, there are 4 new constraints around fiscal stimulus.

First, is the significant rise in global yields. By definition, the cost of incurring debt has grown, in both nominal and real terms. Relative to the pre-pandemic baseline in early 2020, the 10-year US Treasury yield has risen from 1.9% to 5.2%. Similarly in Germany, the 10-year bund yield has risen from -0.2% to 3.6%. Moreover, with yields already elevated, any further borrowing risks crowding out the private sector, and further dampening growth. And for governments, interest spending has ramped up significantly. For instance, US net interest spending as a share of GDP exceeded its recent peak from the early 1990s in the most recent fiscal year, and the CBO forecast this is set to continue.

Figure 6: US Net Interest Costs (% of GDP) with CBO Forecasts



Source : CBO, Deutsche Bank

The second constraint is that public debt itself is now at a higher level. This interacts with the higher yields mentioned above, because even though yields have only returned to their levels from the mid-2000s, that is much more impactful because they are being paid on a much larger debt pile. Using IMF numbers for consistency, net public debt across the advanced economies has risen from 52% of GDP in 2000, to 80% in 2025, with further increases projected to 2030. This means the fiscal space available to governments is more limited than it used to be.



The third constraint on fiscal stimulus is the high starting level of budget deficits. So to generate a meaningful fiscal stimulus, it would require deficits to be pushed up to even higher levels than they are today. But that in turn would add to the risks of bond market disruption that cascades through the economy, leading to higher borrowing costs for corporates and consumers as well.

The fourth constraint is higher inflation. When inflation is high, any fiscal stimulus that boosts demand risks pushing up prices rather than boosting output, and in turn makes monetary tightening more likely that could offset the impact of the stimulus. This trade-off was apparent in 2022, when central banks were hiking rates aggressively, and even further back in the inflationary environment of the 1970s. In a world of high inflation and supply constraints, fiscal stimulus is more likely to overheat the economy.

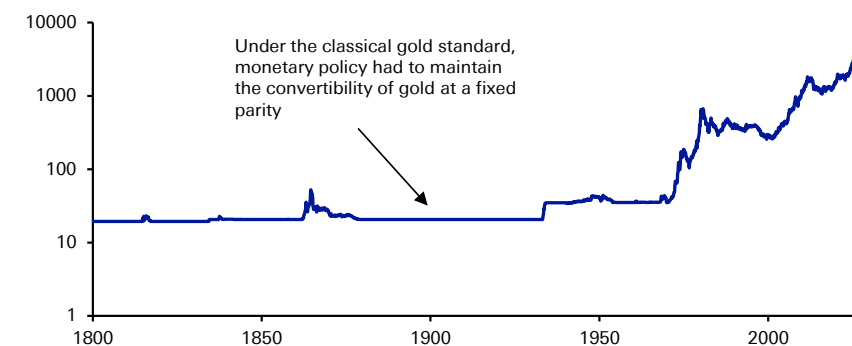
4. Are there any other eras of history where the stabilisers weren't available?

Whilst history never quite repeats exactly, there have been other eras in which the traditional policy stabilisers were either unavailable or severely blunted, which hold some lessons for our own time.

The pre-WWI classical gold standard

Before WWI, the international monetary system was known as the gold standard, in which currencies were convertible into a fixed quantity of gold. In practice however, this acted as a huge constraint on macroeconomic stabilisation. After all, monetary policy had to maintain the convertibility of gold at a fixed parity. This meant contemporary options like QE weren't available in a crisis. Meanwhile on the fiscal side, governments generally adhered to balanced budget orthodoxy. This was before Keynes wrote the General Theory, so the modern understanding of countercyclical deficit spending was not in place. In the US, there wasn't even a central bank until the Federal Reserve's establishment in 1913.

Figure 7: Gold Prices (US Dollars per ounce, log scale)



Source: Finaeon, Deutsche Bank

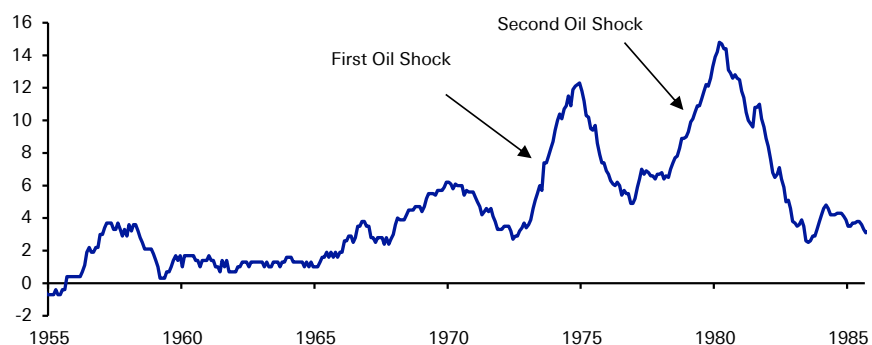


Without modern policy stabilisers however, this was also an era of huge macroeconomic volatility. There was a major financial panic across Europe and the US in 1873, which marked the start of the so-called “Long Depression”. In the US, that contraction lasted until 1879, which in length was even longer than the Great Depression contraction of 1929-33. And this was followed by other major panics, including the 1893 panic that led to another US contraction, along with the 1907 panic that saw the Dow Jones almost halve relative to its 1906 peak.

The stagflationary 1970s

The more recent example comes from the 1970s, another era of multiple supply shocks. The biggest were the two oil shocks in 1973 and 1979, which both represented negative supply shocks that pushed up inflation and hurt growth in the US and Europe. However, this meant policymakers faced a dilemma. If central banks cut rates, that would add to the inflationary pressures and risk worsening the wage-price spirals already underway. Meanwhile, if fiscal stimulus were pursued, that would also push up inflation.

Figure 8: US CPI (year-on-year % change)



Source : Haver Analytics, Deutsche Bank

This context meant that policy couldn’t smooth out the business cycle as successfully as it had in other decades. Indeed, in the decade from the first oil shock in 1973, the US experienced 3 separate recessions. That is significantly more turbulent than the last 40 years, when there have been just 4 recessions in total, or once a decade on average. For financial markets too, the 1970s were a terrible period for asset performance, with both equities and bonds losing ground in real terms across multiple countries.

5. Implications

If economic policymakers can’t use traditional monetary and fiscal policy tools to stabilise output and smooth the business cycle, then this will result in higher macro volatility.

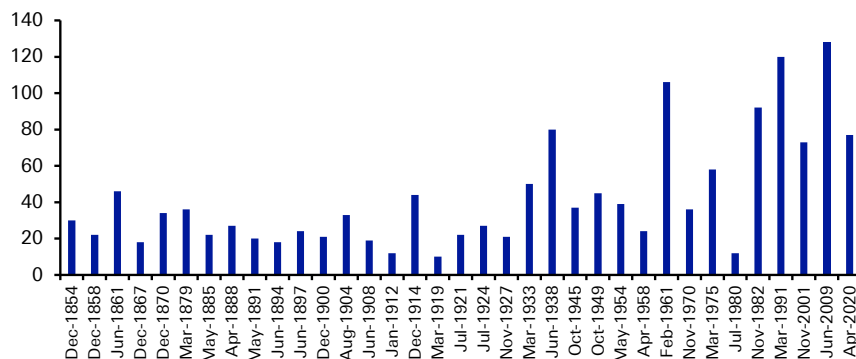
If we look back at the last 40 years, it is notable that the 5 most recent US expansions all rank in the top 7 longest expansions since business-cycle dating began. This has several causes, and some of it reflects a move away from an agriculture-based economy where output was dependent on factors like the weather. But a key role has also been played by pre-emptive policy interventions, such as cutting rates to stop a slowdown turning into a recession. Indeed, recent examples include the dovish shifts in monetary policy after the 1987 stock market crash, the 1997-98



financial crises, and the mid-cycle slowdown of 2015-16.

Beyond the actual policy interventions, these moves also had a confidence effect that supported financial markets. The fact that policymakers stood ready to take action helped create a virtuous circle, as optimism on the economic outlook boosted financial asset values, supported wealth effects, and avoided a tightening in financial conditions that would itself cause growth to slow.

Figure 9: Length of US economic expansions (in months) from the starting point



Source : NBER, Deutsche Bank

Clearly, policy is not as constrained today as in some periods of history. There is no longer a gold standard where the convertibility needs to be maintained. And unlike the 1970s, inflation is not in double-digits. Nevertheless, this is still a more constrained world than we have been used to in the last 4 decades. Higher inflation and more frequent supply shocks are constraining monetary policy. Meanwhile, higher yields and higher public debt levels are constraining fiscal policy. We have already seen the impacts since the pandemic, most notably in 2022, when central banks hiked rates to combat the inflation surge, and there was a simultaneous selloff across bonds and equities of a kind unseen in modern times.

Looking forward, this environment will offer new challenges of the sort we haven't been used to in recent decades. The policy trade-offs are now more acute, and if pre-emptive stabilisation is no longer available, then higher macro volatility lies ahead.



Appendix 1

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German Defense: An Ecosystem in Transformation

September 25, 2026

This is the first section of a report available to clients via our Deutsche Bank Research site [here](#).

Launching Private Company Analysis with the German Defense Sector

Traditional analysis that focused on public companies overlooked a critical market segment: the many important competitors that are privately owned, typically by families or private equity firms. To address this, we are launching dedicated coverage of Europe's private-market giants, starting with Germany, where a traditionally dominant private sector means large firms outnumber public ones almost two to one. We begin with the German defense sector, where understanding the private ecosystem is essential: it is the source of critical innovation, the solution to supply-chain issues, and the venue for a future wave of M&A, consolidation, and IPOs.

From Incremental Modernization to Industrial Scaling

Germany's defense landscape is undergoing its most significant transformation in decades, shifting from a paradigm of 'incremental modernization' to 'industrial scaling'. Triggered by the Zeitenwende in 2022, the new era is underpinned by substantial financial commitments, including a €100bn special fund and constitutional reforms enabling sustained long-term spending. The objective is no longer merely to fill discrete capability gaps but to execute a full 'ecosystem build-out' that addresses structural choke points and enhances industrial capacity. This transformation is creating a more resilient and scalable German Defense Industrial Base (DIB), prepared to meet surging domestic and European demand.

Technology, Integration, and the Rise of New Players

This strategic pivot is also reshaping the industrial and technological environment. EU initiatives (EDF, EDIRPA, ASAP) are fostering joint funding and integrated supply chains, while the market is experiencing a 'next wave' of technological change, with a clear shift in value towards drones, software-defined systems, electronic warfare, and AI-driven capabilities. These dynamics force industry consolidation but also create openings for agile start-ups, which are vital for innovation, as speed and software become paramount. This report provides an in-depth primer on this evolving sector. We identify Germany's industrial strengths and technology gaps, analyze the key private players poised to benefit, and examine the structural themes shaping the market - from funding dynamics and procurement reforms to technology roadmaps.

Authors

Michael Kuhn
Research Analyst

Tim Rokossa
Head of European Equity Research

Christoph Laskawi
Research Analyst

Christophe Menard
Research Analyst

Sriram Krishnan, CFA
Research Analyst



Half the Industry is Invisible

We are Changing that.

Equity research has always looked at the listed half of the economy. That half is shrinking. Companies stay private for longer, the largest capital pools are now private, and in Germany the listed universe was never the industry to begin with. **There are roughly 120 listed German companies with sales above €1bn - and roughly 200 unlisted ones.** In our home market, large private companies outnumber public ones almost two to one.

That gap is not a curiosity. It is an analytical problem.

If a sector's innovation, its supply-chain frictions, its pricing behavior and its eventual consolidation sit largely outside the listed universe, then covering only the listed names means covering only part of the argument. With this report we start closing that gap. We are launching dedicated, ongoing coverage of Europe's private-market giants, beginning in Germany.

Why we are starting with German defense

Defense is the sharpest version of the problem we are trying to solve. Germany is in the middle of its largest rearmament cycle since reunification, but much of the capability being bought is not primarily produced by the listed defense groups. It is produced by an ecosystem: family-owned system houses that have supplied the Bundeswehr for decades, venture-capital-backed companies that did not exist three years ago, and the tier-2 and tier-3 suppliers who sit on the actual constraints - energetics, seekers, qualified electronics, test capacity, cleared labor. That ecosystem is where the innovation happens, where the supply chain either breaks or scales, and where the next wave of M&A, consolidation and IPOs will take place. Understanding it is not a side-project to covering Rheinmetall, Hensoldt or Renk. It is a prerequisite.

Executive Summary - Eight Points Carry this Report

1. **The spending shift is structural, not cyclical.** The €100bn special fund was the headline; the 2025 constitutional reform exempting defense spending above 1% of GDP from the debt brake was the substance. Government budgets now foresee more than €150bn of annual defense spending by 2029, against roughly €52bn of core budget in 2024. NATO's 2025 pledge of 3.5% of GDP on core defense by 2035 puts a floor under the back-end.
2. **Budgets are not revenue.** Between an approved budget and a delivered capability sit parliamentary approval for anything above €25m, tender design, contracting, qualification, test capacity and acceptance. The February 2026 procurement acceleration reform helps, but the conversion chain - not demand - sets the delivery slope.
3. **Value is moving to software, autonomy and effects.** Drones, software-defined systems, electronic warfare and AI-enabled decision layers are taking share from platform-centric spend. That favors companies whose iteration cycle is measured in weeks, and it is the clearest reason private and venture-capital-backed firms now sit inside mainstream force design.
4. **Production is deterrence - and production is where Germany is short.** Scaling breaks at specific points: energetics permits, seekers and guidance electronics, qualification burden, specialized labor, and limited



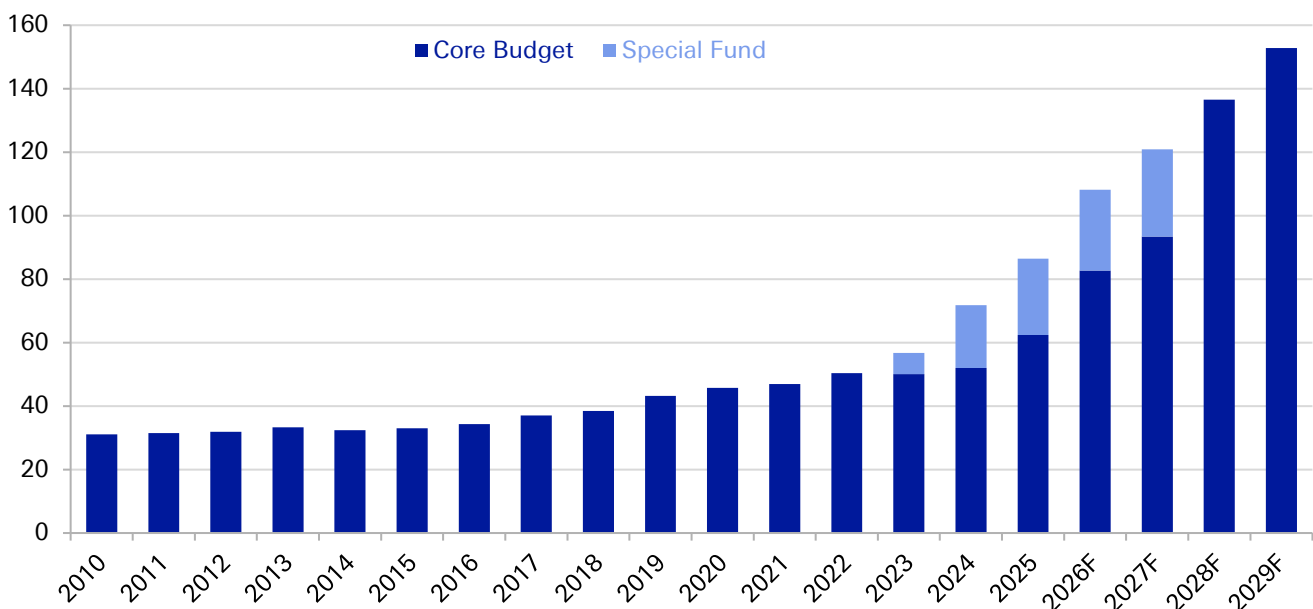
test windows. Almost every one of those nodes is occupied, at least in part, by a private company.

5. **The private ecosystem is now procurement-relevant, not just innovation-relevant.** The February 2026 Bundeswehr loitering-munitions framework, awarded to Helsing and STARK from an initial package of roughly €540m, is the clearest proof point. Sovereign buyers are contracting with companies two to five years old.
6. **Scale still sits with the incumbents.** Diehl generated €5.4bn of sales and €703m of EBIT in FY2025, with Defence at roughly €2.3bn; Rohde & Schwarz passed €3bn of revenue in FY2024/25 with no bank debt. These are not start-ups with a defense angle - they are among Europe's more significant defense industrial assets, and they happen to be family-owned.
7. **A consolidation and financing wave is coming.** Capital intensity is rising precisely as venture-stage companies move from software into hardware and factories. That points to further strategic rounds, industrial partnerships, JV structures, selective M&A and, in time, IPOs. For a bank, this is the most commercially relevant conclusion in the report.
8. **Regulation is the binding constraint, in both directions.** Export licensing, foreign investment screening, NATO interoperability and compliance intensity limit scaling and exits. But policy also accelerates - fast-track procurement, EU joint programs, state-backed vehicles for defense tech.

Why defense, in numbers

The money. Germany's core defense budget has moved from roughly €47bn in 2021 to a projected path above €150bn by 2029, with the €100bn special fund bridging the middle of that ramp. The 2025 debt-brake reform matters more than the fund itself: by exempting defense spending above 1% of GDP, it converts a one-off program into a sustained baseline. Procurement approvals have already responded: 97 major defense projects approved in 2024 against 55 in 2023, with over €58bn allocated.

Figure 1: German defense budget (€bn) - government budgets foresee >€150bn annual spending by the end of the decade

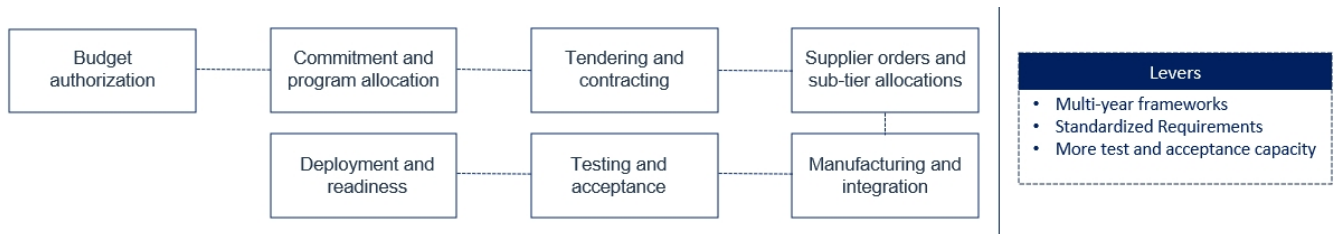


Source : German government, Deutsche Bank research



The conversion chain. What is in doubt is throughput. Contracts above €25m require binding Bundestag budget-committee approval, and tendering, supplier orders, integration, testing and acceptance each add time. Europe's fragmentation compounds it: European states operate around 179 different major weapon systems against 33 in the US, and 19 different main battle tanks against one. Between 2020 and 2024, European arms imports rose 155%, and European NATO members sourced 64% of those imports from the US, up from 52% in 2015-19. Whether 'buy European' becomes policy rather than rhetoric is the single largest swing factor for the domestic industrial base.

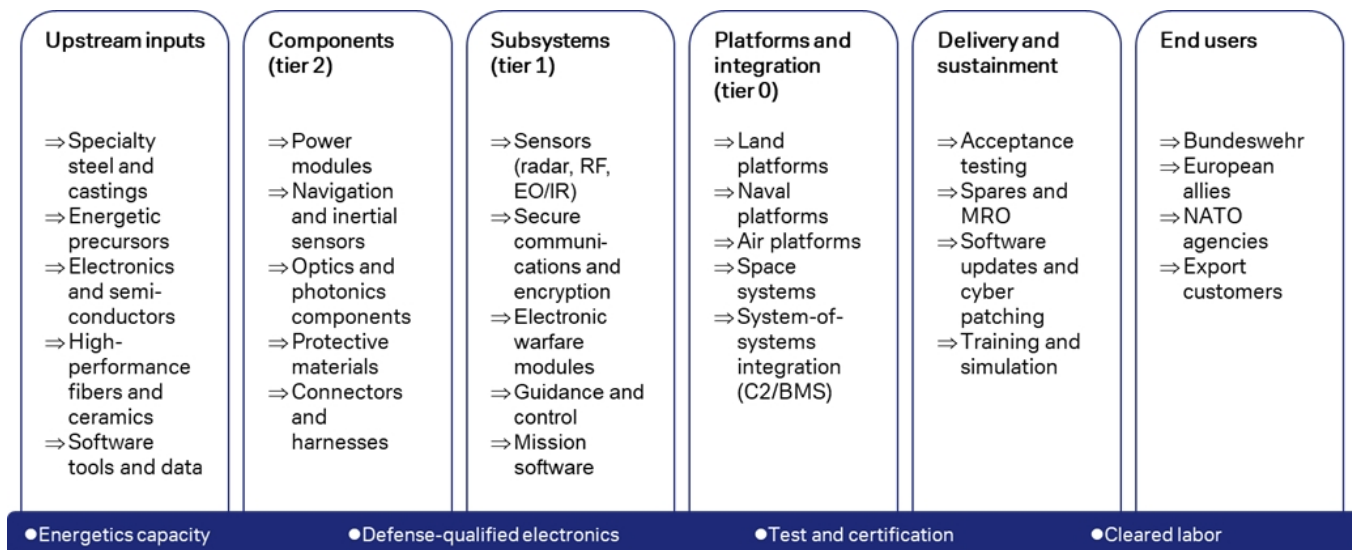
Figure 2: Budget to Output: The Conversion Chain



Source : Deutsche Bank research

Where the private sector sits. The constrained nodes are consistent across almost every program: energetics capacity, defense-qualified electronics, test and certification, and cleared labor. Private companies are disproportionately present in exactly those nodes. Diehl's acquisition of Dynamit Nobel was, in substance, the acquisition of explosives-related operating authorizations it describes as unique in Germany. STARK's dependence on TDW for German-manufactured warheads is the same story one tier down. Rohde & Schwarz occupies the secure-communications and SIGINT/EW layer no platform program can route around.

Figure 3: German Defense Industrial Value Chain

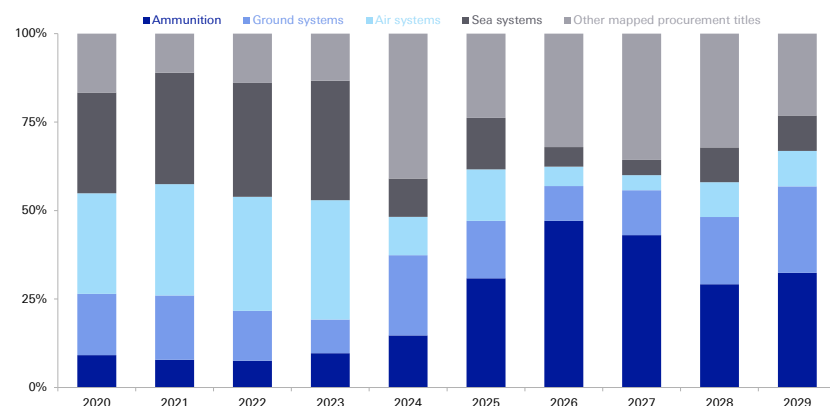


Source : Deutsche Bank research



Where the new money is going. Germany attracted close to €1bn of defense and security venture-capital funding in 2025, second only to the US. The deal mix since 2021: unmanned systems (34 deals), surveillance and space (24), other defense tech (24), electronic warfare (10) and cyber (8). Helsing raised \$1.8bn at an \$18bn valuation in July 2026; STARK Defence raised €500m in June 2026 above €3.5bn. Capital of that scale changes the industrial map.

Figure 4: Spending Percentage of Germany on Equipment by Category



Source : BMVG – Budget

The five debates - and where we go next

1. **Can Germany convert budget into capability fast enough?** Demand is settled; throughput is not. Watch acceptance capacity, qualification timelines and framework design rather than headline budgets.
2. **Do the newcomers industrialize, or stall at the factory door?** Helsing, STARK Defence, ARX Robotics and Quantum Systems are all making the same transition from software and prototypes into serial hardware. That is where defense start-ups historically fail. The next 18 months are the test.
3. **Do incumbents get disrupted, or absorb?** The likelier outcome is a denser partner network - Diehl with Hensoldt and Airbus, Helsing with Saab and Kongsberg, ARX Robotics with RENK and Daimler Truck. Partnership maps are becoming a better guide to competitive position than product catalogs.
4. **Does 'buy European' become real?** A conscious substitution of even part of the 64% US import share would be a structural repricing of the European industrial base. Fragmentation and interoperability standards cut against it.
5. **Who is financed, and how?** Rising capital intensity, thin private disclosure, constrained exit routes under foreign-investment screening, and uneven ESG treatment will all influence the financing and consolidation wave.



The Argument in Five Parts

1. The Strategic Landscape: Russia's 2022 invasion of Ukraine, combined with the distancing of the US as Europe's protector, represents a turning point for the safety of Europe and its defense industry. The shift in policy is creating immense demand across Europe for arms in the decades to come, leading to significant M&A and investing opportunities. German companies are uniquely suited to serve the rising demand due to their immense technical know-how and their already-existing industrial sector. Combined with targeted funding programs, these developments both stimulate existing companies and incentivize start-ups that push the technical frontier forward.

2. New Wars Change the Old Battlefield: Recent wars in Ukraine and the Middle East have reshaped modern warfare, shifting capital from traditional contractors to start-ups that can keep pace with rapid innovation cycles. Key trends defining this new era include: unmanned systems creating a transparent battlefield that demands adaptability; huge munitions consumption proving that production itself is deterrence; the battlespace expanding into commercial space and cyberspace; and societal resilience against disinformation becoming a vital security element. This all occurs amid a new great power competition, heightening the risk of multi-theater wars and the use of nuclear threats to back conventional aggression.

3. How to Gain Exposure – Investment Opportunities in Private Gems: Rapid innovation in modern warfare is creating significant investment opportunities in Germany's defense sector, as start-ups outperform slow, traditional procurement. This operational shift is fueled by a long-term surge in European defense spending targeted at R&D and equipment. Investors can gain exposure through key 'private gems', from established firms like Diehl Group and Rohde & Schwarz to high-growth start-ups like Helsing (AI) and Quantum Systems (drones). These start-ups leverage a mix of venture capital and government grants with innovative business models, making the German private defense market a compelling investment area.

4. Trend – Europe Buys European & Europe Buys between Middle Powers: Recent energy and security shocks have shattered Europe's post-Cold War assumptions, creating an urgent push for strategic self-reliance amid uncertainty over the long-term US security guarantee. This pivot is evident in new security partnerships, such as Germany deepening defense ties in the Indo-Pacific (Japan) and North Atlantic (Canada). Paradoxically, Europe's rearmament has increased its reliance on US hardware (64% of imports '20-'24) and its defense sector remains highly fragmented. The core challenge is whether Europe's quest for autonomy will lead to a 'buy European' policy; doing so would unlock a monumental internal market, driving consolidation and creating a truly independent defense capability.

5. The Regulatory Landscape & Investment Constraints: The defense industry is uniquely constrained and regulated, and investors must understand its landscape. Growth is not driven simply by demand but is filtered through government budgets, procurement laws, and export controls. This creates risks: budgets don't translate to immediate revenue, market access can be denied politically, and production cannot be scaled freely. High compliance costs and national security reviews that limit exit strategies add further complexity. Successful investment, therefore, requires mastering these political and regulatory realities, not just identifying demand.



Appendix 1

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Research

The House View: Snapshot

A broadening tightening cycle

Deutsche Bank

Macro views

World

- Global growth of 3.3% in 2026 and 3.4% in 2027.
- **Rationale:** AI investment supports activity despite the energy shock. Brent is forecast to average \$95/bbl in Q4 2026, with supply normalisation expected from Q1 2027.
- **Key risks:** Prolonged Middle East conflict, higher energy costs and tighter financial conditions. Upside risk from stronger AI investment and productivity.

Europe

- Euro-area GDP upgraded to 0.8% (prev. 0.5%) in 2026; 1.2% (prev. 1.1%) in 2027. Growth has been surprisingly resilient in the face of the energy shock.
- **Rationale:** Consumption, defence and AI spending, modest fiscal easing and bank lending have cushioned the energy shock.
- **Key risks:** A worsening energy shock, higher borrowing costs and vulnerable public finances around elections could lift risk premia.

China

- GDP growth lowered to 4.6% in 2026; 2027 remains 4.5%.
- **Rationale:** Exports and AI support growth, but consumption, investment and property remain weak. Fiscal spending is picking up; PBoC rates remain unchanged.
- **Key risks:** Delayed fiscal support, further property weakness and weaker external or AI-related demand.

United States

- 2026 & 2027 GDP growth forecast at 2.2%.
- **Rationale:** Easy financial conditions, fiscal policy and AI-driven capex support growth. Unemployment is expected near 4.1% into year-end; PCE inflation stays above 3% into 2027.
- **Key risks:** A larger oil shock weakens real income; financial conditions tighten; or tighter labour market and AI productivity surprise on the upside.

Germany

- GDP growth forecast at 1.0% in 2026 and 1.3% in 2027.
- **Rationale:** Strong H1 offset by loss of purchasing power caused by the energy shock, although expansionary fiscal policy continues to provide tailwinds.
- **Key risks:** Energy exposure, higher financing costs and political uncertainty could delay the improvement.

India

- 7.3% y/y GDP for FY27 (prev. 6.7%). FY28 growth outlook unchanged at 7.2%.
- **Rationale:** Growth remains strong even as higher oil prices raise inflation and external-balance sensitivity. Current policy looks too accommodative, with strong growth and rising headline inflation (above 6% through Oct-Dec 2026); 100bps of rate hikes starting in Oct 2026.
- **Key risks:** Persistently high oil prices, weaker global demand and tighter domestic financial conditions.

Key downside risks

- H** **Middle East conflict.** Possibility that the war can extend into 2028 pushes Brent forecast to \$130-150/bbl, leading to higher stagflation concerns.
- H** **Inflation:** Higher energy costs, AI investment, fiscal expansion, and higher defence spending could require more monetary tightening.
- M** **Political and fiscal risks.** European elections and fragile public finances could increase risk premia. US cost-of-living concerns remain important ahead of the midterms.

Marion Laboure
thehouseview@list.db.com

Camilla Siazon
<http://houseview.research.db.com>

Jim Reid

September 2026

Deutsche Bank AG/London

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Market views

	Market Sentiment	– Risk assets remain resilient despite rate hikes and higher yields. – The global economy resilient despite geopolitical conflict and energy risks, supported partly by the AI investment boom. – Key question is how long activity can withstand high energy costs, rising rates and limited fiscal space.
	Equities	– S&P 500 YE-2026 target: 8000; STOXX 600 YE-2026 target: 670. – Earnings recovery has broadened beyond technology. – Overweight mega-cap growth & tech, financials, industrials, and metals & mining.
%	Rates	– YE-2026: 10Y UST at 4.80%; 10Y Bund at 3.20%. – Neutral duration after a long-standing bearish bias. Equilibrium rates and term premia remain structurally higher.
	Credit	– YE 2026: \$IG 85bp; \$HY 315bp; €IG 95bp; €HY 345bp. – Spreads expected to widen modestly. – Prefer US over European credit amid stronger US growth and greater ECB risk.
	FX	– YE: EUR/USD: 1.17; USD/JPY: 150. – Neutral EUR/USD with a mild upside bias.
	Oil & Gold	– Q4-2026: Brent: \$95/bbl; WTI: \$90/bbl. – Supply normalisation expected from Q1 2027, leading Brent to average \$74.bbl in 2027. – Gold: YE-2026 target \$4,600/oz. Higher real yields and a more hawkish Fed weigh near term; 2027 average forecast is \$5,100/oz.
	Monetary Policy	– Fed: Two 25bp hikes in Dec and Mar, taking policy rate to 4.375%. – ECB: 25bp hikes in Dec, taking the deposit rate to 2.75%. – BoJ: Two 25bp hikes in Jan and Apr, bringing policy rate to 1.75%. – BoE: Two 25bp hikes in Nov and Feb, taking Bank Rate to 4.25%. – PBoC: Unchanged at 1.40% through 2026.

Key macro & markets forecasts

GDP growth (%)			Central bank policy rate (%)				Key market forecasts			
	2026F	2027F		Current	Dec 26F	Jun 27F		Current	Dec 26F	Jun 27F
Global	3.3	3.4	US: Federal Funds Rate	3.875	4.125	4.375	US 10Y yield (%)	5.11	4.80	4.75
US	2.2	2.2	Eurozone: Deposit Facility Rate	2.50	2.75	2.75	EUR 10Y yield (%)	3.55	3.20	3.20
Eurozone	0.8	1.2	UK: Bank Rate	3.75	4.00	4.25	S&P 500	7,706	8,000	-
Germany	1.0	1.3	Japan: Policy Balance Rate	1.25	1.25	1.75	Gold	4,288	4,600	5,000
Japan	0.7	0.9	China: 7d OMO Rate	1.40	1.40	1.40	EUR/USD	1.14	1.17	-
UK	1.3	1.4					USD/JPY	158	150	-
China	4.6	4.5					Oil WTI (USD/bbl)	91	90	67
							Oil Brent (USD/bbl)	102	95	70

2026 Macro events calendar

October 2026			November 2026			December 2026		
28	CA	Bank of Canada Decision	3	US	Midterm Elections	9	CA	Bank of Canada Decision
28	US	Fed Decision	5	UK	Bank of England Decision	9	US	Fed Decision
29	EZ	ECB Decision				17	UK	Bank of England Decision
29-30	JN	Bank of Japan Decision				17	EZ	ECB Decision
						18	JN	Bank of Japan Decision

	Recent editions	– A broadening tightening cycle, 23 Sep 2026 – 1999 meets 1990, 9 Jun 2026 – The Impact of the Middle East, 10 Mar 2026
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Sea change: the market implications of the threat to the maritime commons

September 25, 2026

The sea is overwhelmingly the most important conduit for global commerce. It is used to transport 80% of the world's trade in goods and 90% of the trade in oil. Even in a world in which services trade is likely to grow in importance owing to AI, the world's oceans remain supreme, with submarine fibre optic cables carrying close to all of the world's internet traffic.

Painstaking multilateral negotiations after 1945 established a legal architecture to enable the almost frictionless movement of commerce across the seas and delimit nations rights' to restrict freedom of navigation. This process learned from the hugely damaging experience of two world wars, and preceding centuries of maritime warfare which had devastating effect on international trade and finance.

Recent years have seen a growing threat to this paradigm. The frequency of maritime supply shocks is increasing at an alarming rate. After the COVID pandemic generated significant friction in maritime supply chains, geopolitics has come to the fore, including Russia's Black Sea blockade in 2022, Houthi attacks on commercial shipping in the Red Sea in 2023, a resurgence in piracy off the Horn of Africa and most recently the twin blockades of the Strait of Hormuz and Bab el Mandeb Strait. This has dovetailed with an apparently reduced US willingness to uphold freedom of navigation, perhaps as part of a broader trend away from supporting the 'rules of the game' that underpin an open international economy. In short, the sea is increasingly a theatre of geoeconomic competition, a trend that could accelerate further with climate induced changes to Arctic maritime routes.

We argue that economists and market participants should be more alive to these risks. The lexicon of tariffs has dominated the market's attention over recent years, but tariff induced supply shocks have proved far less damaging than physical supply shocks. Disruption to ocean commerce has generated outsized impacts on both the prices of goods and the volumes of cross border trade.

We distinguish between three forms of maritime shocks of increasing intensity. From localised disruption that can generate short lived effects on prices, to more serious regional disruptions like the one at present in the Gulf, to global events (disruptions with the capacity to generate a global recession). In the latter category, we argue that the Indo Pacific Region, especially the South China Sea, East China Sea and Taiwan Strait, increasingly qualify as a major risk. We end our report with an analysis on the sensitivity of different economies to maritime shocks, noting that East Asian and European nations stand out as by far the most vulnerable.

Authors

Oliver Harvey

Macro Strategist

+44(20)754-51947



Introduction

It is difficult to overstate the importance of the sea when it comes to world trade. Maritime transport moves four fifths of all goods traded world wide.¹ Over 90% of exported oil is seaborne.² After energy, coal, iron ore, grain and manufactured goods make up the remaining 60% of all seaborne cargo. The 'container revolution,' starting from the late 1960s, dramatically improved the handling efficiency of cargo, and has been seen as one of the primary drivers of global trade growth during the second half of the 20th century.³ The sea doesn't just ship goods, its floor carries electricity and data, with fiber optic submarine cables carrying 99% of the world's international internet traffic.⁴

If the world's oceans were the arteries for the world's second great globalisation from 1945 onwards, the third decade of the 21st century have seen their disruption. The last six years have seen a dramatic rise in maritime based supply shocks.

The 2020 pandemic challenged supply chains globally. This included container shipping, which faced public health restrictions causing a large share of the global carrier fleet to be held in port. More recently, geopolitical events have come to the fore. Russia's 2022 invasion of Ukraine was accompanied by a maritime exclusion zone in the Sea of Azov and parts of the Black Sea, as well as illegal Russian attacks on merchant shipping, paralysing trade in the region. In response to Israel's military activities in Gaza following the 7/10 attacks, the Houthis began illegal attacks on commercial shipping transiting the Red Sea, which resulted in a severe fall in trade volumes. After an international task force formed in 2009 successfully curbed Somali pirate attacks around the Horn of Africa, recent years have seen a resurgence of hijacking. And this year has seen the twin shocks of Iran's closure of the strait of Hormuz and the Houthi blockade on Saudi shipping.

As a result, risk based analysis of maritime choke points has proliferated. We argue, however, that disruptions of this kind should be seen not as isolated risk events, but as part of a trend. An understanding of the legal and historical basis of maritime commerce suggests that the rules of the game are changing. Near frictionless maritime passage is being disrupted by simultaneous challenges to its legal architecture and, more crucially, the US and its Western partners' willingness to enforce these rules. The Trump administration's recent toying with tariffs on the strait of Hormuz, a break from established international norms, is but one example.

There has been far more oxygen dedicated by markets to discussing the impact of tariffs in recent years by the economic community, but the impact of tariffs on trade are relatively trivial compared to the potential disruption caused by maritime trade blockages. Indeed, in this piece, we argue that physical supply shocks to trade can be among the most disruptive types of economic event. The global shipping industry is highly sophisticated, but its network can also prove fragile. We identify three forms of supply shock of increasing intensity: shocks that increase freight rates, regional blockades with some scope for rerouting or substitution of maritime routes, and global disruptions to the freedom of the seas.

1 [UNCTAD](#) statistical bulletin, April 2025

2 [Stanford Energy](#), March 2025

3 Bernhofen et al, "Estimating the effects of the container revolution on world trade." [Journal of international Economics](#), January 2016

4 [United Nations](#)



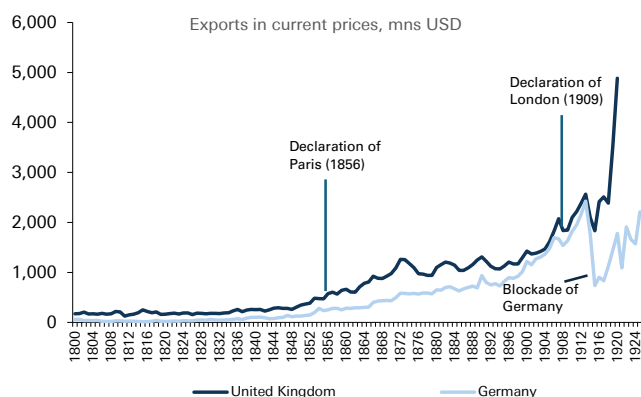
Our work over the last eighteen months has built a framework to understand the implications of shifts in US political economy for the international economy. In particular, we have examined how the US's increasing unwillingness to act as a 'benevolent hegemon' upholding the international order could affect markets, focusing on the dollar, global imbalances and interest rates, and the more frequent use of economic coercion.⁵ Shifts in the maritime arena are yet another example of this shift. Increasingly, the sea is becoming a theatre of geoeconomic competition, while the 'Pax Americana' that has thus far mostly guaranteed free movement across the seas is coming under threat. We end the report with specific case studies on the potential implications of a disruption to commerce around critical maritime routes.

A short history of maritime law and war

"The bare principles of international maritime law are no more than a reasonable compromise between a belligerent's contention, that he should be permitted to stop an enemy's commerce, wherever it is to be found, and the neutral's rejoinder, that he is no-one's enemy, and he has a right to trade and traffic with any country with which he is at peace." **"Official Foreign Office Account of Britain's Economic Blockade of Germany, 1914-1919," published 1937**

The maritime blockade dates back to the Peloponnesian Wars in 405 BC, but it was not until the early 19th century that a recognisably modern framework of the rights of belligerents and neutrals, and legal basis around maritime commerce, started to be drawn up. The blockade and counter-blockade of Napoleonic France saw precedents established for what was acceptable for 'prizes,' namely the ships and goods that warring parties were allowed to seize and confiscate, the law of 'blockade,' and other important concepts.

Figure 1: Growing trade in the 19th century incentivised the creation of the legal architecture of maritime commerce, which was sharply disrupted by the First World war



Source : Deutsche Bank, Federico, G. and Tena-Junguito A. (2019): World trade, 1800-1938: a new synthesis. Revista de Historia Económica-Journal of Iberian and Latin America Economic History, Vol 37, n.1. <https://doi.org/10.1017/S0212610918000216>

Figure 2: The Union's blockade on confederate shipping devastated the South's cotton exports



Source : Deutsche Bank, data from Powell, J. "Cotton, Liverpool and the American Civil War," PhD thesis October 2018

5 See, [FX Special Report](#), "The Unruly Hegemon: what a decline in US leadership means for markets," 13th March 2025, [FX Special Report](#), "How Defence Spending will reduce global imbalances and weaken USD," 6th May 2026 and [EM Special Publication](#), "Geoeconomics 101: implications for the dollar, emerging markets and beyond," 21st January 2026



These were developed further during the mid-19th century as the rapid rise in global trade led merchants, manufacturers and governments to push for greater protections for mercantile shipping. The 1856 Declaration of Paris provided more stringent legal definitions of blockade and what constituted the contraband of war (trade between neutrals and belligerents that assisted the war effort and could therefore legitimately be seized). This was developed upon by the 1907 Hague Conference. Britain, then the preeminent economic and maritime power of the day, took the lead at these conferences, but also pursued a strategy of Imperial Defence in the late 19th century based on the fortification of its coaling stations and ports across the globe, as well as developing a 'two-power' naval standard,' whereby its naval strength would be at least as powerful as its next two rivals combined.

This underscores that while the establishment of legal norms helped maritime commerce become more predictable and orderly, ultimately their survival depended on the willingness of their signatories to enforce them. Maritime economic warfare during the 19th and 20th centuries often saw these norms suspended. The results could be devastating. Britain's economic blockade of Germany during WW1 debilitated the German war effort as well as causing mass starvation. It saw a ~65% collapse in German exports based on trade data from the Federico Tena database (figure 1). The Union's blockade of Confederate Ports during the American Civil War crippled the confederacy's war effort, with export volumes to Britain falling by close to 100%.

After the disruption caused by the second world war, the growth in global trade over the second half of the 20th century once again motivated the codification of rules of the sea. Successive UN conferences from the mid 1950s onwards culminated in the 1982 UN Convention on the Law of the Sea (UNCLOS), which settled several ambiguities in international law. Among these were geographically defined territorial rights for maritime states, including the twelve mile territorial limit, and the concept of innocent passage, as well as the status of international straits and transit passage. Another increasingly important provision of UNCLOS was the establishment of a legal framework governing the laying and protection of undersea cables.

Although the Convention was ratified by almost every industrial nation, the United States has remained a notable exception. Initially, this was due to concerns in the Reagan presidency that to do so would transfer US claims on seabed ownership. Since then, the US has come close to ratification under Presidents Bush and Obama. Nevertheless, as we shall see below, up until recently the US has largely upheld and enforced the broad provisions of UNCLOS through its Navy and Coast Guard as accepted customary law, with the exception of Part XI of the agreement which regulates the mineral resources of the seabed. Indeed, the US has been described as the 'Convention's best supporter,' despite not ratifying the treaty.⁶

Some important principles in present day maritime law

Territorial waters. The sovereign territory of coastal states. Under the 1982 UN Convention on the Law of the Sea this was defined as up to 12 nautical miles from the low water mark of a state's coast. Some ambiguities exist with respect to defining this limit, including whether states utilise continuous baselines to define their coast, or straight baselines (mapping coastal points from a to b in a straight

⁶ Leah Cole, Lawfare, Not Warfare, [Harvard Kennedy School report](#)



line).⁷

International strait. A passage of water used for international navigation between one part of the high seas or exclusive economic zone and another. Where the strait is narrow enough such that it is entirely made up of certain states' territorial waters, the law of transit passage applies. This law provides freedom of movement for ships and aircraft, provided their journey is for the purpose of continuous and expeditious transit. The concept of transit passage was recognized in the 1982 UN Convention in the Law of the Sea.⁸

Innocent passage. The concept of innocent passage is more restrictive than transit passage. It provides a right of passage through the territorial sea of all states, "so long as it is not prejudicial to the peace, good order, or security of the coastal State." Innocent passage can be suspended or safe routes established if there are risks to pollution, smuggling, security concerns etc. The regime of innocent passage was recognised in the 1958 Convention on the Territorial Sea.⁹

Blockade. A blockade constitutes a maritime or airborne operation to prevent enemy and neutrals from entering an airport, port or coastal area. There is no established body of international law that defines a military blockade, but military custom generally agrees on several principles.¹⁰ A blockade must be declared and notified to all states, including details such as geographical scope. A blockade needs to be 'effective,' in other words appropriately policed by military or other means. Finally, it must be impartial, and not applied on an ad hoc or discriminatory basis to neutral vessels of different states. A 'maritime exclusion zone' or 'maritime interdiction zone' constitute less severe versions of blockades. In the former case, they are announced by states as part of military operations whereby enemy ships in the area are liable to be fired upon. In the latter case, they are usually introduced as part of counter-terrorism, piracy or smuggling operations, whereby ships are liable to be boarded and searched as part of operations.

Freedom of the seas and the Pax Americana

Despite certain flaws, UNCLOS is widely regarded by practitioners, governments and scholars as one of the most successful multilateral treaties ever created. Among its achievements have been the regulation of international maritime trade, the de-limitation of coastal territorial and economic claims and ultimately the preservation of the right to free navigation on the high seas.

The United States, along with other Western partners, have been active enforcers of UNCLOS. In the late 1980s, the Reagan administration launched Operation Earnest Will to protect commercial shipping from mines and missiles fired by Iran in the Gulf of Hormuz.¹¹ More recently, the United States was the largest contributor to the multinational anti piracy Operation Ocean Shield in the Horn of Africa between 2009-2016. Outside of military operations, one of the main ways through which the United States has freedom of the seas is through Freedom of Navigation Operations (FONOPS), which involve the transiting of naval vessels through certain stretches of water such as the Taiwan strait in order to push back

⁷ More details on legal [definition](#).

⁸ More details on legal [definition](#).

⁹ More details on legal [definition](#).

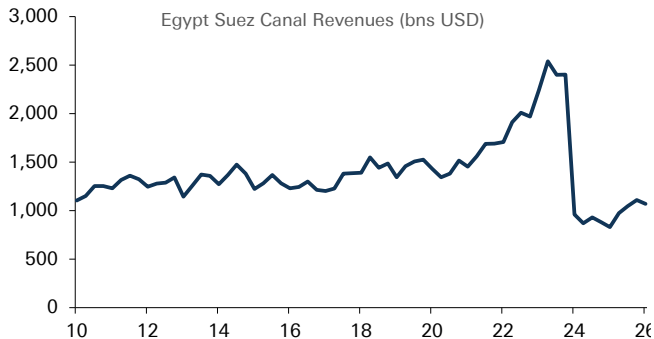
¹⁰ See [Naval College Review of Theories of Blockade](#), Winter, 2021

¹¹ [Revisiting the Tanker War](#), WOTR, March 2024



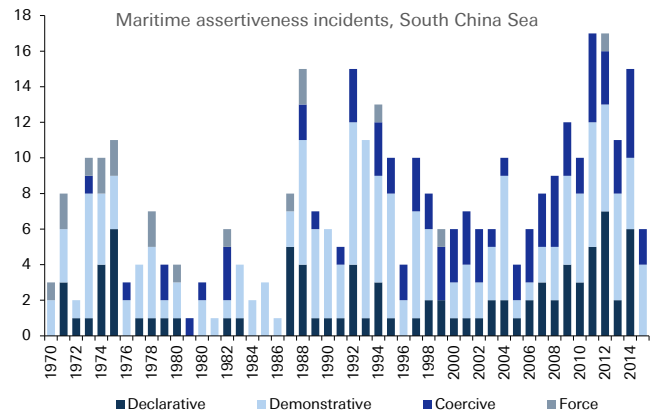
against territorial claims or other navigational restrictions seen as out of line with international law.¹²

Figure 3: Egypt's Suez Canal revenues show little recovery from their collapse in late 2023 due to Houthi attacks in the Red Sea



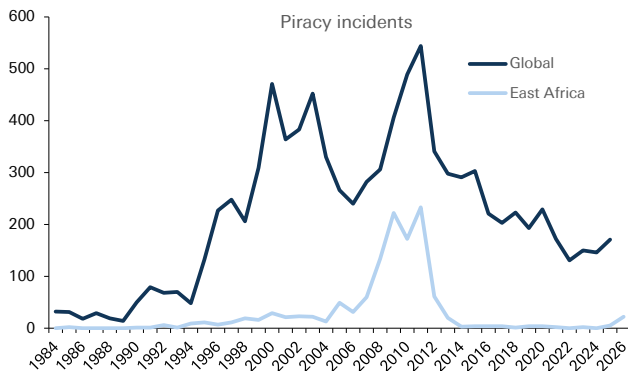
Source : Deutsche Bank, Haver Analytics

Figure 4: Territorial incidents in the South China Sea have picked up markedly in recent years



Source : Deutsche Bank, [Maritime Awareness Project](#)

Figure 5: After a decade of calm, piracy incidents in East Africa are picking up again



Source : Deutsche Bank, Piracy Reports compiled by IMO

Figure 6: Damage to the C-Lion 1 submarine cable has been ascribed to sabotage



Source : TeleGeography Submarine Cable resource

Prior to the Iran war, the most visible challenge to freedom of navigation in the Middle East region was the start of Houthi attacks on commercial shipping in the Red Sea. Although initially focused on ships with unspecified 'links' to Israel, these attacks broadened out to incorporate a wide range of ships, dramatically reducing traffic flows through Suez since 2023. A useful visual manifestation of this are Egyptian Suez revenue numbers which have still to meaningfully recover from the drop of traffic (figure 3). Operation Prosperity Guardian, the taskforce launched by the United States was largely ineffective in restoring commercial traffic. This

¹² for an example of FONOPS, see US Navy [website](#). The US, and other international navies, also enforce UNCLOS through the conduct of counter-terrorism, counter-drug smuggling and environmental operations



blockade was subsequently expanded by the Houthis in July this year to incorporate attacks on Saudi vessels and energy infrastructure.

Another important development has been the growing assertiveness of actors in the South China Sea. Data from the Center for Strategic and International Studies suggests that occurrences of maritime assertiveness have dramatically increased over the last decade (figure 4). The CSIS notes that even if incidents have moved away from the outright use of force towards more coercive and demonstrative actions, China, the Philippines and Vietnam have each sought to both claim and counter each other's military/security, fisheries and natural resource rights.¹³ Tension has been especially pronounced in the Taiwan strait, with China objecting to FONOPs from the US, UK, New Zealand, Australia and Germany in recent years, while both Taiwan and China have conducted military exercises in the strait.

Still further threats to maritime security have come from non-state actors or as a result of asymmetric activities designed to undermine or challenge commercial networks. One of these has been the resurgence of pirate attacks off the Horn of Africa, with 15 reported incidents so far this year after joint-coalition operations had all but quashed Somali pirate attacks over a decade ago (figure 5).¹⁴ Another extremely important area of maritime connectivity that is seemingly under threat concerns undersea cables and pipelines. After incidents such as the sabotage of the Nord Stream 2 pipeline in 2022 and more recently telecommunications cables such as the C-Lion in the Baltic (figure 6), there is increasing focus among policymakers as to the vulnerability and economic cost of attacks on sub-sea infrastructure.

Why is Hormuz so important? The blockade of the straits of Hormuz began in early March this year, shortly after the instigation of US and Israeli military attacks on Iran. Iran's blockade, the US counter-blockade and the subsequent MoU adopted in June this year (in practice suspended later that month), represent major departures from the rules based maritime framework enforced by the US and other Western allies.

First, by deliberately targeting neutral vessels, Iran's 'blockade' was a clear violation of both UNCLOS and the established military practice of blockade. This is simply the latest of numerous examples of Iran's flouting of international law and transit rights through the strait, albeit probably the most severe recent example.

Second, while the US's initial blockade on Iranian ships headed to or departing from Iranian ports had a solid legal basis, the subsequent expansion of this blockade at a global level, including ships of any flag state deemed to be carrying contraband of war, has been widely considered to be legal overreach.¹⁵

Third, Iran's plans to impose a transit toll on vessels through the strait of up to 7% of cargo value is a violation of transit rights established under UNCLOS. Iran is not a signatory to the convention, and it has argued that its actions are permissible under the narrower definition of the rights of innocent passage. Nevertheless, even under the narrower 1958 Convention on the Territorial Sea to which Iran is a signatory, this action is considered by maritime law experts to be illegal.¹⁶

¹³ [NBR Special Report 99](#), May 2022

¹⁴ Al Jazeera, 31st August 2026

¹⁵ See the following [legal analysis](#).

¹⁶ For a detailed analysis, see July 2026 [Al Jazeera report](#)



Most surprising, however, has been President Trump's apparent endorsement of a toll based system, to apparently enable the US to recoup the cost of military operations on Iran. Although some analysts have pointed to navigational or tug fees as offering a potential precedent for these actions, this is misleading. Navigational assistance or tugs are supplied for the safe passage of commercial vessels (e.g. in busy shipping lanes with physical obstacles) and consistent with UNCLOS.¹⁷ By contrast, the US's willingness to countenance transit tolls as an acceptable resolution mechanism to the crisis in turn calls into question the White House's support for the post 1945 architecture. This commitment has also been challenged by other actions taken by the Trump administration, including the recent push for deep sea mining on the US continental shelf, in violation of UNCLOS's provisions for the deep sea bed being a common global good.¹⁸ Still a further concern with respect to the US's posture towards maritime security has been the dramatic reduction in US Navy FONOPS in the strait of Taiwan, which have fallen more than 50% from the previous Biden and Trump administrations.¹⁹

Looking ahead, it seems increasingly likely that the sea becomes another theatre of geoeconomic competition, which may in turn be motivating US policy actions. When it comes to Greenland and the Arctic, for example, climate induced changes to maritime routes, especially the possible emergence of viable routes through the Northwest Passage and Transpolar Sea Route, could have informed recent US policy shifts.²⁰ Similarly, there have been reports that Russia is considering withdrawing from UNCLOS on account of territorial claims over the Arctic. UNCLOS governs the Arctic, although a number of states have recently sought to extend the limit of their claims on the continental shelf in that region.²¹

Further examples of increased US attention on strategic maritime considerations in its hemisphere were the respective export blockade on Venezuela earlier this year, whose legality was a subject of discussion although ultimately fell under the umbrella of sanctions enforcement, fuel blockade in Cuba and attention directed towards the Panama Canal.

From global commons to economic warfare?

Markets should care about these developments for the simple reason that supply shocks emanating from major disruptions to maritime trade and digital routes can have severe global economic ramifications. In previous research we have discussed how shifts in US policy over recent years have called into question the maintenance of 'global public goods,' such an open international economic environment. Much in the same way as the Trump administration's increased use of economic coercion using tariffs has undermined confidence in multilateral trade institutions and the 'rules of the game' when it comes to settling trade disputes, should the US cease to enforce international maritime law, the risk could be that a 'race to the bottom' ensues in which state actors can leverage maritime coercion for political or economic purposes.

17 FT Alphaville's coverage of this issue in an August 3rd report appears to disregard widely accepted international customary maritime law as well as UNCLOS

18 See, [Lowy Institute Report](#), 24th October 2025

19 See "[Breaking Defence](#)" August 2026

20 Greenland will be strategically important for both of these routes, which may have motivated US territorial claims. See [Belfer Centre](#), January 2025

21 See [Arctic Institute](#) report April 2024

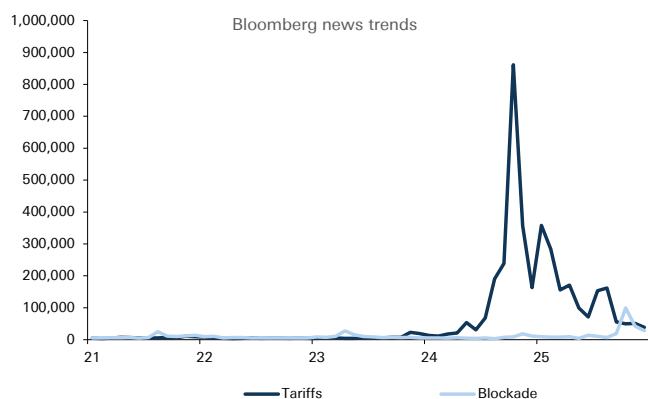


Surprisingly, maritime trade disruptions has still meaningfully lagged behind tariffs in the market lexicon (see Bloomberg news trend for 'tariffs' and 'blockade' figure 7). Recent years has shown that physical supply shocks have larger economic effects than tariffs, however. It is difficult to detect a protracted impact on US import prices, for example, from US tariffs imposed either in the first or second Trump administration, but the impact from the pandemic, Russian invasion of Ukraine and more recently the Hormuz crisis has been much more pronounced (figure 8). More formally, the IMF recently modelled the impact of port delays on US inflation at port and product level, finding that a one hundred hour delay in ports lifts US inflation by 50bp over the following five months.²²

The economic impact extends well beyond price effects. For countries that are especially exposed to maritime imports, with limited substitutability in the near term for affected products, import volumes will also be hit. This was evident during the recent disruption to energy markets in Asia, which saw Bangladesh, South Korea, Thailand and Vietnam imposing energy rationing, and work from home requirements ordered in Indonesia and Malaysia. Figures 9 and 10 track import volumes in Korea and China and clearly demonstrate the impact both on energy imports (first chart) and overall import volumes (second chart) of recent maritime disruptions. In Pakistan, the force majeure declared by Qatar on LNG deliveries has forced Pakistan to instigate widespread load shedding on its electrical grid.

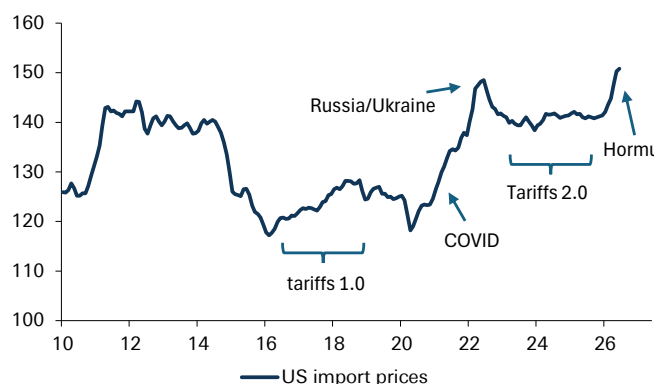
At a global level, the IMF's downgrades to global growth on account of tariffs last year amounted to 0.2% over two years, but its downgrade in the aftermath of the Ukraine war was a cumulative 1%.²³ In short, then, in a world in which supply shocks have become the dominant concern for economic policymakers, the risks to maritime commerce should occupy the first rank of worry.

Figure 7: Maritime disruption has featured a lot less than tariffs in the market's mind



Source : Deutsche Bank, Bloomberg Finance LP

Figure 8: Even as physical supply disruptions have had larger economic effects than tariffs



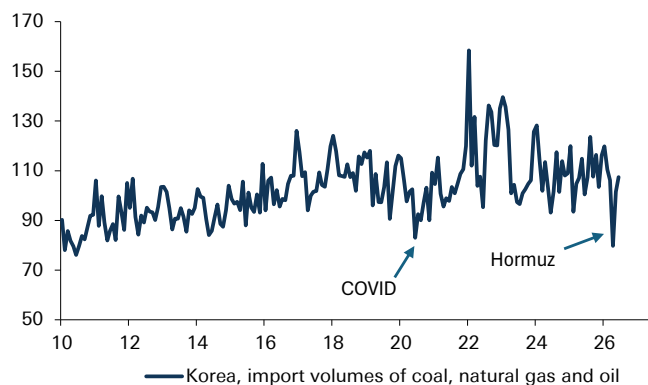
Source : Deutsche Bank, Haver Analytics

22 IMF [Working Paper](#), "From Ports to Prices," February 2026

23 IMF WEOs April 2022 and October 2025.

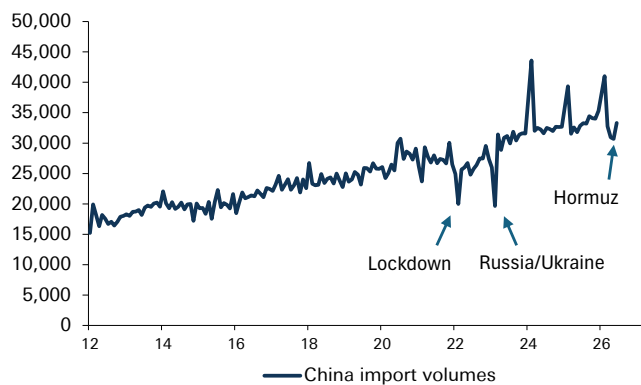


Figure 9: Physical supply shocks have also had material effects on import volumes, especially in Asian economies such as Korea...



Source : Deutsche Bank, Haver Analytics

Figure 10: ...or China



Source : Deutsche Bank, Haver Analytics

Assessing the importance of maritime shocks

With this in mind, the first step for market participants to be better prepared for maritime shocks is to distinguish between their severity. We distinguish between three broad kinds of shocks, and identify some salient characteristics of each.

1. Localised maritime disruption, factors generating higher insurance costs, country specific sanctions.

Examples of the least severe, but still relevant, maritime supply shock would include the sanctioning of Russian seaborne oil by the United States and EU after 2022, Houthi attacks on commercial shipping in the Red Sea in 2023 or pirate attacks off the Horn of Africa.

More limited maritime shocks of this nature can be disruptive, but their economic effects are generally short lived owing to the capacity of the international shipping industry to substitute routes or exporters or otherwise manage risks. They are more likely to manifest themselves in terms of **price** rather than **volume** effects.

Important transmission mechanisms for prices are increased freight rates, which in turn owe to higher insurance costs or demand bottlenecks. This was evident, for example, during the surge in Somali piracy attacks in 2011 which resulted in increased costs for the shipping industry due to ransom and kidnap insurance, but in many cases did not stop ships transiting close to the horn of Africa.²⁴ In these more localised disruptions, shipping companies can choose to re-route. This was the case following the Houthi attacks in 2023 which saw numerous vessels choose to take the lengthier Cape of Good Hope route from Europe to Asia and is reflected in the recent rise in ship ton miles relative to ship tons transported internationally according to UN data (figure 12). Similarly, in response to Russia's blockade of the Black Sea, Ukraine was able to successively reroute a proportion of its grain exports via the Danube or overland routes.²⁵ Still a third option available to ships in these more limited disruptions include re-flagging vessels. Russia has employed re-flagging of oil carriers as part of its 'shadow fleet' following Western sanctions.

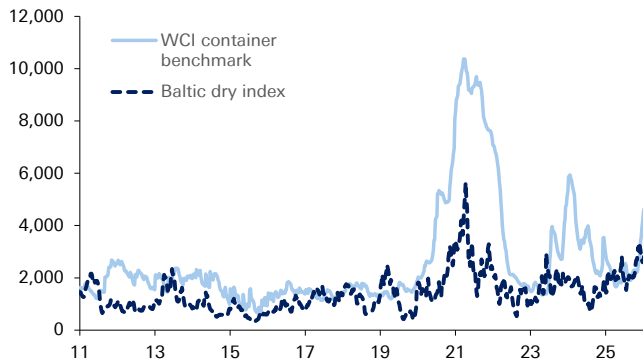
²⁴ See 'From the Oceans' report [here](#)

²⁵ [Council for Foreign Relations Report](#), February 2024



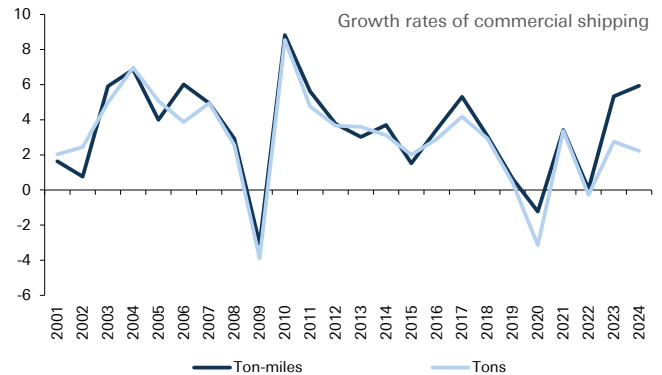
Indeed, 'flags of convenience' are often employed to avoid both the threat of interception by belligerent states, regulatory requirements and in certain cases tax and registration costs.²⁶

Figure 11: Container freight rates tend to be more sensitive to geopolitical shocks, but both container and bulk shipping have been affected



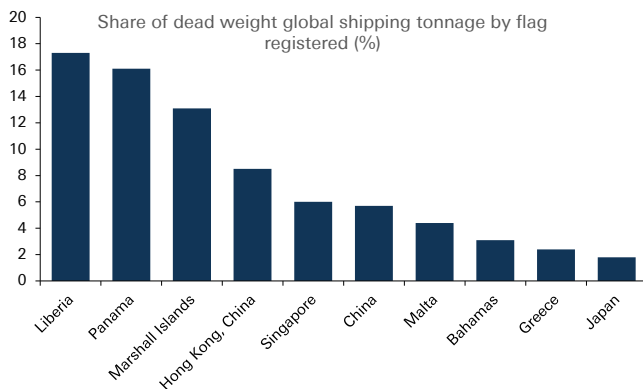
Source : Deutsche Bank, Bloomberg Finance LP

Figure 12: Distances travelled are outstripping tons transported in terms of growth rates of maritime trade, owing to recent disruptions



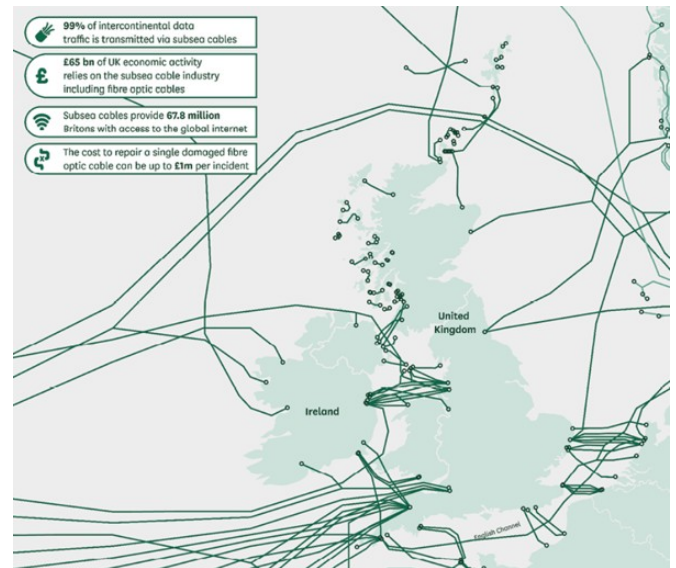
Source : Deutsche Bank, UNCTAD Review of Maritime Transport

Figure 13: 'Flags of convenience' dominate global commercial shipping



Source : Deutsche Bank, UN Maritime factsheet

Figure 14: The UK is one of the most vulnerable countries globally to disruption to sub sea cables



Source : House of Lords

Another example of more limited maritime disruption would be to undersea cables. A recent UK House of Lords Joint Committee addresses the risks associated with the UK's extensive network of over 60 undersea cables. It concluded that the severing of a number of cables was manageable from a communication continuity perspective due to redundancy and rerouting, but that a severe disruption (large

²⁶ [PBS report](#)



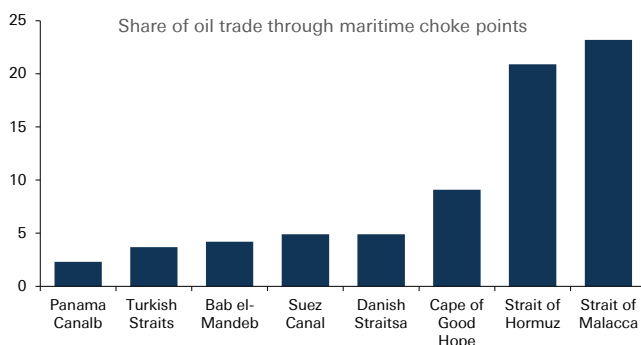
scale attacks on cable infrastructure) could have more severe economic and financial consequences.²⁷

2. Regional disruption or blockade with limited capacity for trade substitution.

An obvious example of this more severe form of maritime supply shock is the current crisis in the Gulf. Owing to the share of crude oil transported via the Strait of Hormuz (21%) and Bab el Mandeb (4.2%), as well as the share of LNG (20%) transported through the region, and its importance for other commodities such as nitrates, the effect of the twin blockades of the Straits of Hormuz and Bab el Mandeb has proved much more difficult for suppliers and the international shipping industry to work around. Moreover, war risk premiums from the insurance market have become prohibitively high, or in certain cases cancelled altogether for high risk vessels.²⁸ Re-flagging of vessels can be of limited use in the event of maritime blockade, where interdiction extends to all commercial vessels regardless of flag.

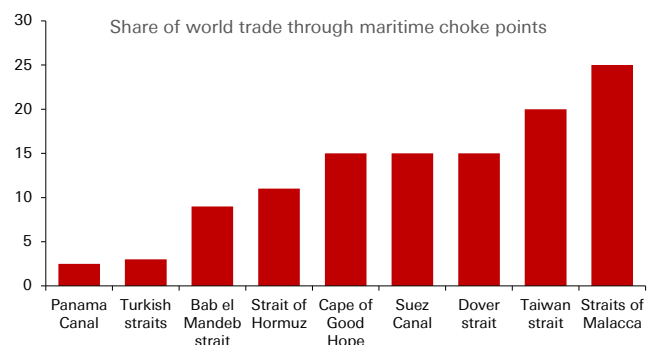
Figures 15 and 16 illustrate the relative importance of different maritime choke points for oil and world trade respectively. It should be noted, however, that the share of trade transiting through these locations does not, on its own, determine the severity of maritime shocks. Geography matters. There are alternatives to transit through the strait of Malacca for commercial vessels transiting to East Asia, even if these involve higher costs and lengthier journeys. By contrast, the 'twin blockade' of Hormuz and Bab el Mandeb has proved problematic due to the fact that it effectively has disrupted both of the main routes of supply for Saudi oil even if, over time, land based alternatives will become available. Correspondingly, the economic impact of these developments has been more severe, and manifested themselves in physical supply shortages, in other words **volume** as well as **price** effects.

Figure 15: The straits of Malacca and Hormuz are overwhelmingly the world's most important transit points for oil



Source : Deutsche Bank, IEA

Figure 16: But there are multiple choke points for world trade, including the Taiwan strait, Dover strait and Suez Canal



Source : Deutsche Bank, various sources including the UN, IEA, think tank reports

²⁷ [House of Lords Report](#) September 2025

²⁸ [Lloyds List](#) March 2026



Figure 17: The 'twin blockades' of Hormuz and Bab El Mandeb have meaningfully hindered global oil exports



Source : Deutsche Bank, map generated by Copilot

3. Global disruption or an extremely severe regional disruption.

The third category of severity has not occurred in market participants' living memory with the only analogies being the shocks to international commerce occurring during the First and Second World Wars. Maritime blockades imposed during these conflicts, and reciprocal attacks on commercial shipping both of belligerents and neutrals across the world, devastated international trade and financial flows.

Outside of a world war, however, another form of severe maritime shocks would involve disruption in the Indo Pacific maritime region, more specifically the South China Sea, East China Sea and Taiwan Strait. First, the region is immensely important from a global trade perspective. The South China Sea carries over a third of global maritime trade and hosts the world's six busiest container ports, while the wider Indo Pacific region host two thirds of global commerce. The straits of Taiwan and Malacca are the two single most important global maritime choke points. Second, economies in the region are highly dependent on maritime commerce for export and energy needs, with economies such as Japan, Korea and Singapore almost entirely dependent on maritime trade of their energy needs. Third, the region carries a complex and dangerous set of competing territorial, economic and strategic claims.



Figure 18: The two most important choke points for global trade are in the South China Sea and Taiwan Strait



Source : Deutsche Bank, map generated by Co-Pilot

Fully elucidating the complexities of this region are beyond the scope of this paper, but briefly these revolve around a series of bilateral and multilateral territorial claims. First, in the East China Sea between China, Japan and Taiwan, principally focusing on the Senkaku/Diaoyu islands and to a lesser extent the contested atoll of Okinotori. Second in the South China Sea between China, Malaysia, Vietnam and the Philippines predominantly focusing on the Spratly Islands, Scarborough Shoal and Paracel Islands. Third, with respect to contested sovereignty over Taiwan. These are in turn related to economic claims concerning natural resources (especially the exploitation of potential oil and gas reserves) and fisheries.²⁹

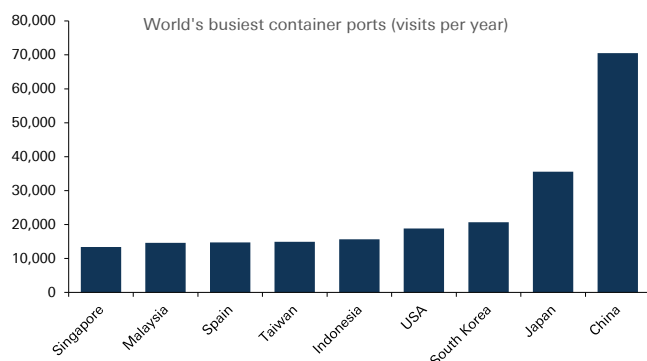
The legal architecture under UNCLOS has been used as an arbitration mechanism over many of these claims (for example recently with respect to both Okintori and in a dispute between the Philippines and China over the nine dash line which the latter claims provides wide ranging sovereign and economic rights over the South China Sea). Even more importantly, UNCLOS has provided the legal basis for the United States and other allies to assert freedom of navigation rights in the region. Among these has been the Taiwan Strait, where UNCLOS's definition of an international strait has underpinned the US and other nations' rights to both freedom of navigation and overflight, as well as military exercises.³⁰ This illustrates once again the importance of the existing legal architecture governing maritime rights, and the potential negative spillovers should the architecture or states' willingness to enforce it, unravel.

29 For a useful discussion of these claims, see Triki, ".[Understanding the South China Sea Crisis: State Claims, International Interventions and Implications](#)," Asia and the Pacific Policy Studies, January 2026. See Also, [War on the Rocks](#), One Sea, Near Sea, Whose Seas? May 2024

30 See, for example, [CIMSEC](#), Naval Operations and the Right to Operate Freely in the Taiwan Strait, 2022

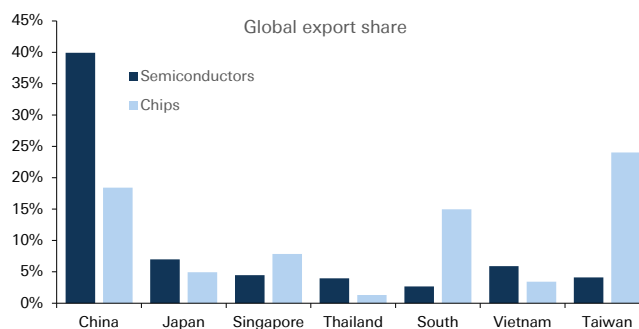


Figure 19: Almost all of the world's busiest container ports operate in the Eastern Pacific



Source : Deutsche Bank, Atlantic Council

Figure 20: The region also dominates market share in electronics



Source : Deutsche Bank, Observatory of Economic Complexity

In this context it is disquieting that both military planners and policy analysts have been dedicating increasing attention to the possibility of China adopting an economic approach, largely through maritime channels, to attempt to settle the long standing dispute over Taiwanese sovereignty.³¹ On the one hand, a blockade or quarantine of Taiwan might be seen as a positive relative to the alternative of a more kinetic military approach, by carrying less human costs, as well as reducing the scope for intervention by other powers.³² On the other hand, such an action is designed to maximise the economic costs for Taiwan, with major consequences for the global economy.

Of even greater concern would be a Taiwan crisis that spills over into a broader regional maritime crisis. Recently, rivals to China, especially Japan and the Philippines, have sought to enhance security cooperation with Taiwan, which risks creating what Chinese military planners have called a 'Three Seas Problem' in which the entire eastern seaboard of China becomes one theatre of operations.³³ Another concern among analysts relates to a potential counter-blockade by the United States of the Strait of Malacca. Still a further potential axis of escalation concerns the Sunda Strait. Indonesia has been increasingly assertive in its sovereign claims over its archipelagic waters, leading some US analysts to worry a non-aligned Indonesia could close the strait as part of its non-alignment policy.³⁴

The importance of Taiwan in the semi-conductor supply chain is well understood, as illustrated by the proliferation of reports on the subject, and even a crisis that was limited to simply a blockade of that country, in the absence of any wider escalation, could lead to significant losses of global output.

The knock on consequences of wider regional maritime disruption, even if this were confined to the purely economic sphere, would be far more disruptive. Consider a scenario in which a Chinese maritime blockade or quarantine on Taiwan was met

31 [Atlantic Council Strategy Paper](#), "A Maritime Blockade of Taiwan By the People's Republic of China, December 2023

32 The difference between a blockade and quarantine largely boils down to whether it is executed via military actors like naval forces or non-military forces, in China's case the People's Armed Forces Maritime Militia. See [here](#)

33 Luo, Shuxian "A Rock and China's Three Seas Problem," [War on the Rocks](#), August 2026

34 [CSIS Report](#), "Why the US Navy Must Resume FONOPS in Indonesia," December 2025



with a counter-blockade from the United States and its allies Japan and Korea extending from the Taiwanese Strait to the Strait of Malacca. Such an approach may be an appealing alternative to direct military confrontation with China. This would have a chilling effect on maritime trade throughout the region, leading to sharp rises in insurance premiums, port closures and the practical cessation of commercial shipping. With the region dominating global container trade (figure 19), electronics exports (figure 20), as well as myriad other consumer and intermediate goods, the effect on international trade would only be comparable to twentieth century world wars, and would almost certainly be consistent with a global recession.

Assessing country vulnerability

The final port of call for this report is to assess how threats to the freedom of the seas impact financial variables.

First, the increased frequency of maritime disruption is a negative supply shock, and as such likely to put upward pressure on inflation and downward pressure on demand. The former is likely to dominate when it comes to interest rates. Under a severe maritime disruption, this may be exacerbated by the pressure placed on exchange rates, especially if central banks seek to curb volatility by defending currencies. For example, in the event of a severe disruption to Taiwanese trade, it is possible to envisage the central bank stepping in to prevent financial markets from exacerbating the shock, utilising foreign exchange reserves and therefore principally treasuries in doing so.

Figure 21 provides an illustrative scorecard of relative vulnerability to maritime shocks. Economies are more vulnerable if they have a larger share of maritime trade or energy imports relative to GDP, or if they are an island state. They are less vulnerable if they have a stronger navy (measured by WDMMW database). Strikingly, the most vulnerable economies are all located in Asia or Europe, with Singapore, the Philippines, Vietnam, the Netherlands and Korea all highly open economies with a heavy reliance on maritime trade, but with limited naval resources. In contrast, larger economies with multiple land borders like the US, Russia, China and Brazil score as much less vulnerable. Translating this into actionable trade or investment ideas is not the point of the table below, but it should serve as a guide for investors and corporates as to the potential vulnerability by country to severe maritime disruption, and the increasing use of the sea as a theatre for geoeconomic competition.



Figure 21: Maritime vulnerability scorecard

	Maritime imports as share of GDP	Seaborne imports of oil and gas % GDP	Naval power index	Island or limited land alternatives?
Singapore	17%	8%	43.2	Yes
Philippines	11%	4%	37	Yes
Vietnam	20%	3%	2	-
Netherlands	14%	3%	37.3	-
Korea	11%	3%	122.9	Yes
Australia	10%	0%	48.9	Yes
Belgium	17%	0%	7.6	-
Spain	7%	2%	56	-
Japan	6%	1%	121.3	Yes
Germany	12%	0%	54.7	-
Indonesia	5%	2%	137.3	Yes
UK	4%	1%	88.3	Yes
India	4%	4%	100.5	-
Italy	6%	2%	80.7	-
France	7%	1%	92.9	-
Turkiye	8%	0%	80.5	-
Brazil	7%	0%	39.9	-
China	3%	2%	319.8	-
Russia	4%	0%	242.3	-
USA	4%	0%	323.9	-

Source : Deutsche Bank, Bloomberg Finance LP, WDMWW database



Appendix 1

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UNCERTAINTY



Deutsche Bank
Research

M&A Spotlight:

When rates rise ...

17 September 2026

Luke Templeman | Thematic Strategist | Luke.Templeman@db.com | +44 207 541 0130
Galina Pozdnyakova | Research Analyst | Galina.Pozdnyakova@db.com | +44 207 547 4994

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

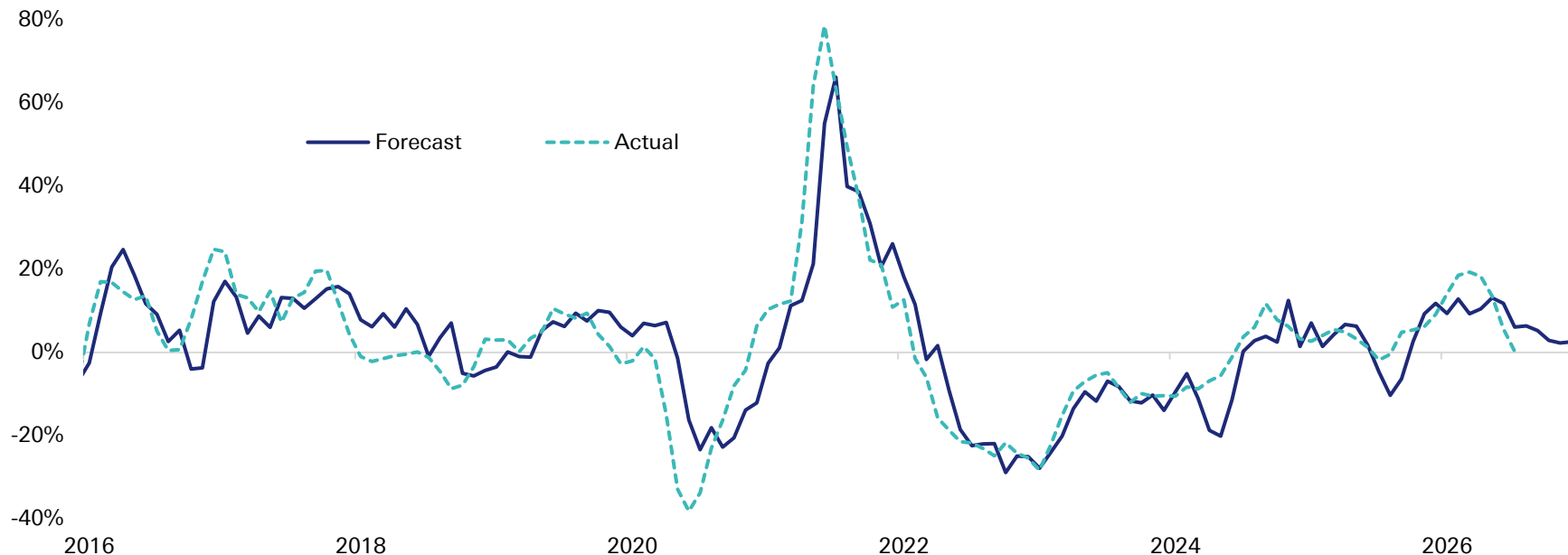
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GEOPOLITICS

Our M&A forecast model: In the US we see low but positive growth over the coming 3 months
Mid-term elections are incentivising some CEOs to wait for greater legislative certainty. That is combining with the reemergence of greater geopolitical uncertainty and the second round of rising oil prices to dampen near-term forecasts



Our 3m-forward forecast for growth in US M&A deal numbers

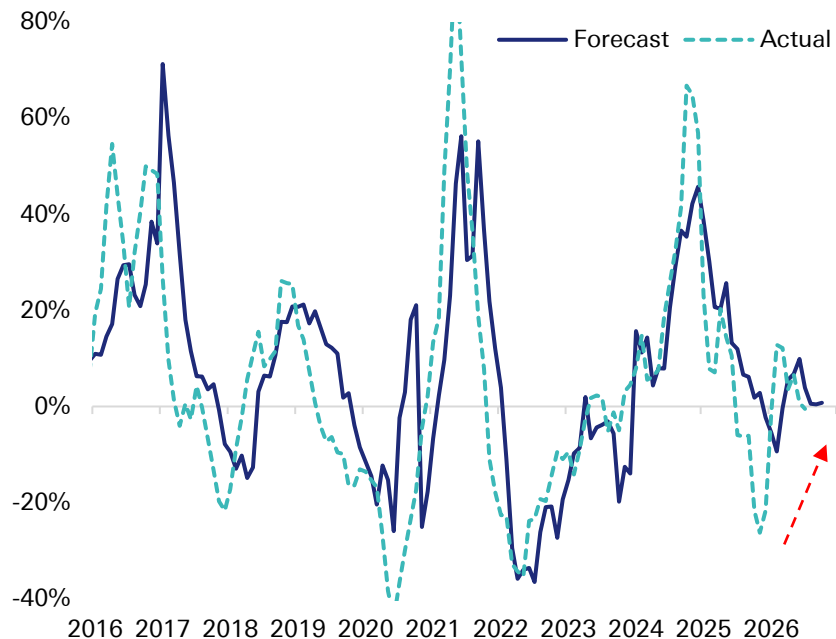


Source: Bloomberg Finance L.P., Dealogic, Deutsche Bank

Our M&A forecast model: In the UK and Europe, our model shows similar low but positive growth
In the UK, this will be encouraging given the market is recovering from a significant drop over the last 24 months

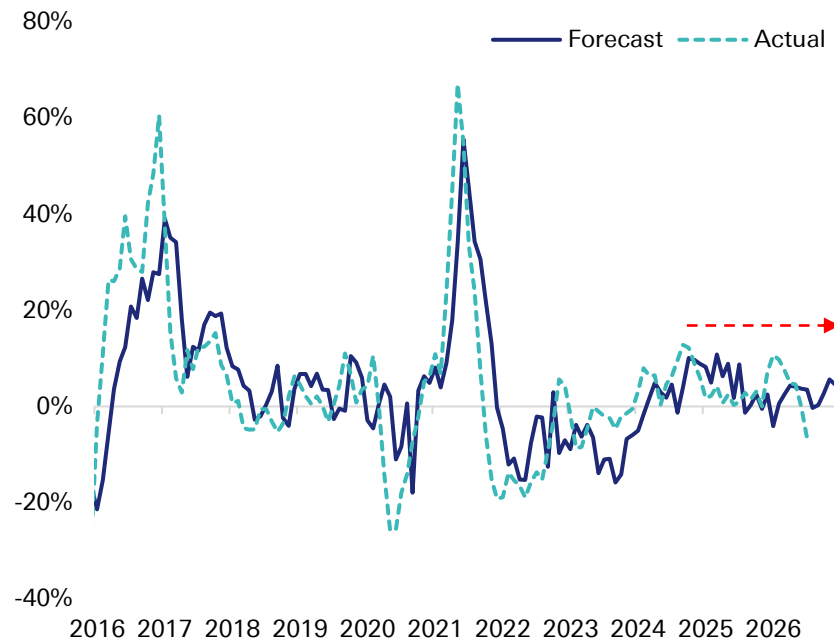


Our 3m-forward forecast for growth in UK M&A deal numbers



Source: Bloomberg Finance L.P., Dealogic, Deutsche Bank

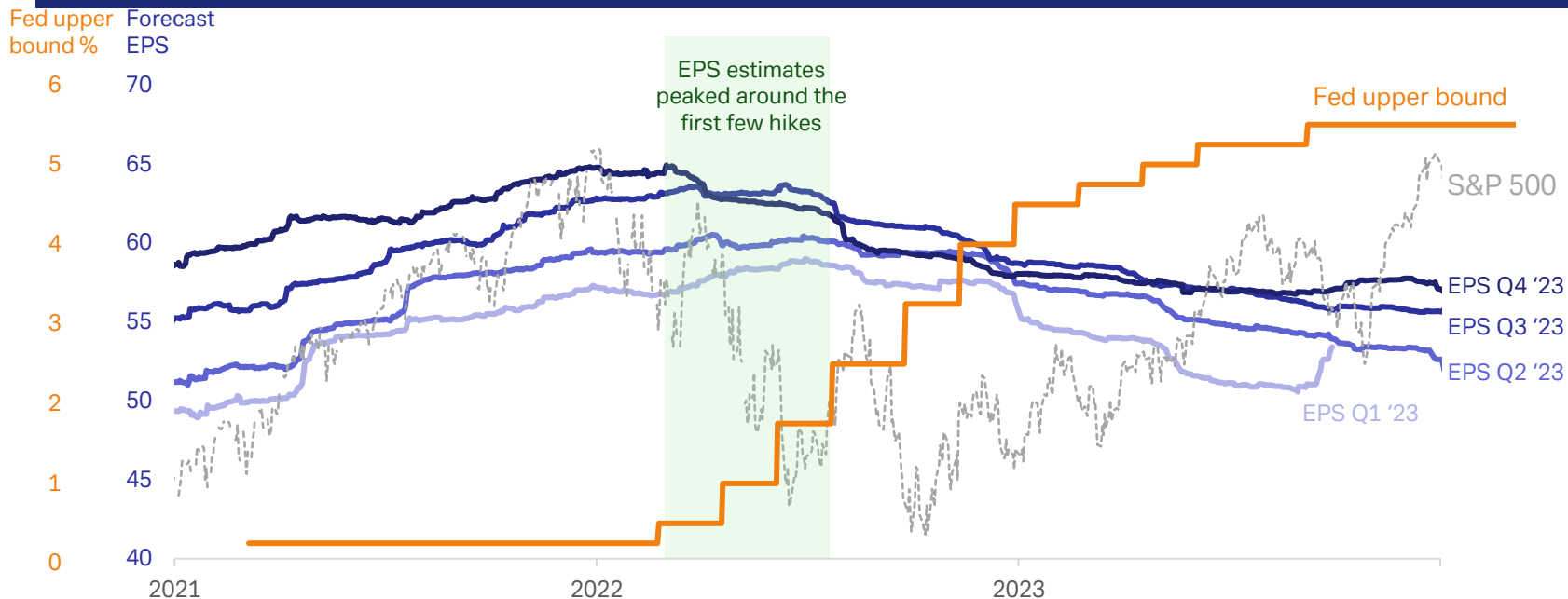
Our 3m-forward forecast for growth in Eurozone M&A deal numbers



Last time around ... Earnings estimates peaked just after the 2022-23 hiking cycle began
It took a few rate rises for investors to materially cut their EPS estimates. However, the extent of the hikes in that cycle were significantly higher than they are likely to be this time around.



EPS estimates for the S&P 500 during the 2022-23 hiking cycle



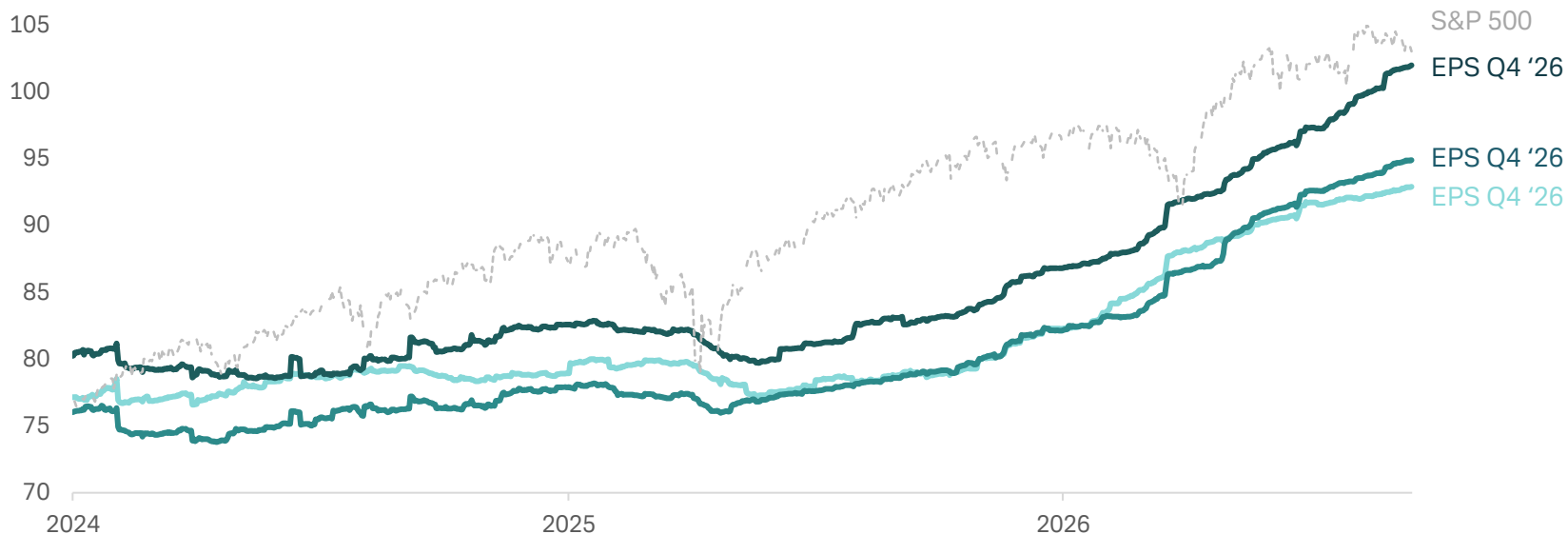
Source: Bloomberg Finance L.P., Deutsche Bank

This time around ... Earnings estimates have accelerated into the rate hike

So far, expectations for rate rises have not caused a substantial rerating of EPS estimates, including in tech companies. The effect may become more pronounced if bond yields breach the 5.00% barrier for a longer period of time.



EPS estimates for the S&P 500 for selected future quarters



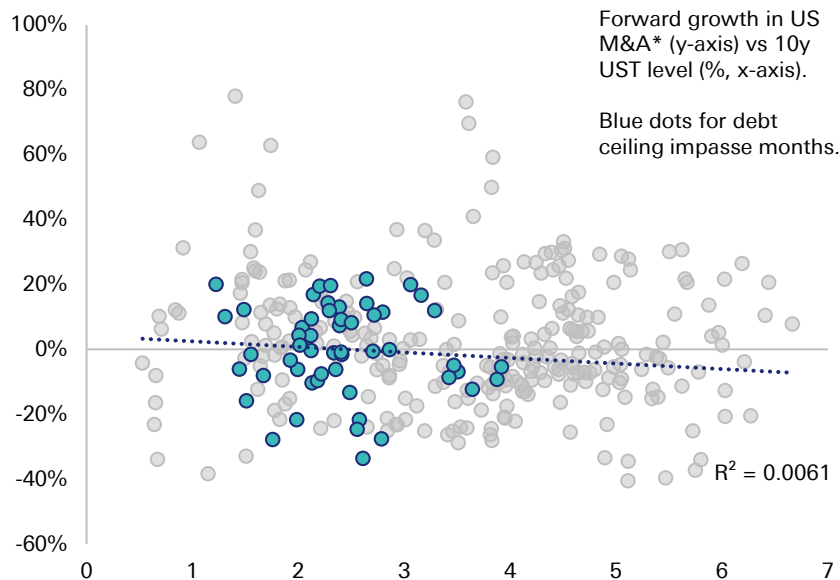
Source: Bloomberg Finance L.P., Deutsche Bank

Positive news: Throughout recent history, M&A is relatively unaffected by treasury yields

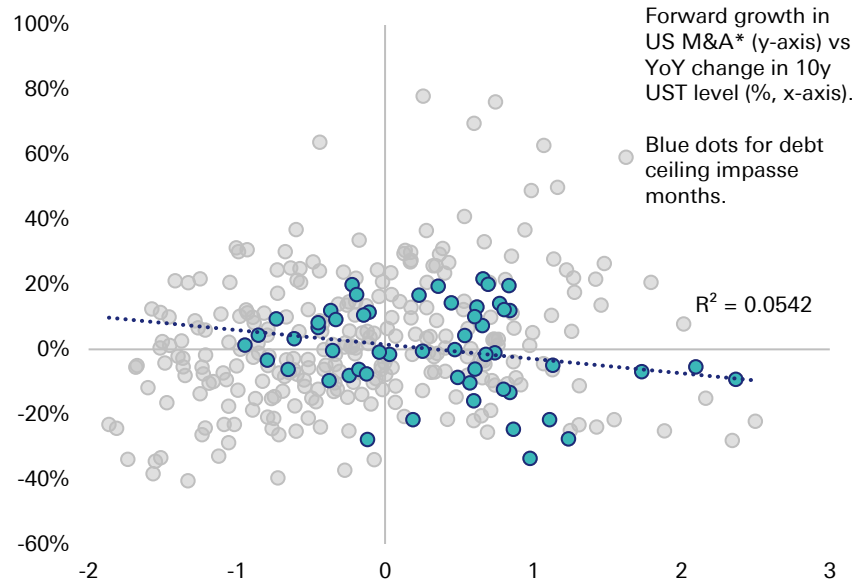
The only material exception to this is when there is an extreme change in the steepness of the 2-10 yield curve – ie. When the yield curve steepens >150 basis points in 12 months.



M&A v the level of the 10yr yield



M&A v the change in the 10yr yield



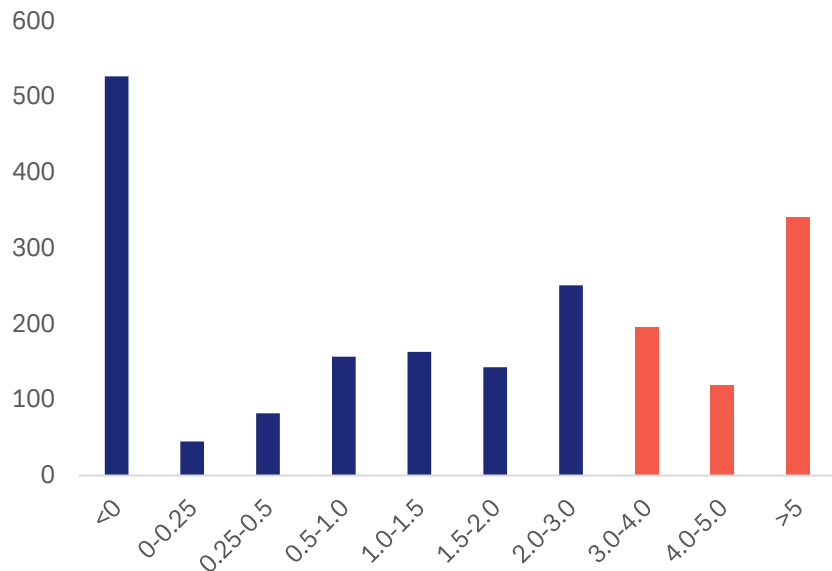
Source: Dealogic, Bloomberg Finance L.P., Deutsche Bank

Make no mistake, higher nominal rates will have a material impact on some firms

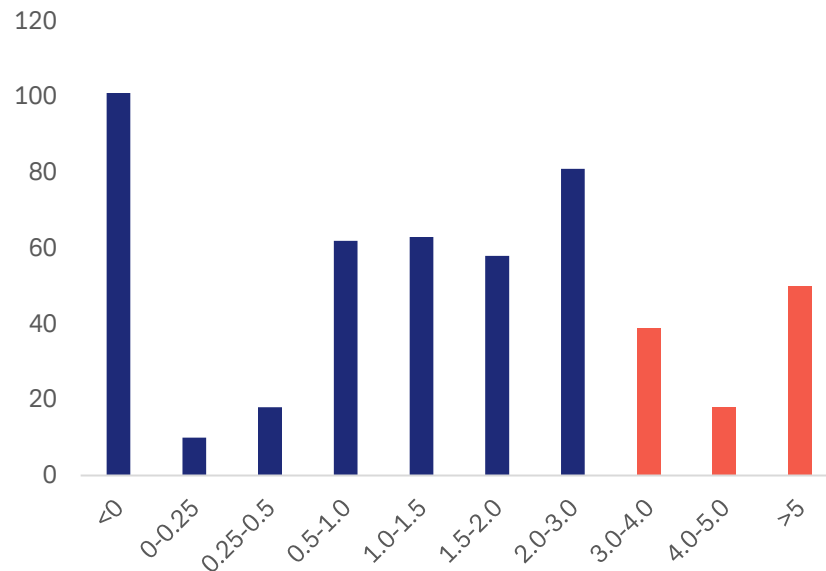
Firms with high debt loads may be more open to M&A opportunities if expectations move towards a more drawn out rate hike cycle



Number of companies with net debt/ebitda in each bucket (Russell 3000)



Number of companies with net debt/ebitda in each bucket (Stoxx 600)

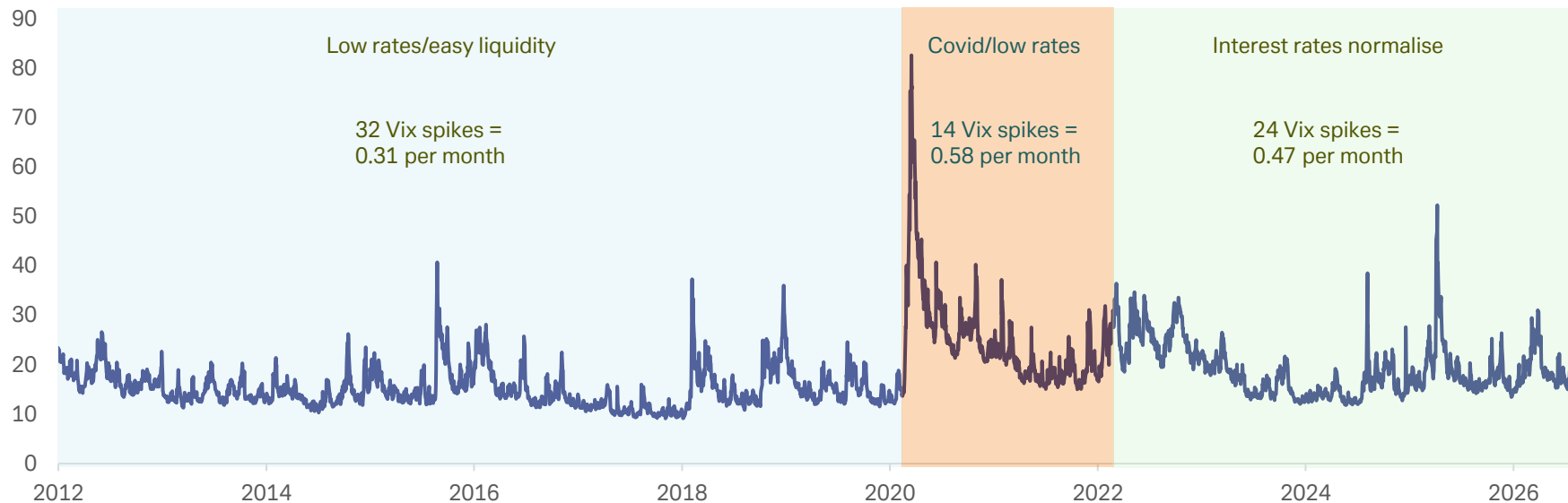


Source: Bloomberg Finance L.P., Factset, Deutsche Bank

The impact of higher rates will be amplified by a market that is showing more volatility
Higher rates tend to bring pre-existing problems to the surface. Given there has been a higher frequency of “volatility events” since rates normalised in 2022, higher rates today could catalyse even more volatility events, even if they are contained.



Vix volatility gauge



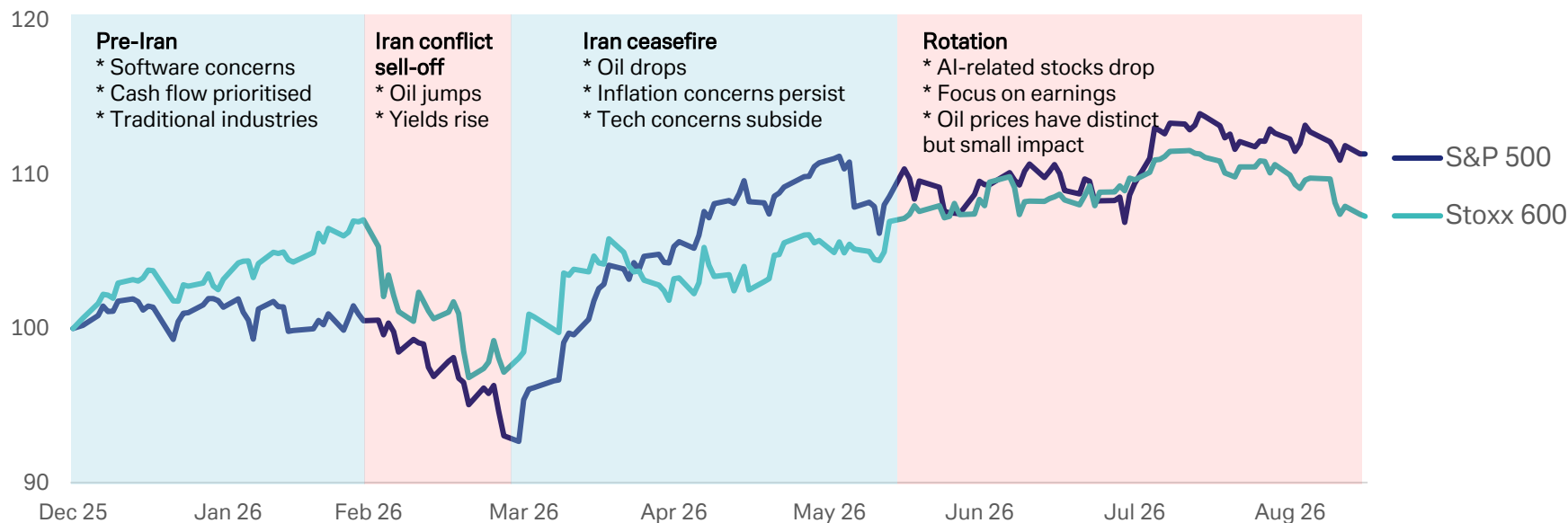
Source: Factset, Deutsche Bank

Already this year, equity markets have lurched between different narratives

Volatility in 2026 has been less dramatic than it was in 2025, however, the changing narratives indicate investors remain skittish



Four stages in equity markets in 2026



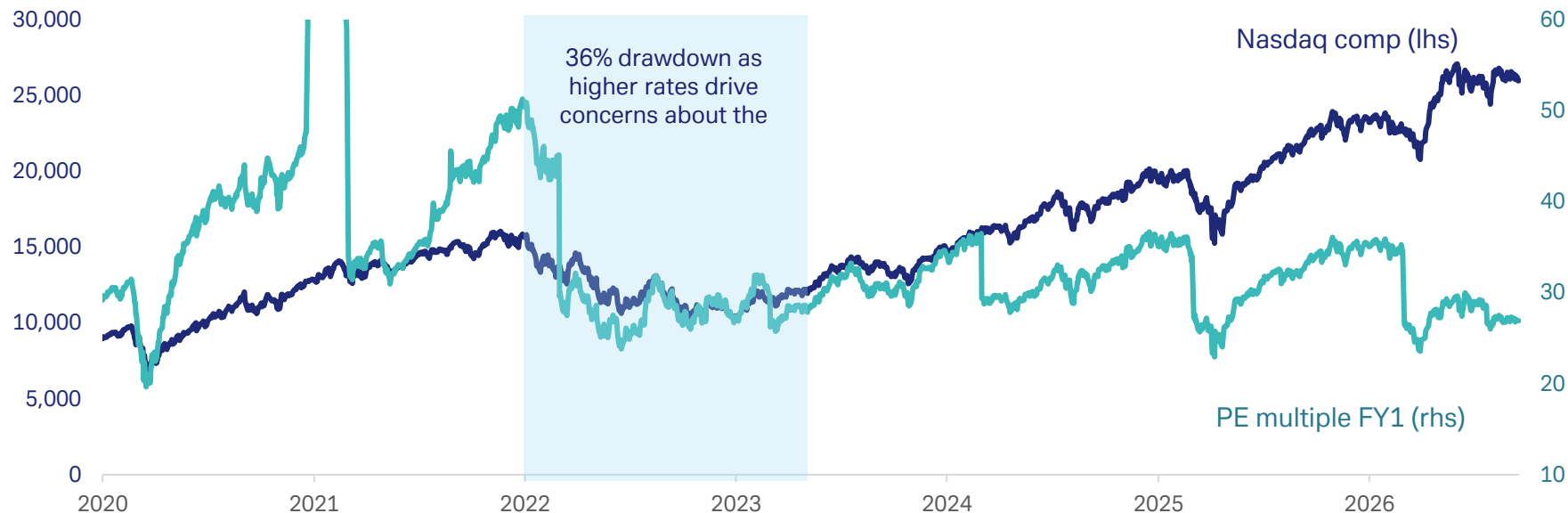
Source: Factset, Deutsche Bank

Broader markets are benefiting from the lack of a tech sell-off, despite higher bond yields

That is a stark contrast to the 2022-23 rate-rise period and is partly due to the strong earning growth at Big-Tech firms. If current earnings forecasts remain intact, current multiples are undemanding compared with recent history. The key thing is expectations about future rate rises and how they affect terminal rates.



Nasdaq composite (price and valuation)



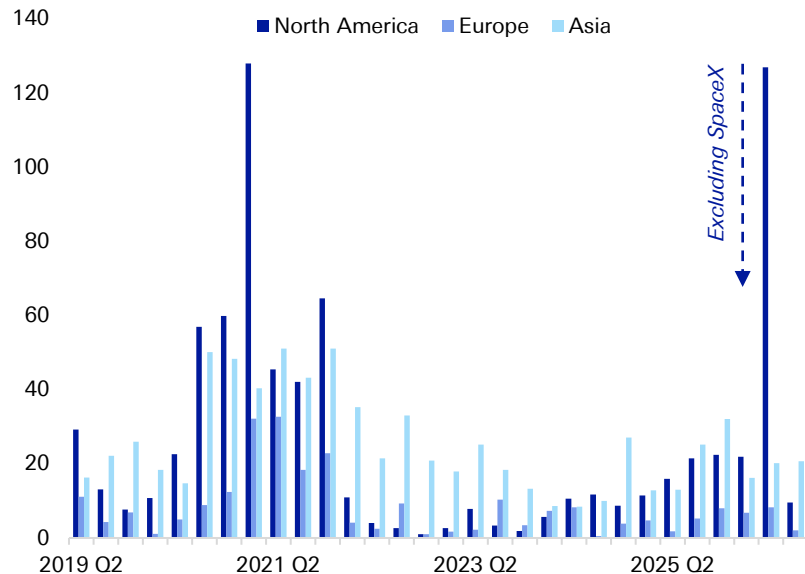
Source: Factset, Deutsche Bank

IPOs are recovering but are still well below 2021 levels ... excluding the very largest

More AI-related businesses are likely to come to the market even after larger firms go public. Tech IPOs have been relatively muted despite the boom in financing this decade. Investors' appetite for AI and big capital markets issuances may push more managers to exit tech holdings while the mood lasts. There have been just c\$7bn in pure-play AI IPOs in the US since 2025.

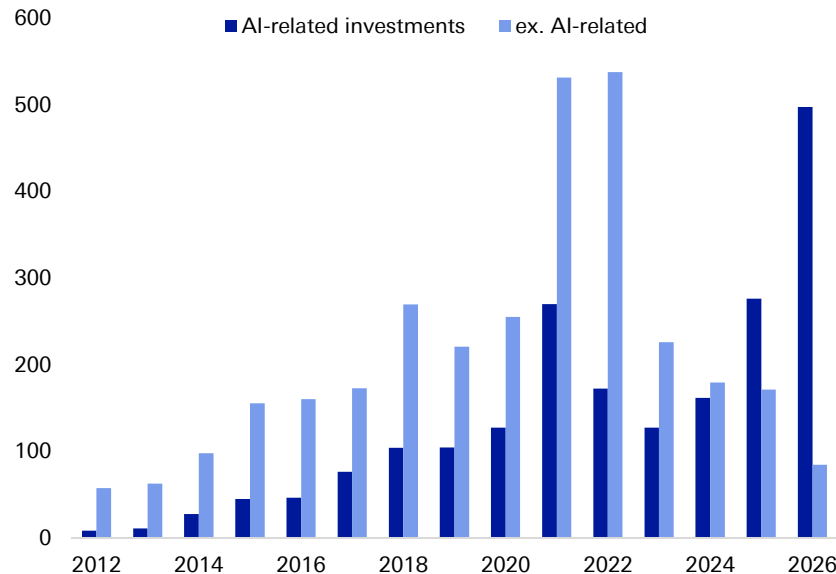


IPO deal values by quarter \$bn



Source: Dealogic, Deutsche Bank

Worldwide VC investment in AI (\$bn)

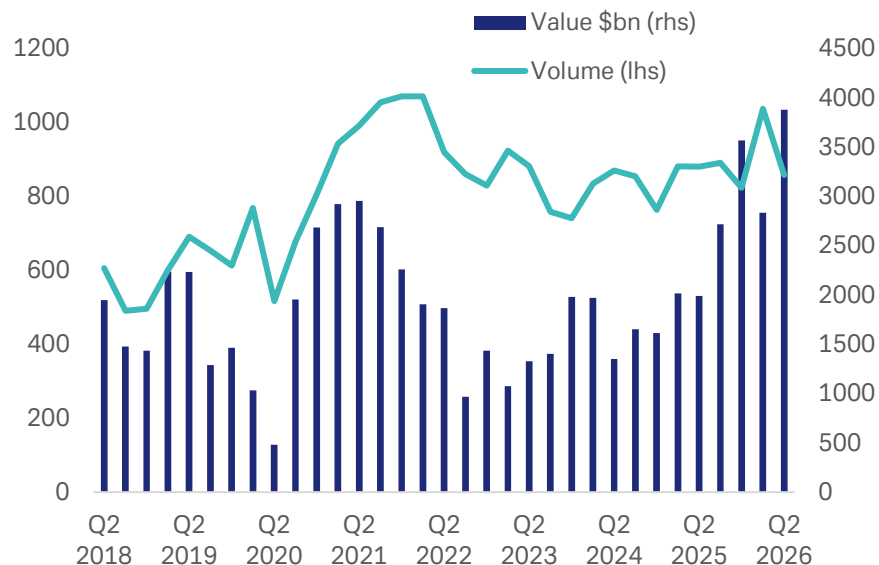


M&A continues to be dominated by big deals

Deal values continue to climb even though deal numbers remain in a tight range. We believe a period of background economic calm is needed for the CEOs of smaller firms to gain the confidence needed to pursue deals.

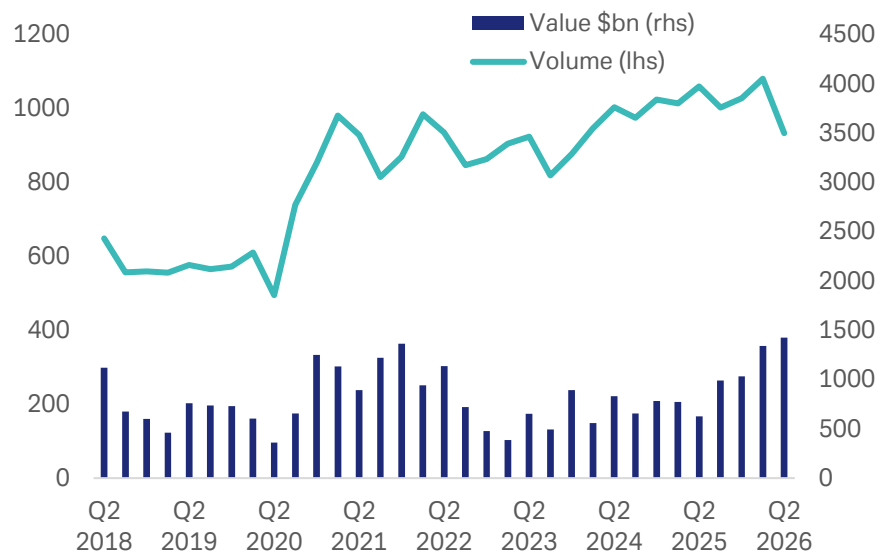


North American M&A



Source: White & Case, Deutsche Bank

Western European M&A



CEO confidence: Leaders expect the economy to be resilient to the current Iran conflict

Still, surveys show that although CEOs are confident about their own company, they are less confident in their industry, and less still in the economy more broadly. That is keeping a lid on some M&A for now.



CEO confidence in the economy 1yr from now



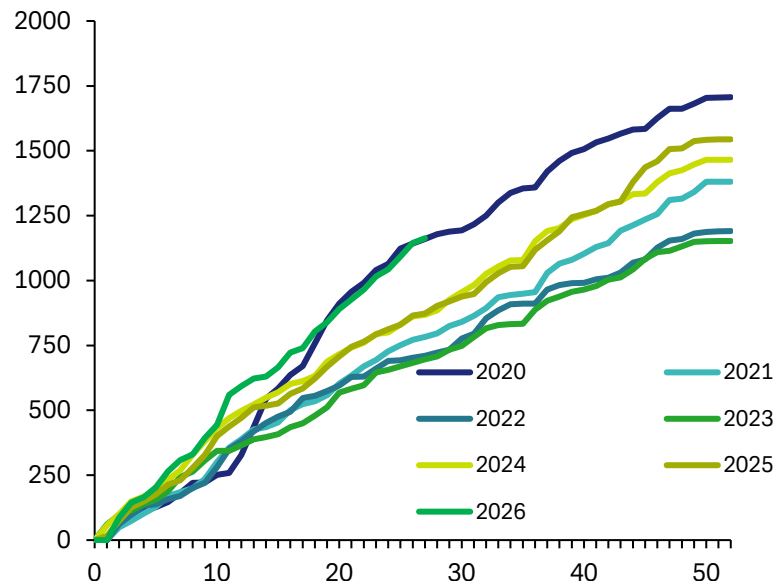
Source: Bloomberg Finance L.P., Deutsche Bank

Financing is abundant ... but can be selective

Debt issuance is still strong and tracking alongside the strongest level this decade. Private capital dry powder continues to push to record levels

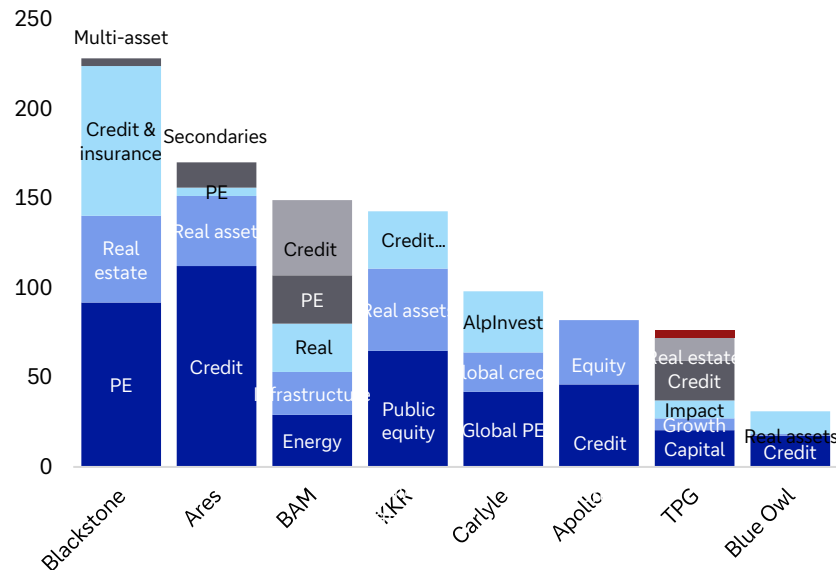


\$IG gross supply (\$bn) by week



Source: Bloomberg Finance L.P., Deutsche Bank

Dry powder by manager and segment, Q2 2026 (\$bn)

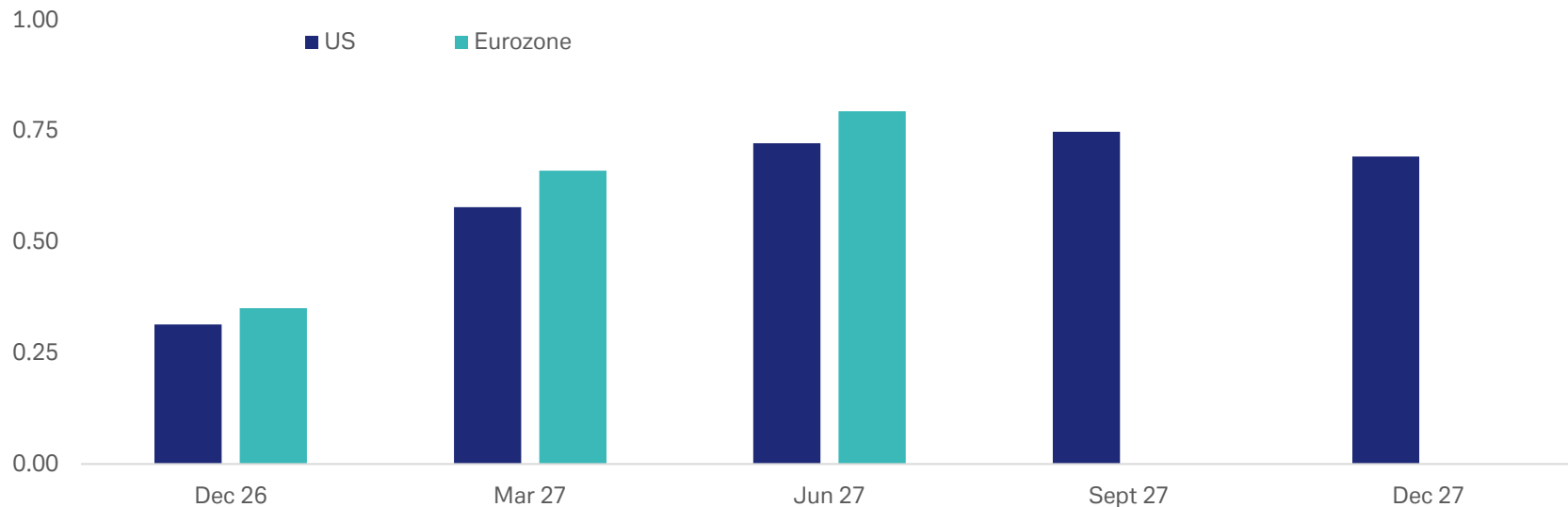


Markets are forecasting an easing of rate-rise (and thus inflation) expectations in 2027

The risks in the current environment may skew towards higher yields as expectations for the conflict in the Middle East lengthen and its impact becomes more severe



Implied interest rate rises by the end of each period (pp)



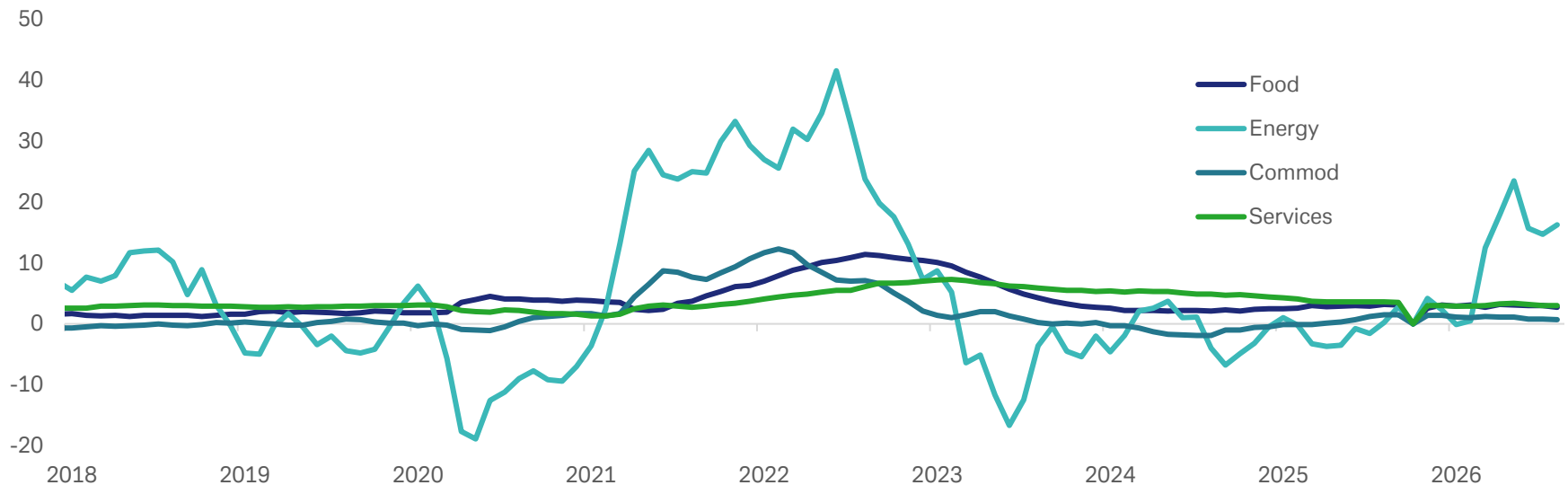
Source: Bloomberg Finance L.P., Deutsche Bank

The last energy shock lasted about 2 years

Iran's assertive push to control energy flows around the Middle East makes it unfeasible to project the length of energy shock



US CPI components (yoy % change)



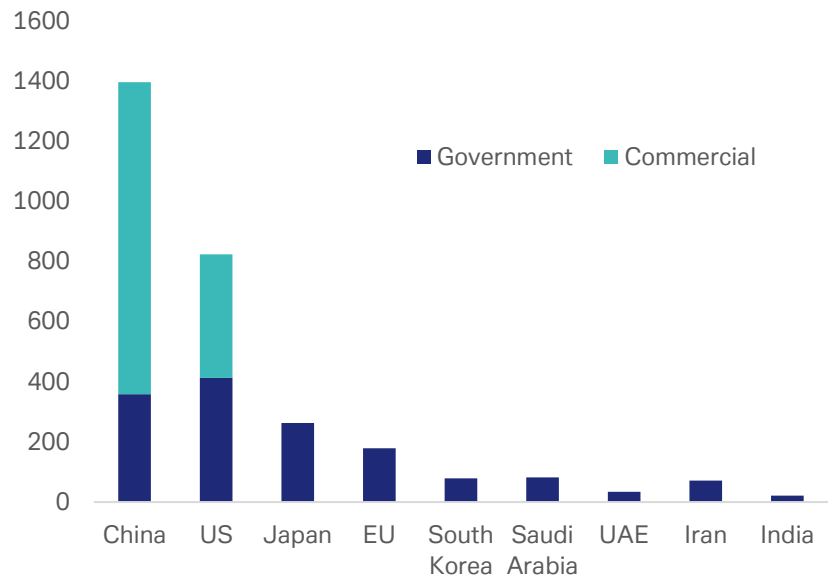
Source: Haver Analytics, Deutsche Bank

The trajectory of the energy shock will depend, in part, on oil inventory buffers

Industry executives have warned over the last two weeks that energy inventories have buffered the world economy for the last 6 months but that those buffers are approaching a cliff edge



Crude oil inventories pre-Iran conflict (m barrels)



Source: RAND, Deutsche Bank

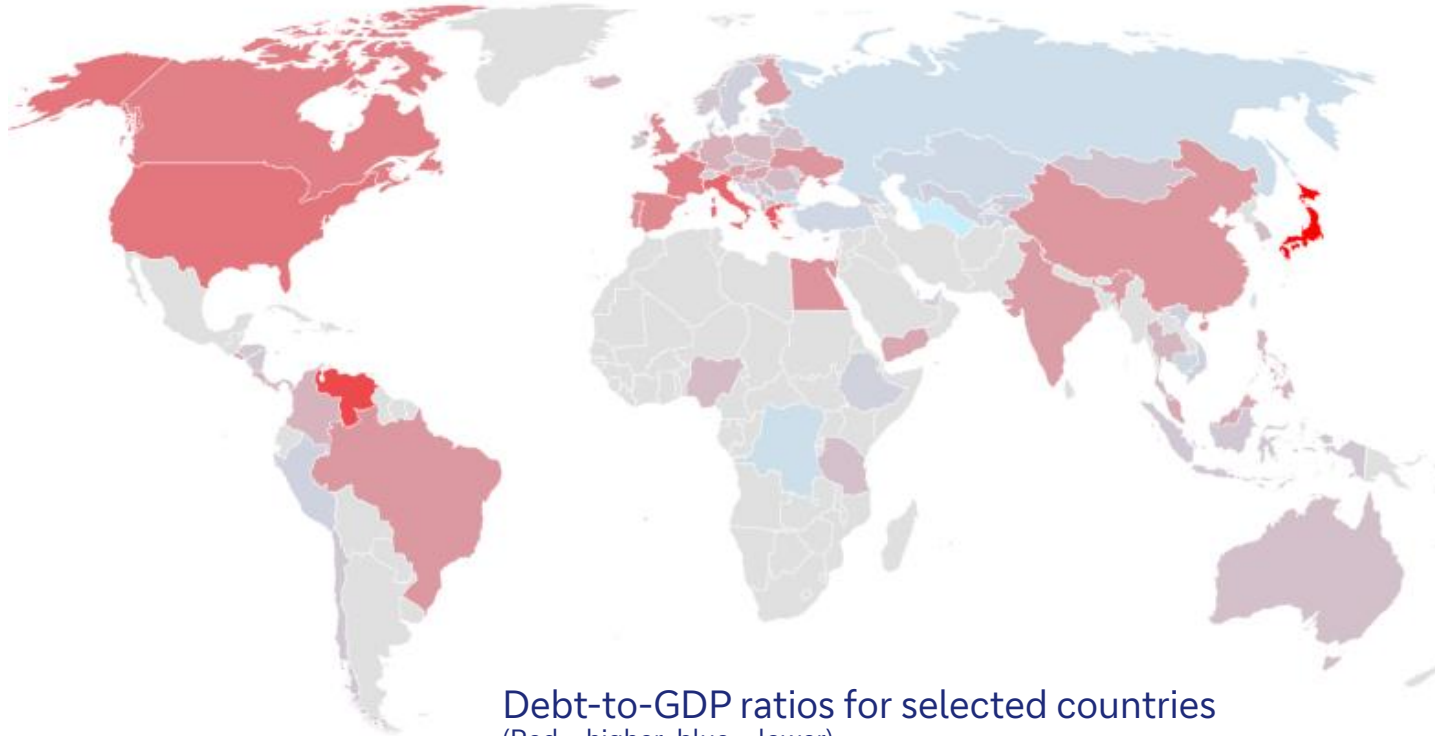
The way China uses its Strategic Petroleum Reserve is critical for world energy prices

RAND estimates China held between 1.1bn and 1.4bn barrels as of May this year – equivalent to 110-140 days of net imports.

That compares with the 400m barrel release coordinated by the IEA in March which had only a limited impact on markets.

Sovereign debt around the world

Confidence in indebted sovereigns tends to ebb and flow and markets usually experience at least one major 'risk-off' moment regarding sovereign debt each year. Upcoming elections in Europe will be a major focus point for investors over the coming months.



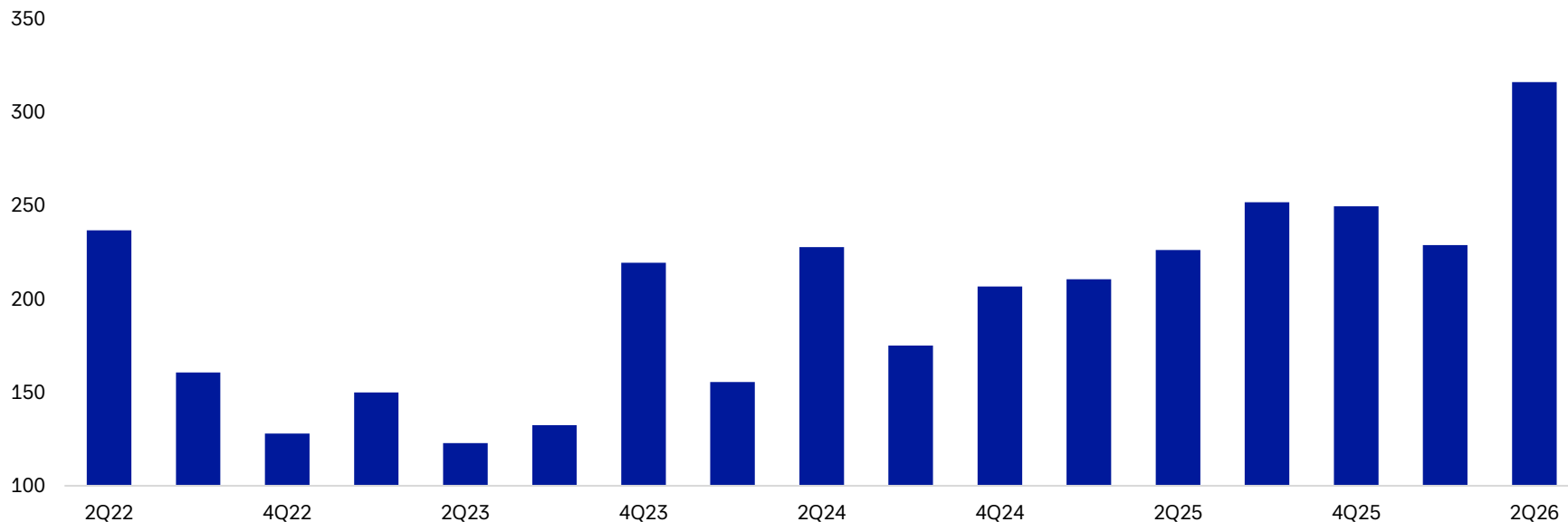
Source: IMF, Deutsche Bank

Private capital is searching for deals 1/2

Aggregate fundraising growth has shot up (in part due to one-off factors) and managers are confident in continued strength



Fundraising by selected large, listed private capital managers (\$bn)



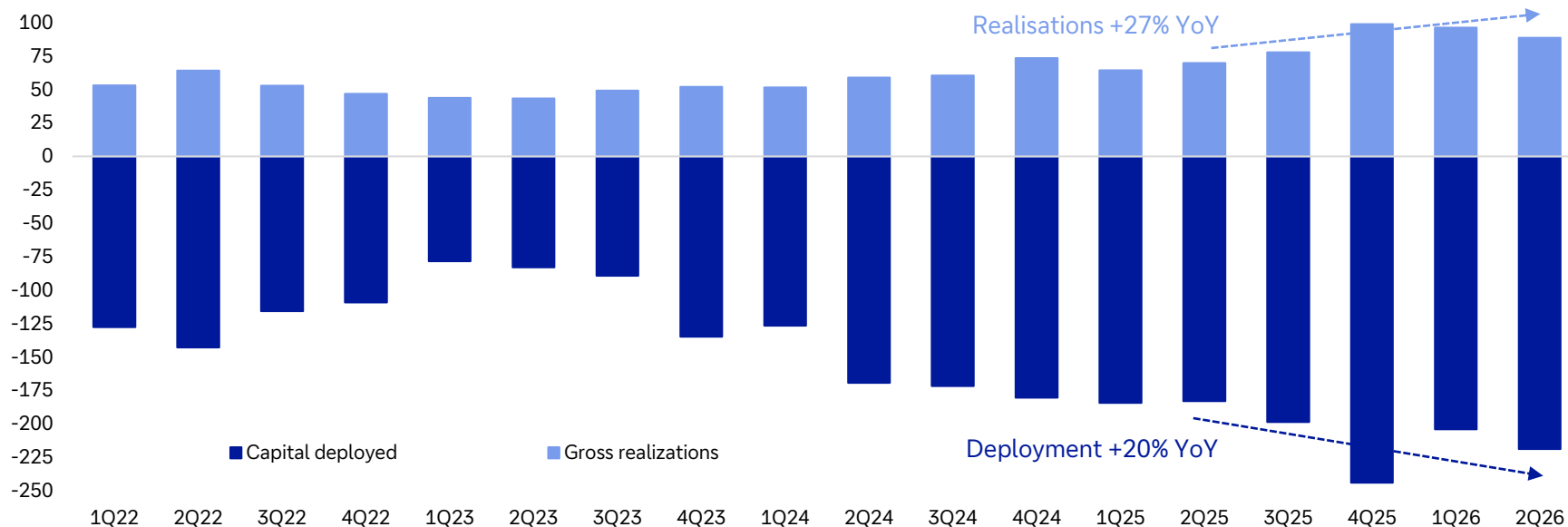
Source: Deutsche Bank company research - [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank

Private capital is searching for deals 2/2

Manager confidence has been helped by the growing rate of deployment and realisations. Through to the end of the year, realisations growth will be in focus given the growing macro uncertainty



Selected private managers, deployment v realisations (\$bn)



Source: Deutsche Bank company research - [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank

Appendix 1



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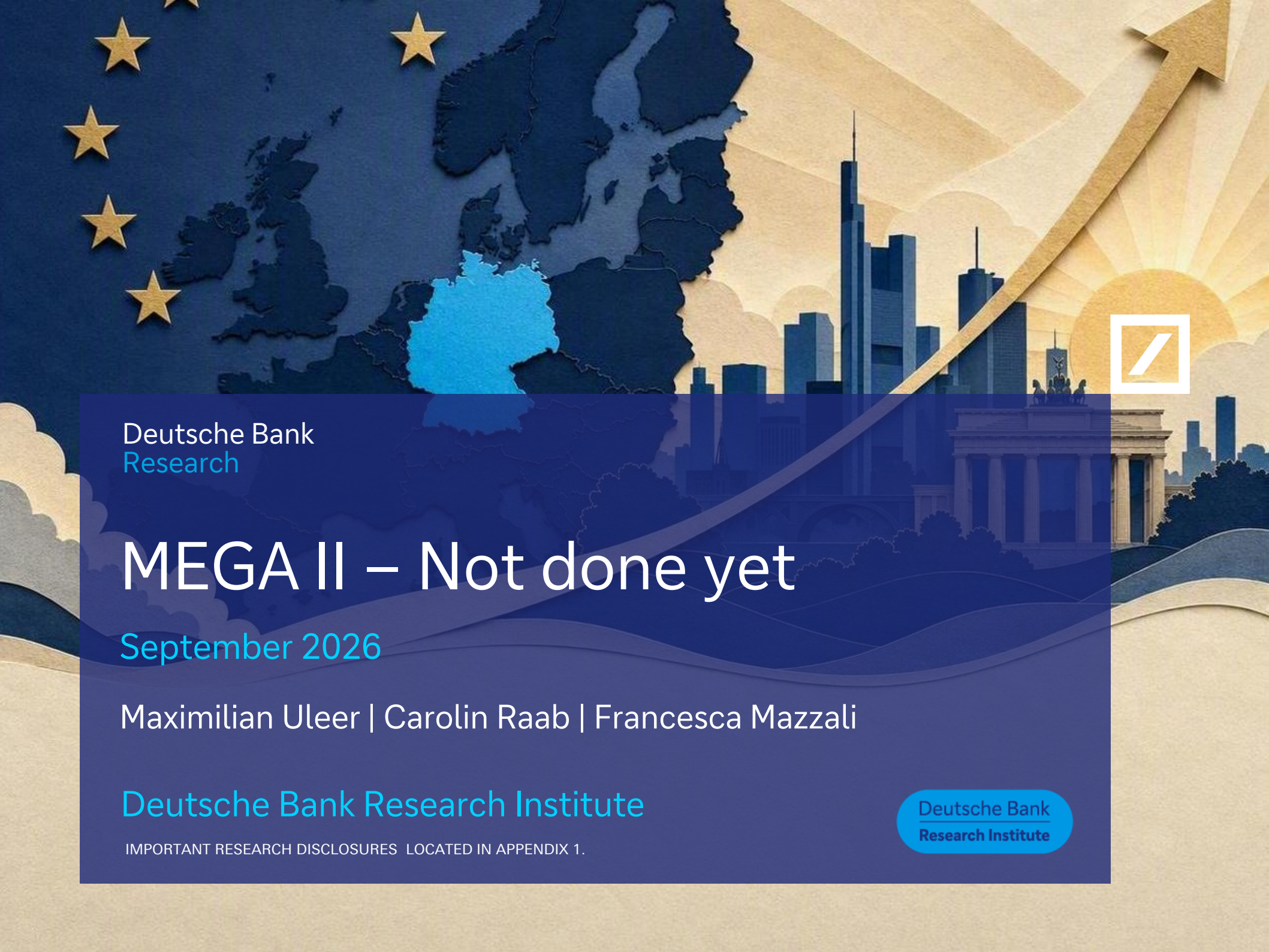
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Deutsche Bank
Research

MEGA II – Not done yet

September 2026

Maximilian Uleer | Carolin Raab | Francesca Mazzali

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Europe is back and Germany's potential seems not yet fully appreciated by markets



In early 2025, we made a bold call: we argued that Europe's 15-year history of underperformance has come to an end (MEGA and [here](#)). We projected the P/E valuation gap between US and European equities to narrow from a record 9.0x at the end of 2024 to 4.3x. Since then, the gap has indeed tightened to 4.7x. With Germany's potential still being underestimated by markets, we expect more to come for Europe.

Looking ahead, we see several structural factors supporting Europe's relative attractiveness:

- **Valuations:** After the tightening, we expect the gap to stabilize. Higher US growth justifies a premium but most of the below arguments argue against a renewed widening.
- **Earnings:** We expect the European earnings recovery to continue, delivering double-digit growth in 2026. While US earnings are still expected to grow faster, uncertainty around the sustainability of future growth, particularly in the Tech sector, remains elevated.
- **Germany:** The US and China are likely to continue growing faster than Europe, but much of that growth has been accompanied by widening deficits and rising debt. US fiscal deficit and debt metrics now exceed European levels. At the same time, Germany's fiscal pivot could support a relative recovery in European growth, an opportunity that equity markets have yet to fully embrace. A recovery in global manufacturing activity provides an additional boost to Europe's economy. Germany could become the powertrain of Europe's economic recovery. Having reduced its debt-to-GDP ratio over the past decade despite macro headwinds, it now has the fiscal capacity to support growth through a substantial spending plan including EUR 1tr in new debt. The economy is showing greater resilience than many expected, manufacturing activity is improving, international investor interest is beginning to re-emerge and reforms are advancing. We estimate the pension reform alone could unlock €10-15bn of annual inflows into European equities.
- **Diversification:** Europe's equity market is well diversified across sectors, providing broader return drivers than more concentrated markets such as the US and Emerging Markets. This helps cushion performance during sharp market downturns.
- **AI:** While Europe is unlikely to catch up with the leading AI developers, it is also less exposed to risks surrounding the monetisation of heavy AI investment. Instead, it should benefit indirectly from AI spending through its advanced manufacturing capabilities, its exposure to AI infrastructure and through growing AI adoption itself.
- **Political stability:** While Europe's fragmented political landscape is a structural disadvantage, external pressures from US and Chinese trade policies and ongoing geopolitical challenges have strengthened coordination across the EU on trade, defence and fiscal matters.

Closing valuation differential



In March 2025, we called for a decline in the valuation gap between US and EU equities. We have now moved significantly closer to our target.

P/E difference S&P 500 and STOXX 600 (12m fwd)



*DB forecast based on the P/E difference before the start of the Russia-Ukraine war (02/24/2022). Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

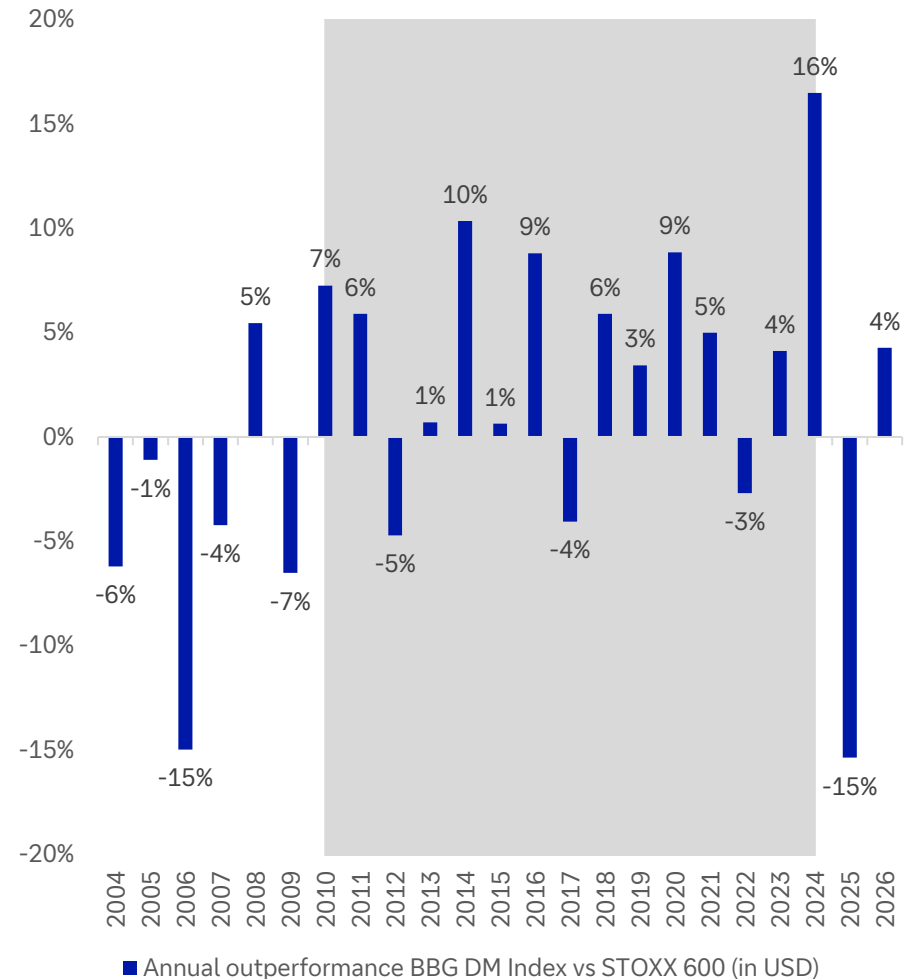
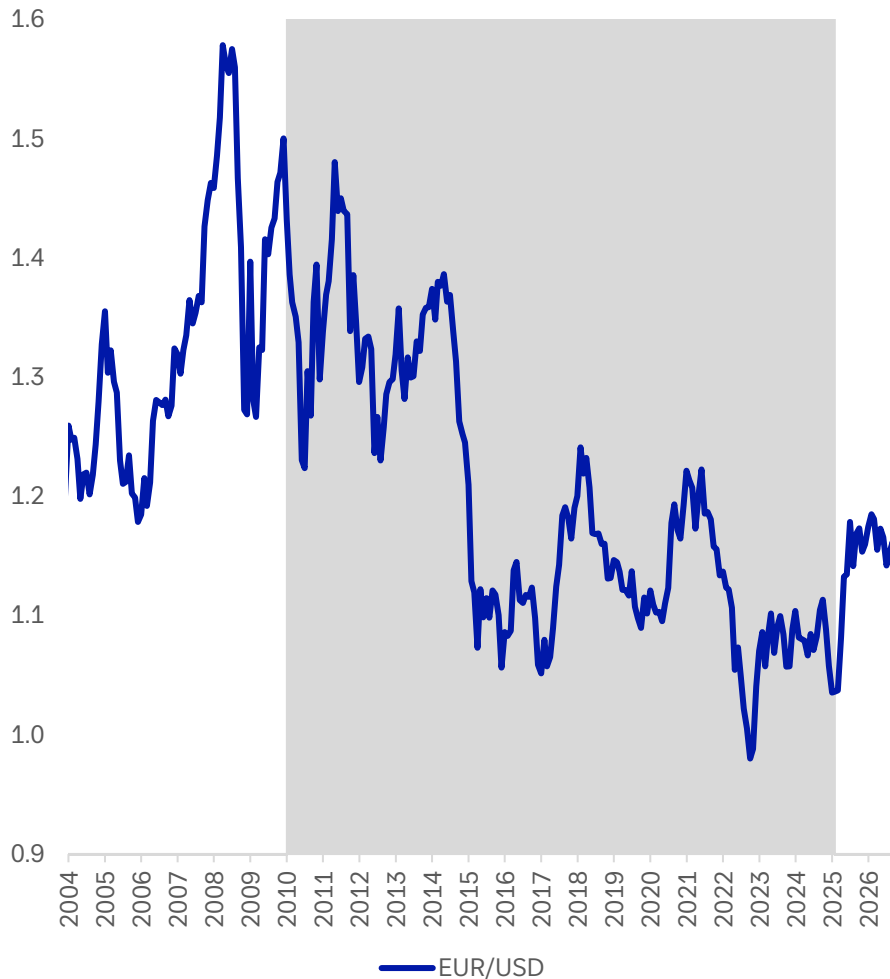
US Dominance... from 2010 to 2024?



European underperformance coincided with a 15-year USD rally. Will the USD continue to grow in its importance over the next 10 years?

In 2010-2024, the USD strengthened...

...supporting RoW outperformance



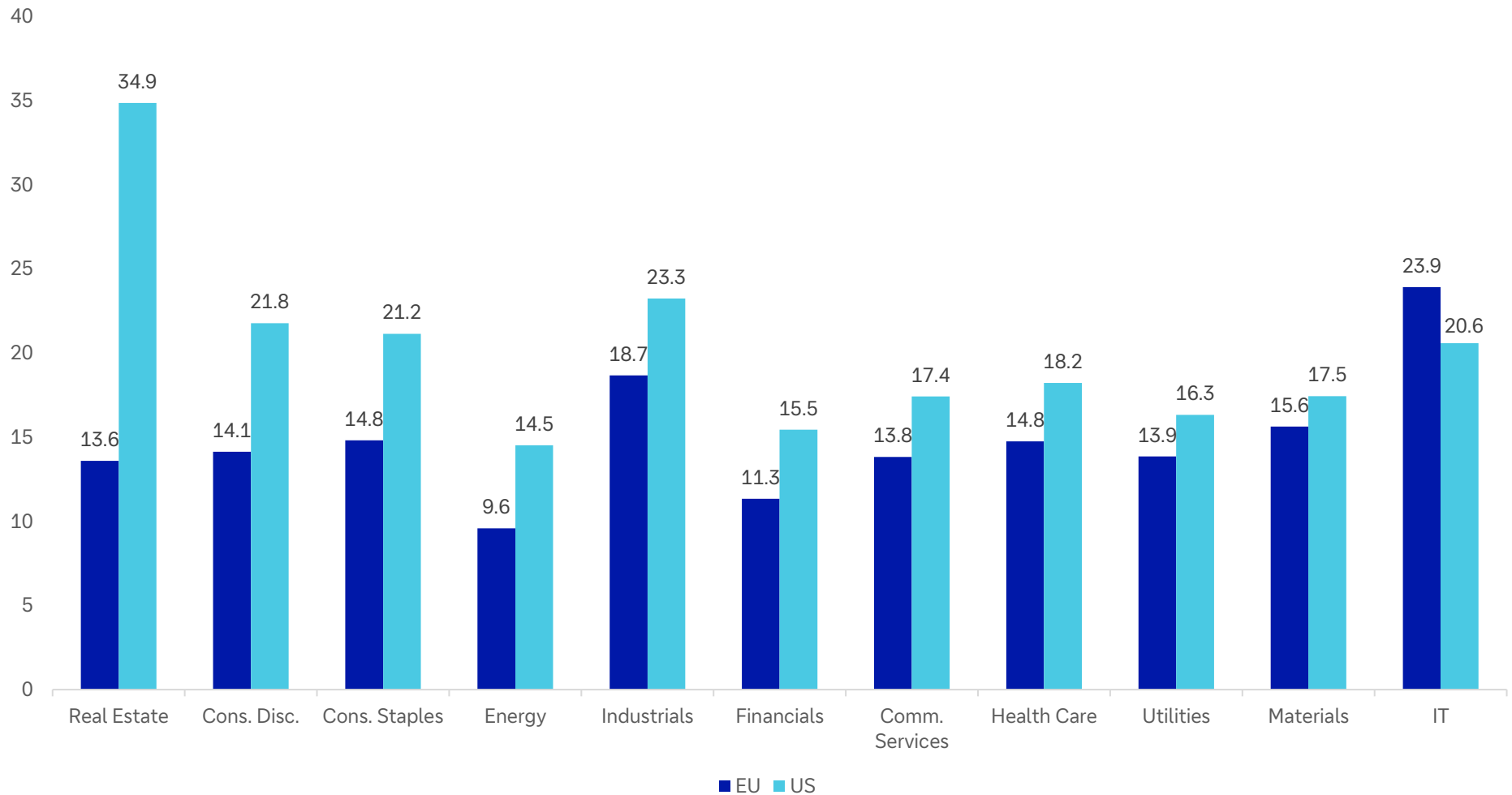
Note: Right chart is total return, 2026 is year-to-date. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Valuation Gap: applies across sectors



The valuation premium no longer stems from the IT sector. Still, every US sector, ex IT, trades at a premium versus the European sector.

Current sector P/E (12m fwd), sorted P/E difference US vs EU sectors



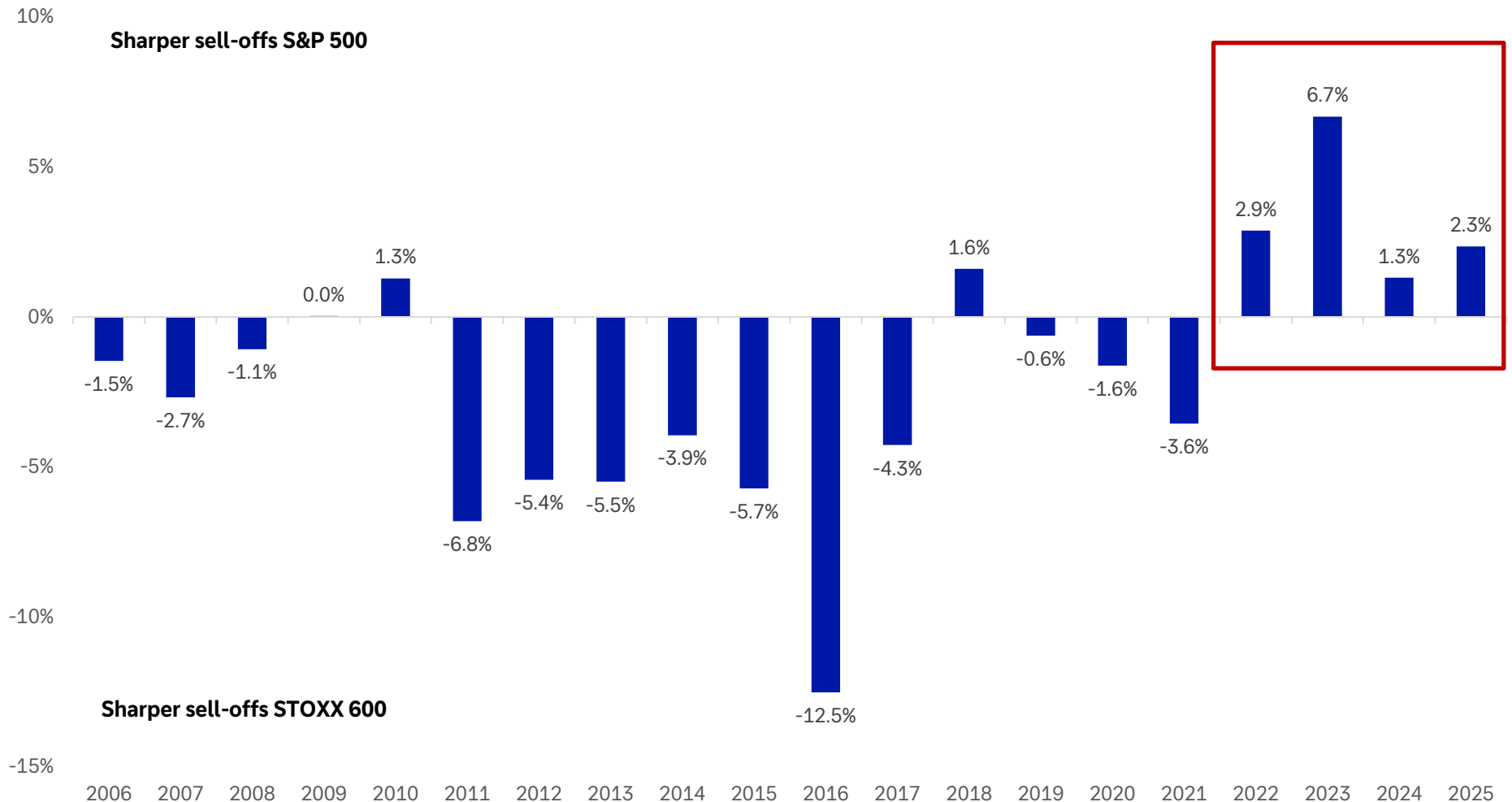
Reference indices: S&P 500 and STOXX 600, GICS sectors. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Concentration: leads to sharper sell-offs



„When the US sneezes, Europe catches a cold” no longer holds true. Drawdowns are significantly sharper in US equities. “When the US sneezes, Europe says Gesundheit”.

Difference in maximum drawdown of S&P 500 vs STOXX 600 (annual, local currency)



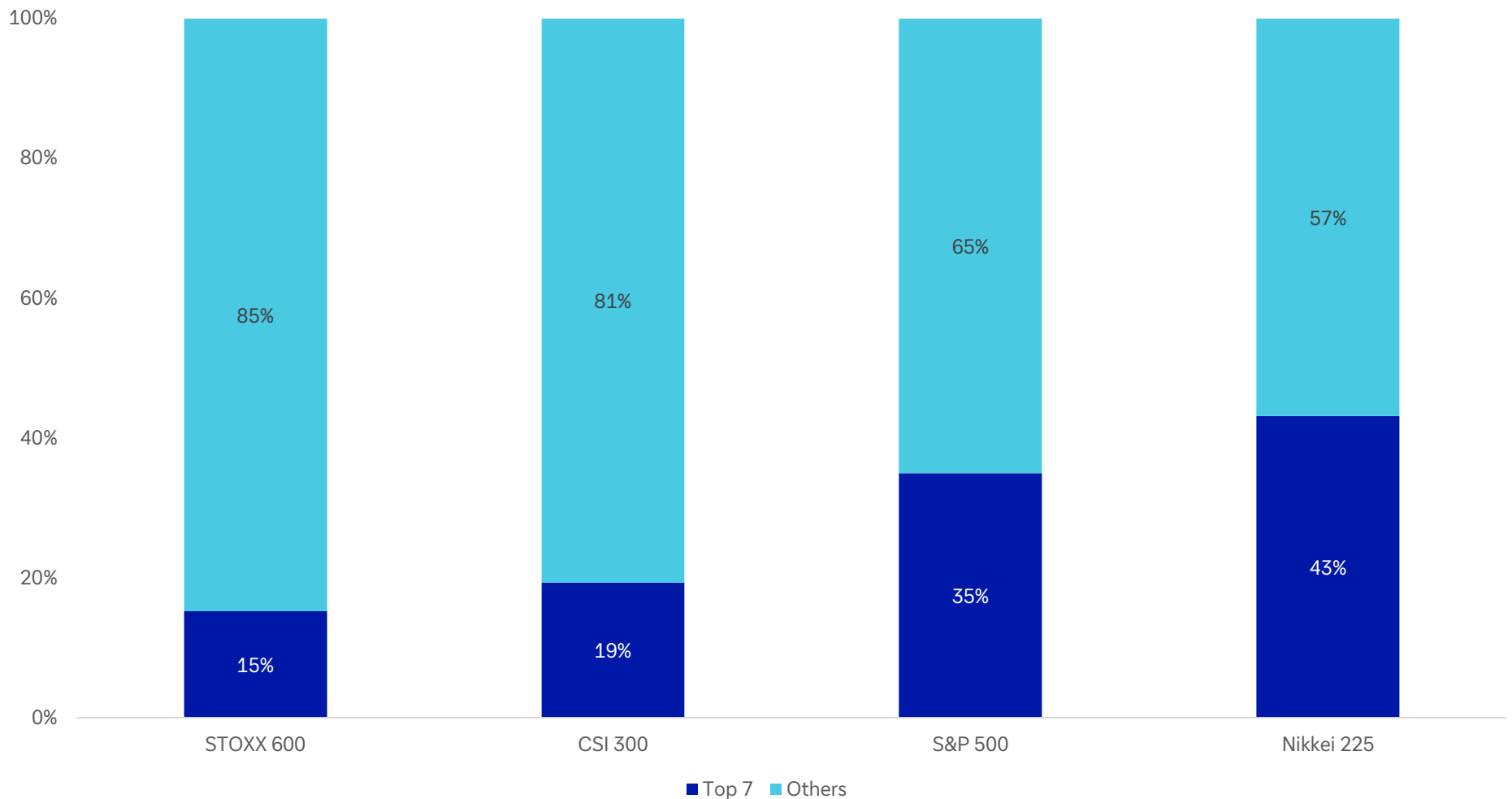
Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Concentration: in index weights



The European STOXX 600 is well diversified. The 7 largest corporates make up only 15% of the index weight vs 43% in the Nikkei and 35% in the S&P 500.

Current index weights



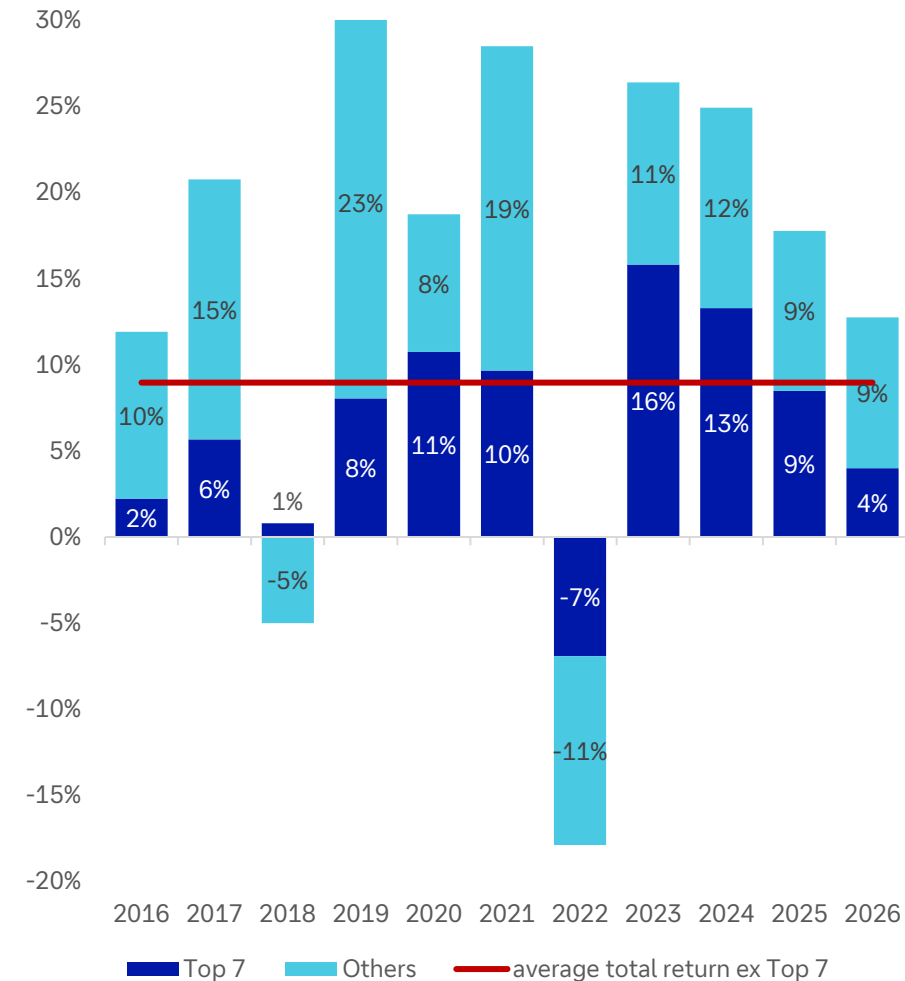
Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Concentration: in performance

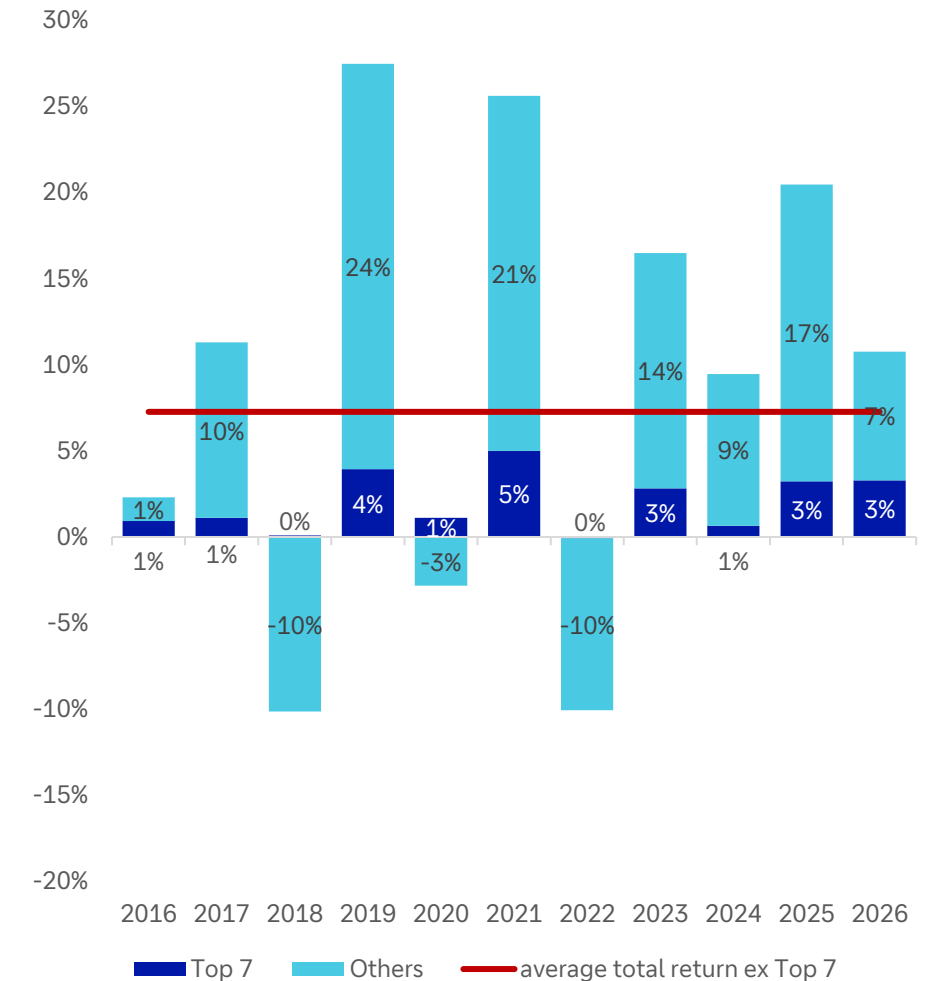


On average over the past 5 years, the S&P returned 16% per year vs 12% in the STOXX 600. Without the top 7 companies, the S&P would have delivered an average return of 8% vs 10% for the STOXX ex top 7.

S&P 500 – total return contributions



STOXX 600 – total return contributions



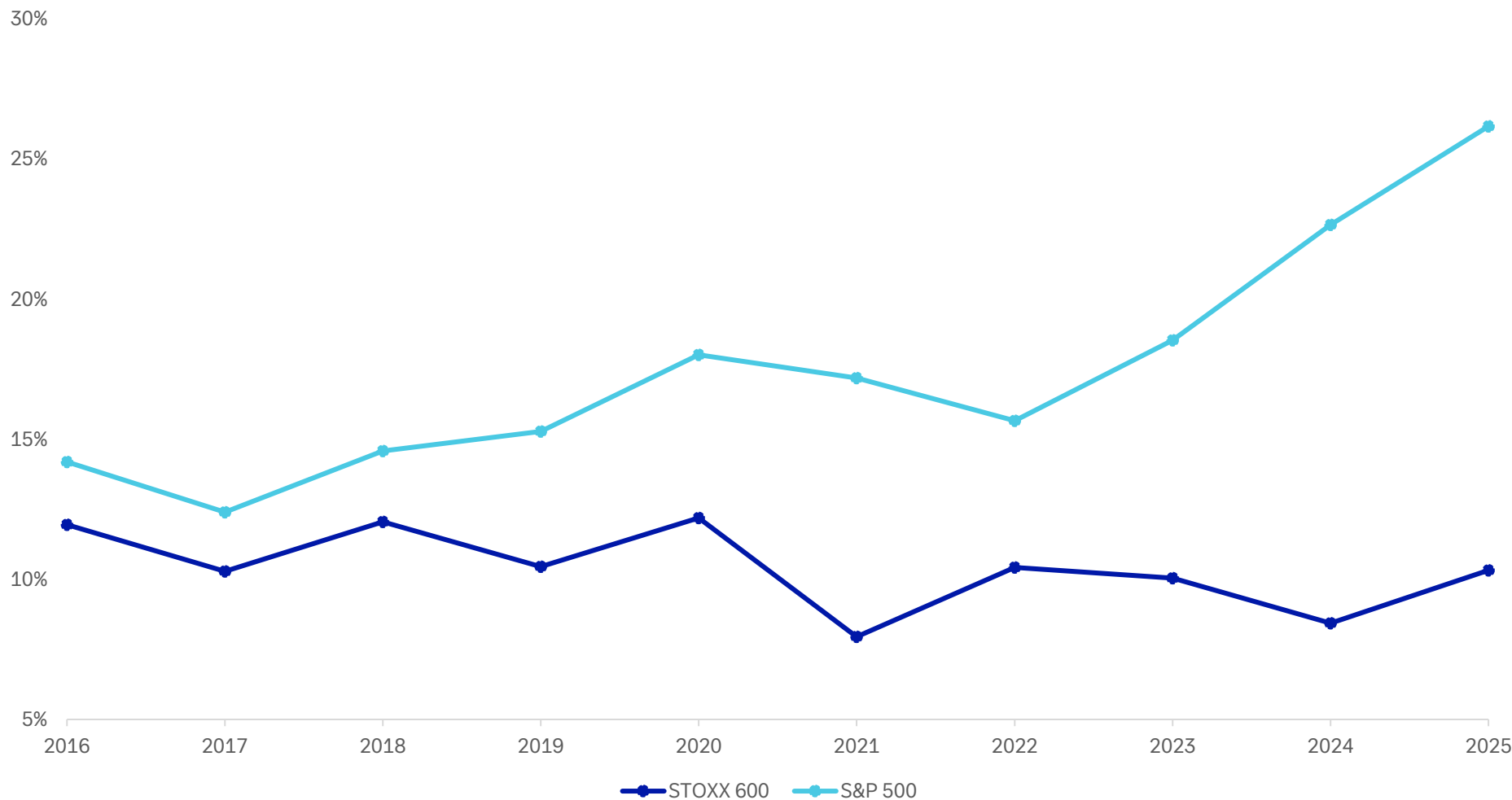
Note: 2026 is YTD. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Concentration: in earnings



The top 7 US companies contributed 53% to earnings growth since 2021. In the STOXX 600, the top 7 only accounted for 33% of profit growth over the same period.

Top 7 companies: share of aggregate index earnings



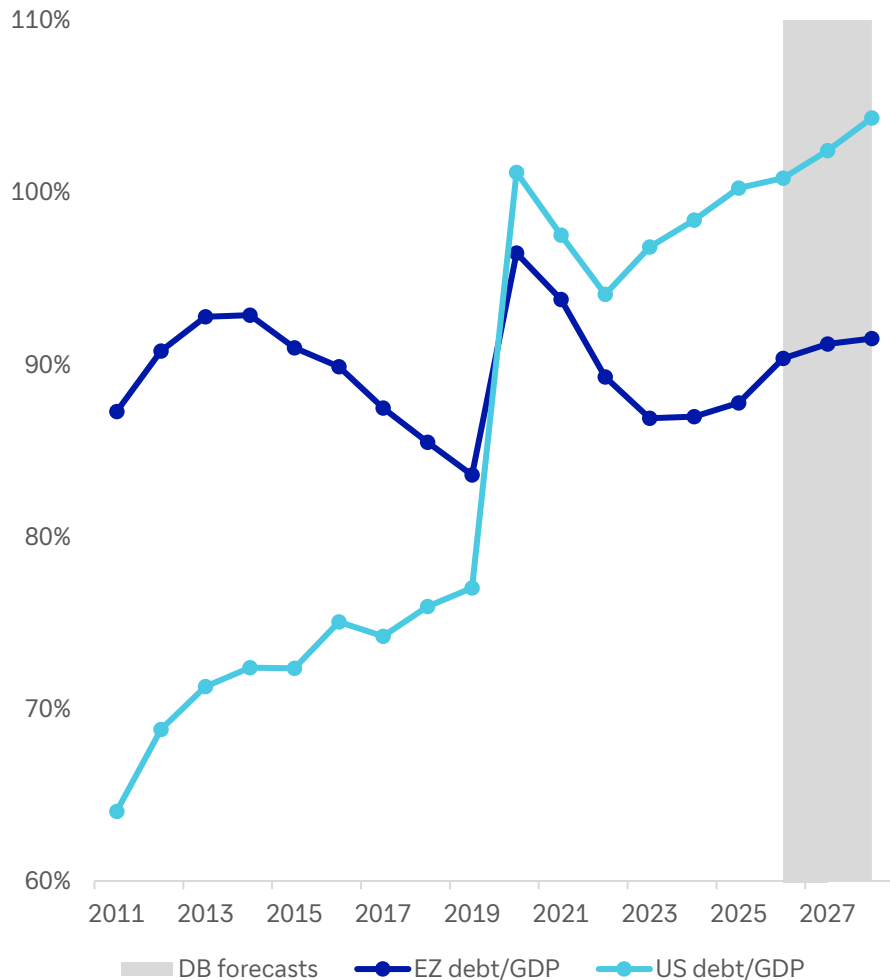
Note: based on current index members and current Top 7. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Fiscal Risks – look increasingly similar

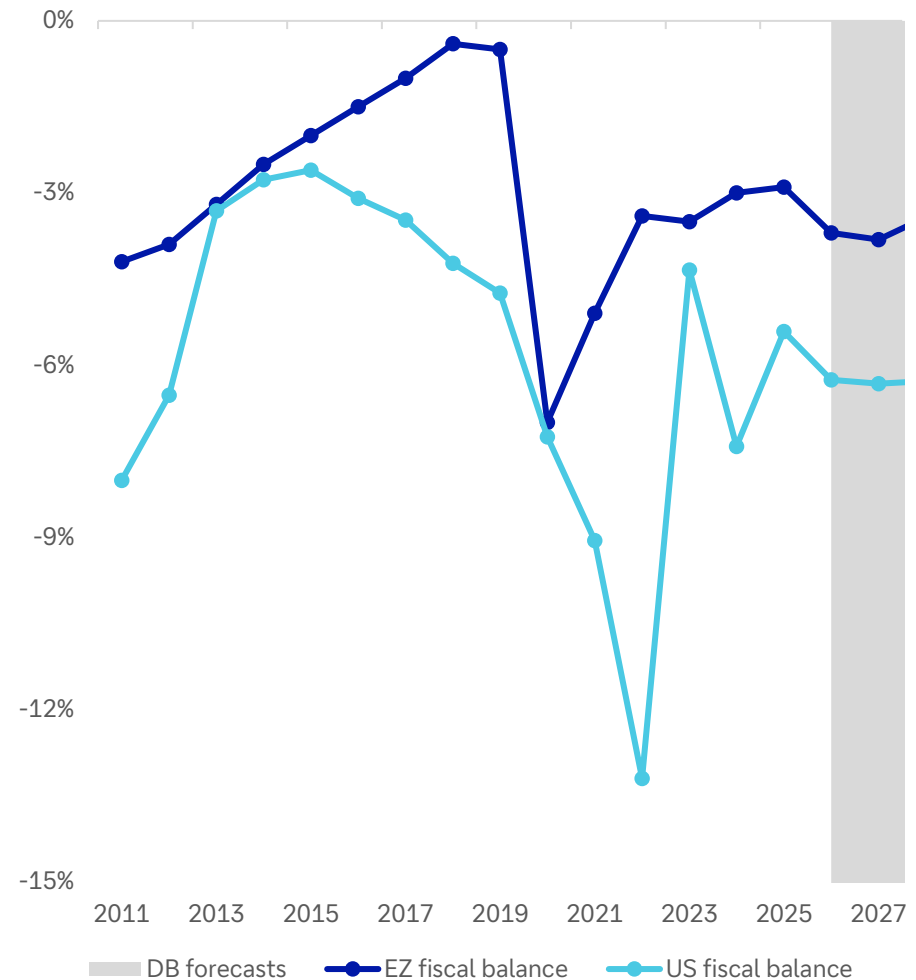


Despite the German fiscal expansion, debt ratios are expected to worsen more in the US than in the Eurozone.

Debt-to-GDP: Eurozone vs US



Fiscal balance (% GDP): Eurozone vs US



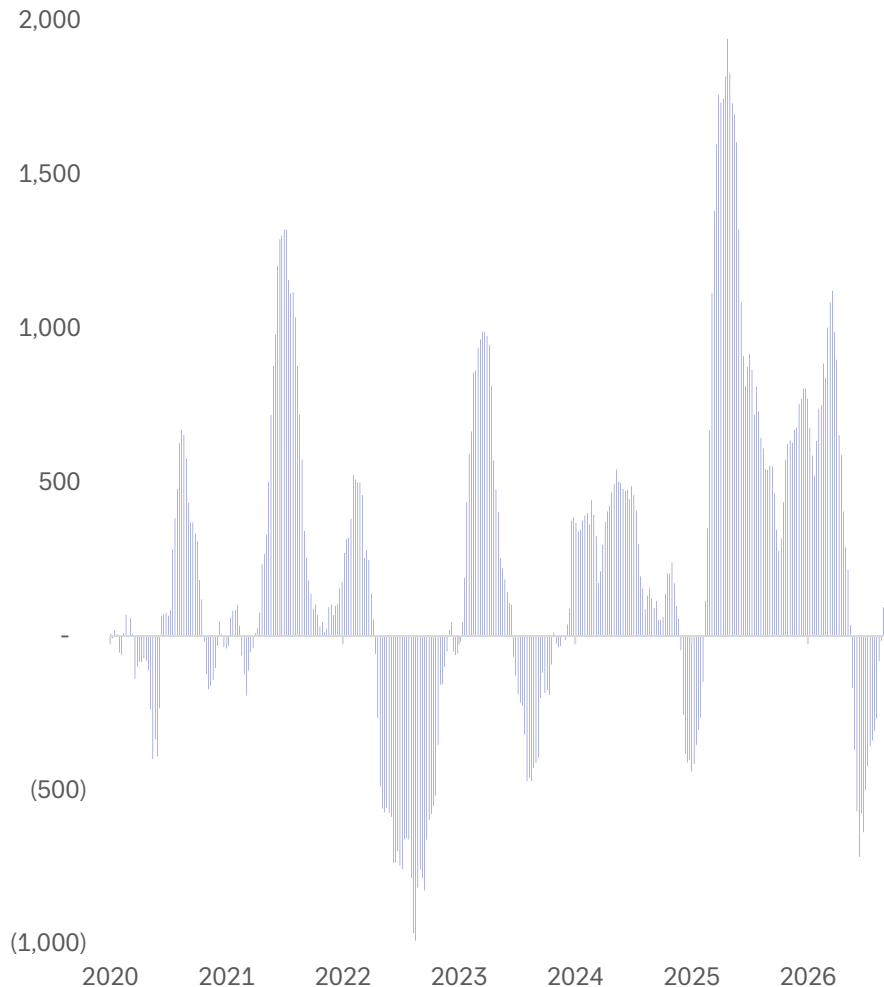
Note: Debt-to-GDP is net for US and gross for EZ to align with DB forecasts. Source: Deutsche Bank Research 09/15/2026

Flows – US dominates, but less than it did

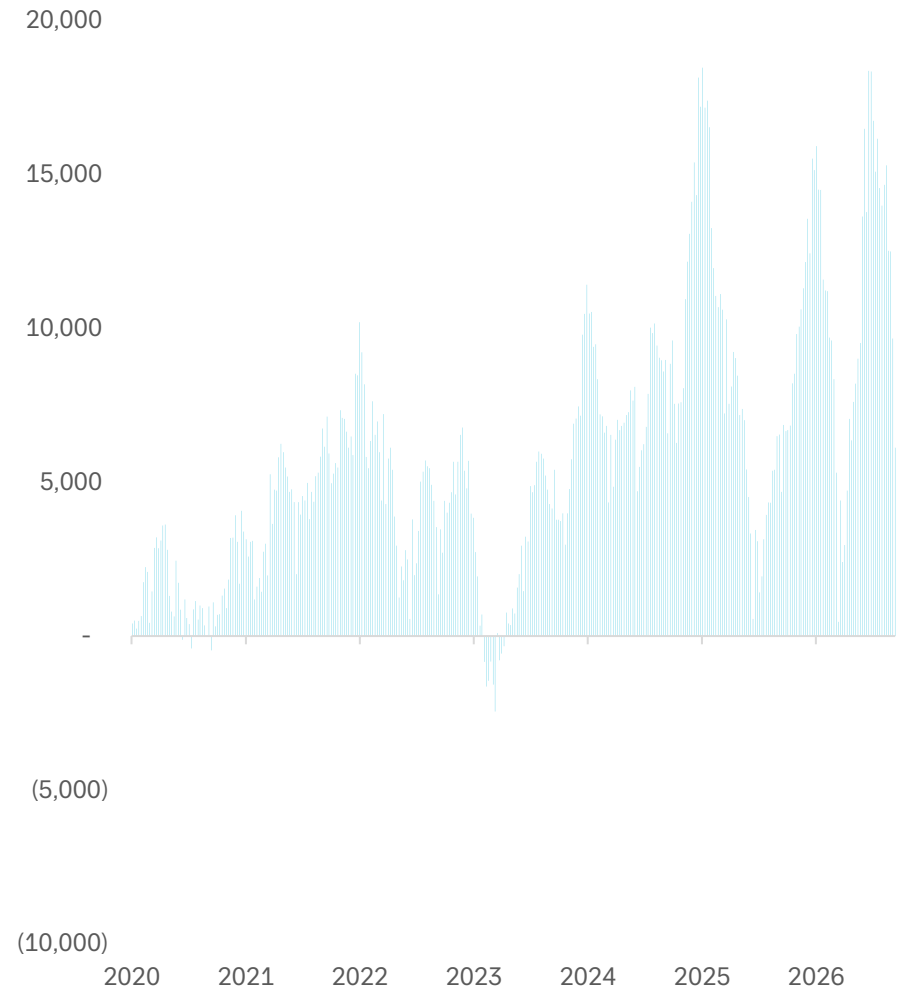


Inflows into European-focused ETFs have picked up significantly over the past two years but remain much smaller and more volatile than US flows.

Europe



USA



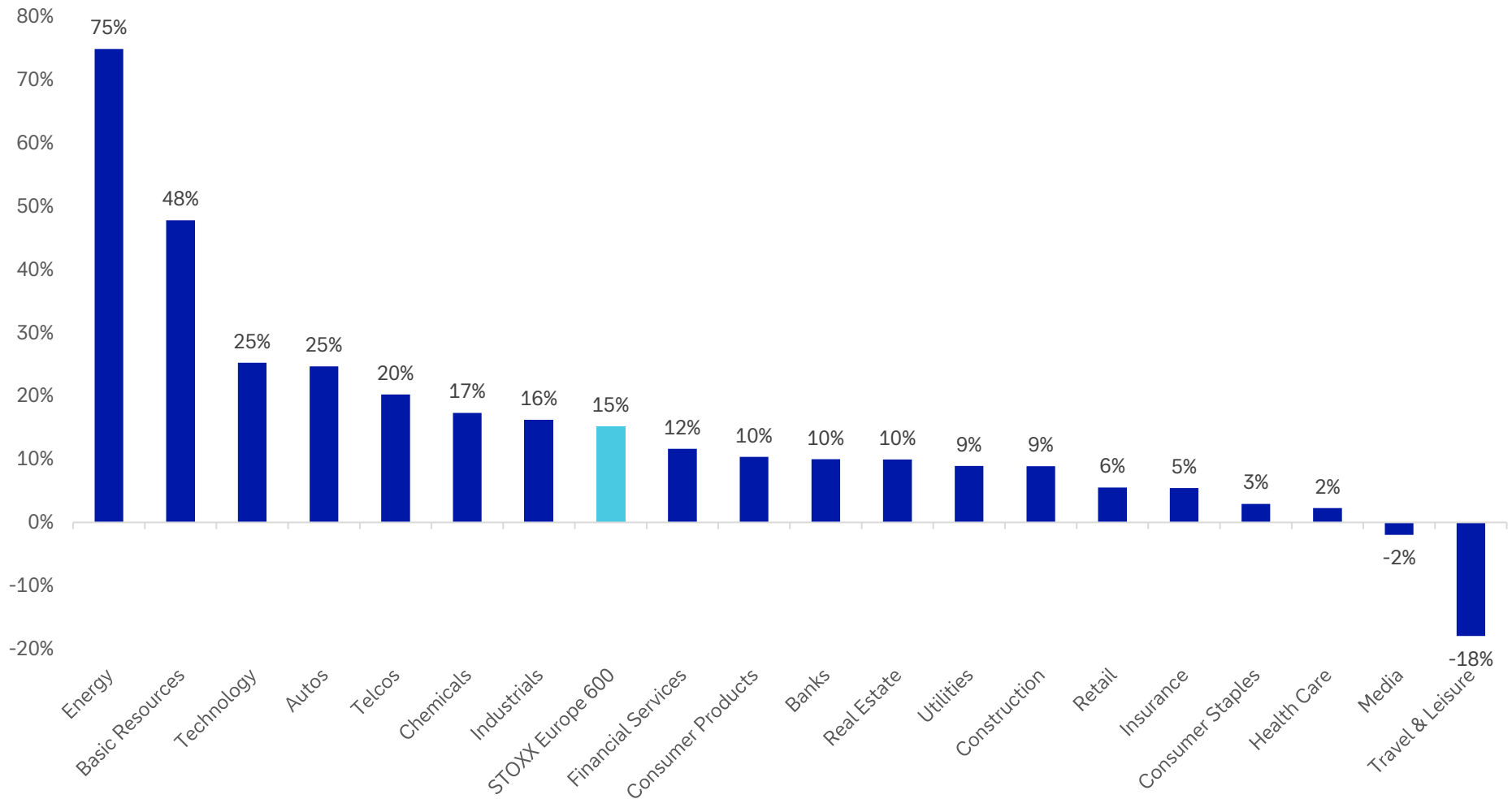
Note: Aggregate fund flow, 12w moving average, based on 50 largest ETFs for the region, in mn USD. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Earnings – Broad-based growth



The outlook for European earnings is encouraging. Almost all sectors should see positive earnings growth this year.

STOXX 600: Consensus expected earnings growth, FY26



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026



Deep dive: Germany

Don't underestimate Germany's capacity to change



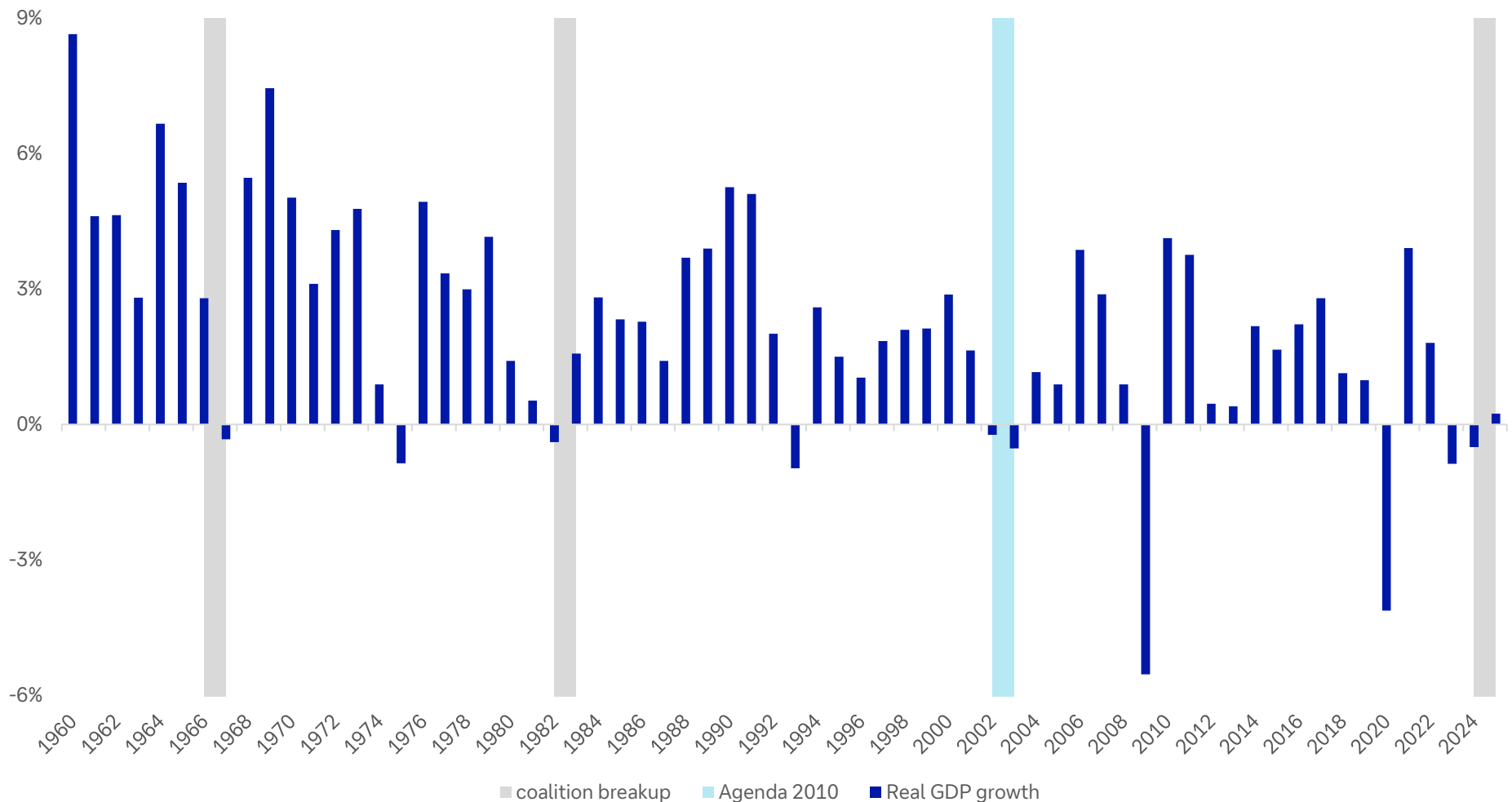
Germany has the potential to become the driving force behind Europe's economic revival. Despite years of macro headwinds, it reduced its debt ratio over the past decade, creating the fiscal space for a major spending package. Economic activity is stabilising, manufacturing conditions are improving, reforms are progressing, and global investor interest is returning. Germany has repeatedly demonstrated an ability to adapt and reinvent itself, with major reform waves following the coalition collapses of 1966 and 1982 and the implementation of Agenda 2010 in the early 2000s.

- **Fiscal:** Few countries have faced as many challenges as Germany in recent years: rising Chinese competition in key industries, the energy shock following Russia's invasion of Ukraine, large-scale immigration, US tariffs, and a three-year manufacturing downturn. Despite these headwinds, Germany is the only G7 economy to have reduced its debt-to-GDP ratio over the past decade. The government plans to raise around €1tn of additional debt by 2030, on top of the current debt stock of €2.8tn. Net credit borrowing is expected to increase from €180bn in 2026 to more than €200bn in 2027, while spending on defence and investment is set to reach roughly €120bn each this year, up more than 30% year-on-year.
- **Macro:** Germany's economy is showing greater resilience towards the ongoing geopolitical headwinds than expected. The manufacturing PMI climbed above 54 in August, its strongest reading in over four years, and both the Ifo and ZEW surveys surprised positively. Factory orders are gradually improving and GDP growth in H1 exceeded expectations, prompting consensus GDP forecasts to be revised up. Although higher energy prices may create short-term headwinds, the burden remains manageable, with energy costs representing just slightly more than 1% of sales for listed German companies.
- **Politics:** Germany's reform momentum is strengthening. The government is combining fiscal stimulus with measures to lower energy costs, boost corporate investment, accelerate infrastructure projects, reform the labour market, and modernise the pension system. We estimate the pension reform alone could unlock €60-100bn of annual capital market inflows. At the same time, political risks appear manageable. Regional elections are unlikely to affect federal policymaking, and we see little risk to the government's reform agenda. If anything, weaker results for the coalition parties could increase pressure to accelerate reforms.
- **Equities:** Markets have yet to price in Germany's improving outlook. While economic indicators continue to strengthen, investor flows into German equities remain weak and the MDAX has lagged the DAX by 7% since late June. As the index with the greatest exposure to the domestic economy, the MDAX is the purest listed play on Germany's recovery and has historically shown a much stronger relationship with German GDP growth. A combination of fiscal stimulus, structural reforms, and more industry-friendly policies creates substantial catch-up potential following years of underperformance. Importantly, valuations remain attractive, with the MDAX trading at its cheapest relative level in history.

BIG PICTURE: Recession → Reforms → Growth

Don't underestimate Germany's capacity to change. After the coalition break-ups of '66 and '82, Germany introduced reforms and re-emerged stronger. Reforms were also implemented when growth turned in 2003 under the Agenda 2010.

Germany was in recession in '66 & '82, but recovered quickly after the political crisis was resolved



Source: Bloomberg Finance LP, Bundesbank, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion

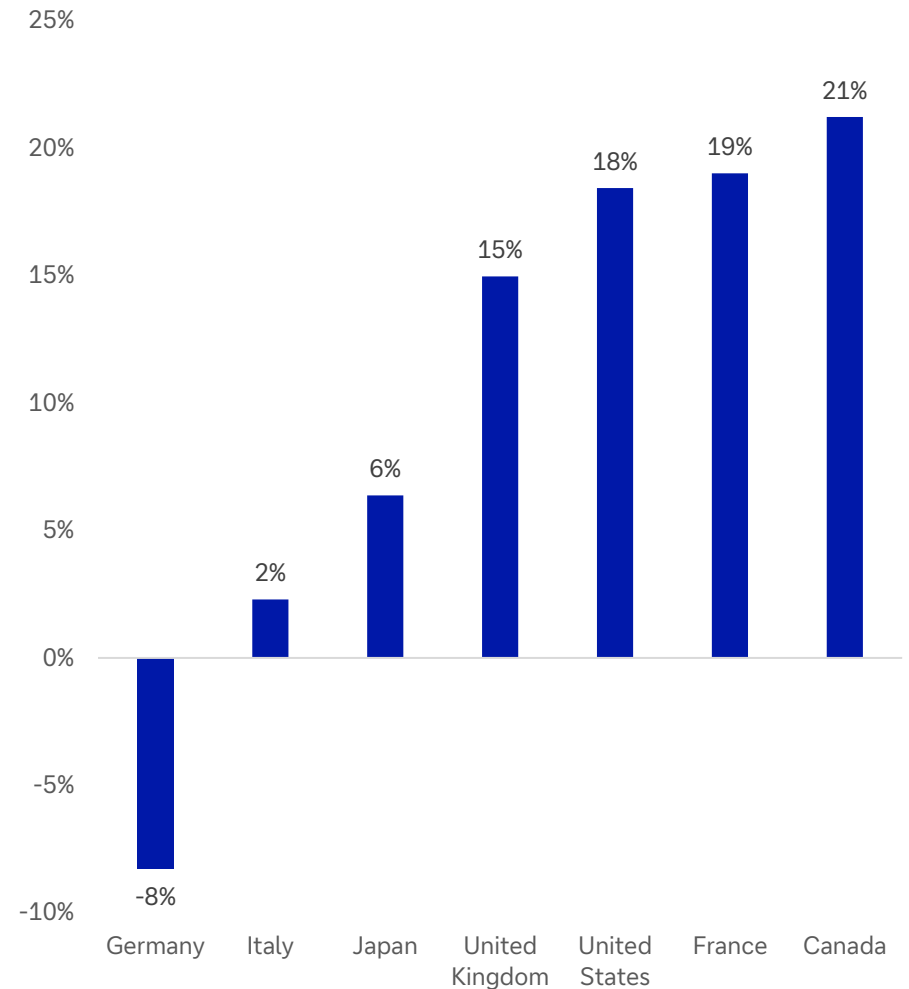


Germany is the only G7 country to have reduced its debt relative to GDP over the past decade.

Key takeaways

- Over the past decade, few countries were facing as many headwinds as Germany: growing competition from China in its key industries, the energy transformation following the war in Ukraine, mass immigration, US tariffs and a 3-year manufacturing crisis.
- Despite all of this, Germany managed to be the only G7 country to have reduced its debt relative to GDP in the last 10 years.
- German debt-to-GDP dropped by 8 percentage points while e.g. US growth was largely debt-financed, with the US' debt-to-GDP ratio increasing by 18 percentage points over the same period.

Change in debt-to-GDP over the past 10y (pp)



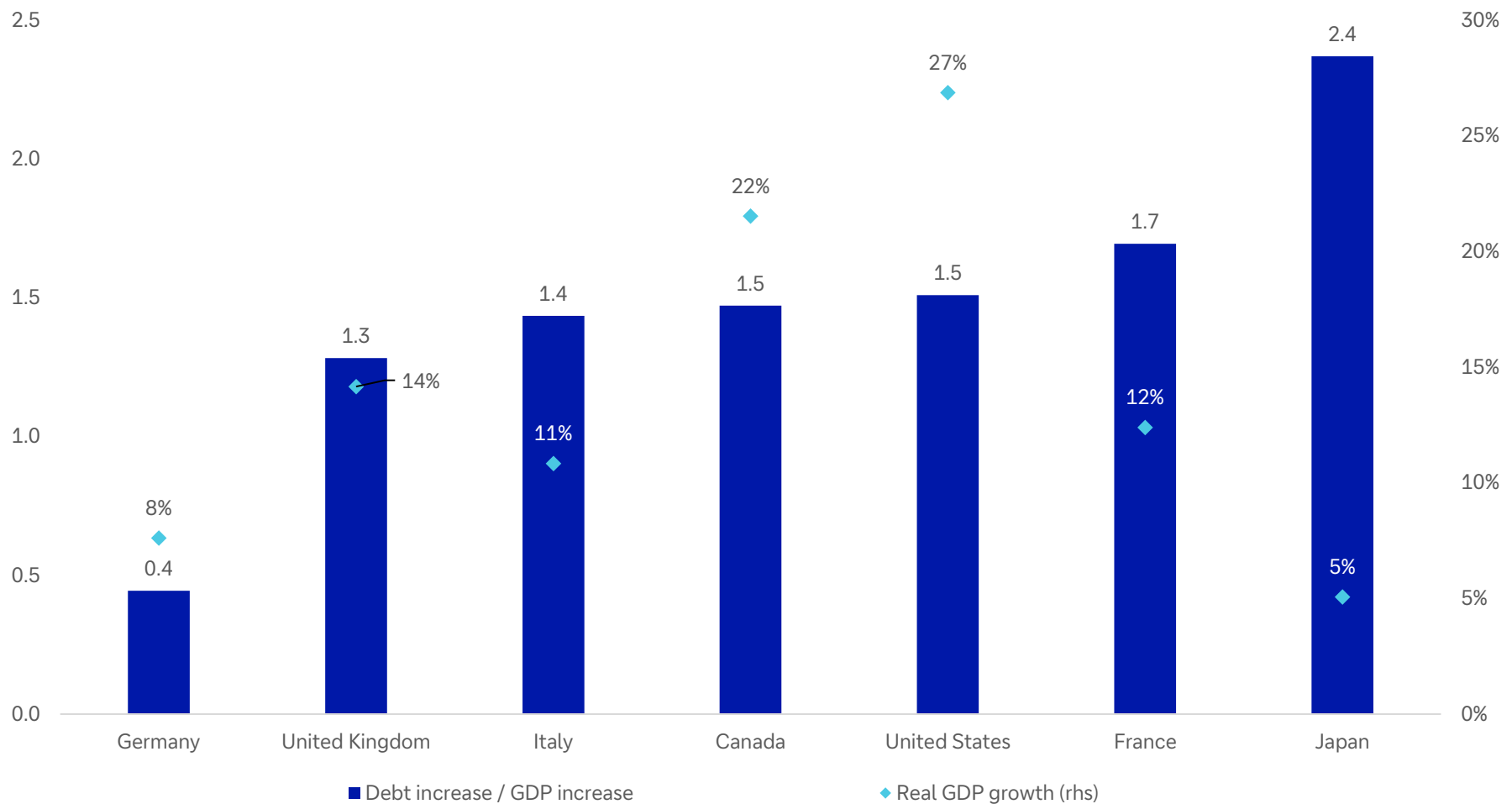
Note: All statistics in local currency, 2025 vs 2015, debt-to-GDP based on nominal GDP and gross general government debt. Source: IMF, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion



Germany is the only G7 country that increased debt by significantly less than 1 Euro for every Euro of GDP growth.

Bang for the buck: Germany financed its growth at a lower cost (10y)



Note: All statistics in local currency, 2025 vs 2015, debt increase / GDP increase in nominal terms. Source: IMF, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion



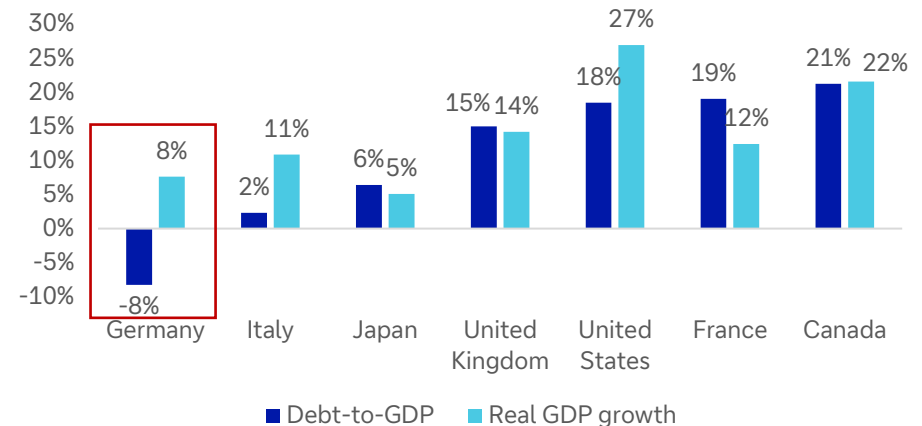
Germany managed to grow its GDP despite fiscal consolidation.

An additional €1 trillion in debt by 2030 could unlock significant growth.

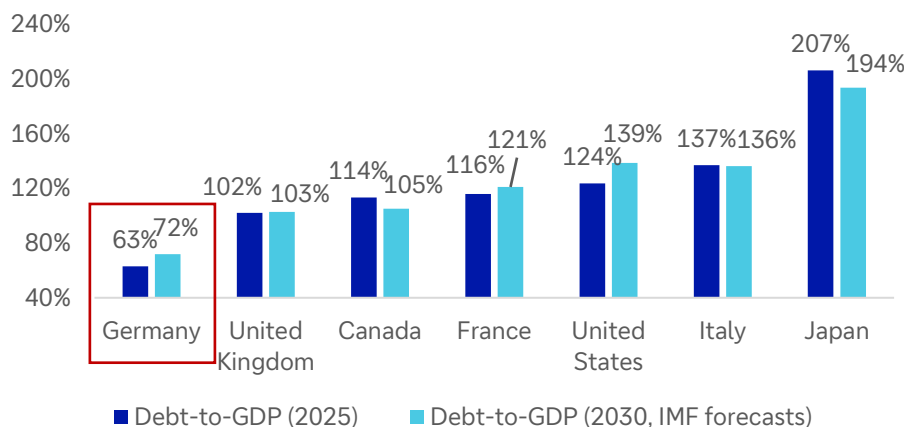
Key takeaways

- The German government plans to increase debt by more than €1 trillion by 2030, coming from €2.8 trillion in 2025 (gross general government debt). Almost 70% of the debt increase will be related to the core budget, vs 30% in off-budget funds.
- A large share of the additional debt (>€600bn) will be allocated to defence and security-related spending, almost €300bn to the special infrastructure fund (especially transportation and digitalization), and the remaining amount to other core-budget borrowing.
- For more information, see [here](#).

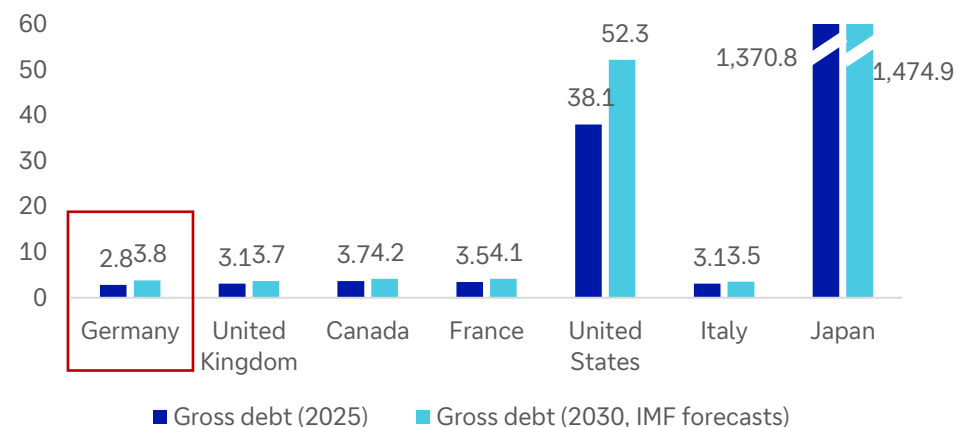
10y changes in debt and GDP across the G7



IMF projections of debt-to-GDP



Gross debt (trillions, local currency)



Note: All statistics in local currency, 10y changes are 2025 vs 2015, debt-to-GDP in nominal terms based on gross general government debt.
Source: IMF, Bundestag, Deutsche Bank Research 09/14/2026



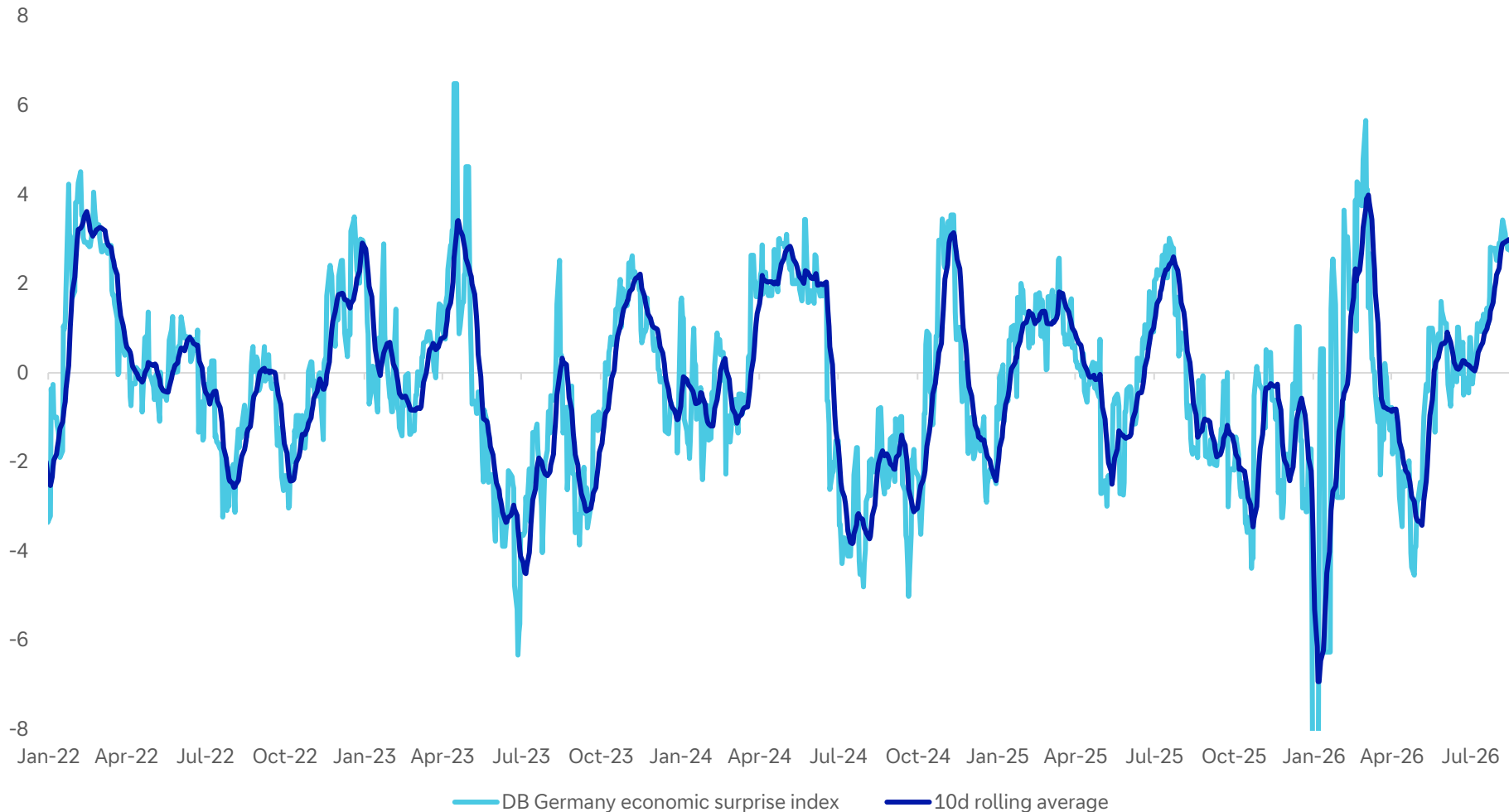
Macro & political backdrop

Macro backdrop – back on track



Most recent macro data came in much better than expected in Germany.

DB Germany economic surprise index has reached its highest level since 2021



Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

Macro backdrop – back on track

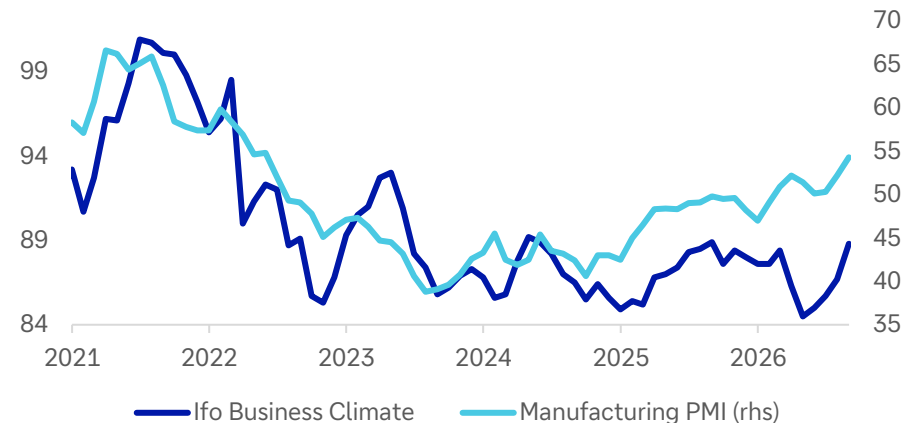


Green shoots have emerged in the German economy over recent months.

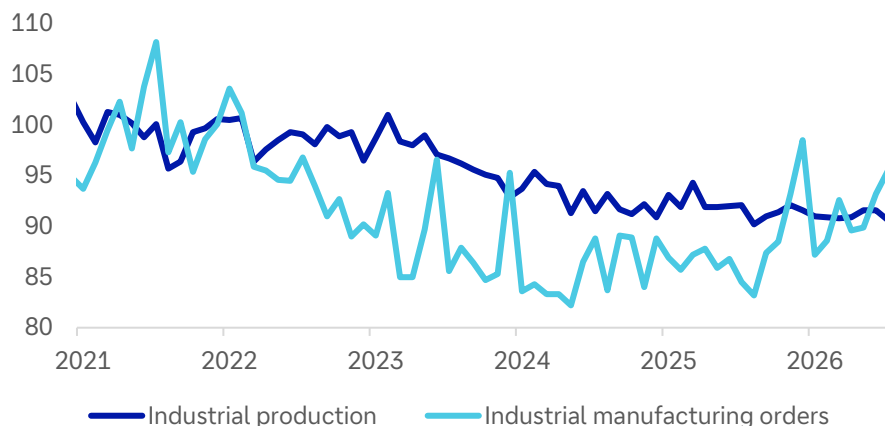
Key takeaways

- The German economy is more resilient than expected. The manufacturing PMI rose above 54 in August, its highest level in more than four years. Ifo and ZEW surveys also both surprised to the upside in August.
- Factory orders continue to edge higher, albeit still driven mainly by large orders.
- After GDP proved stronger than expected in H1, we recently upgraded our '26 GDP forecast to 1%.

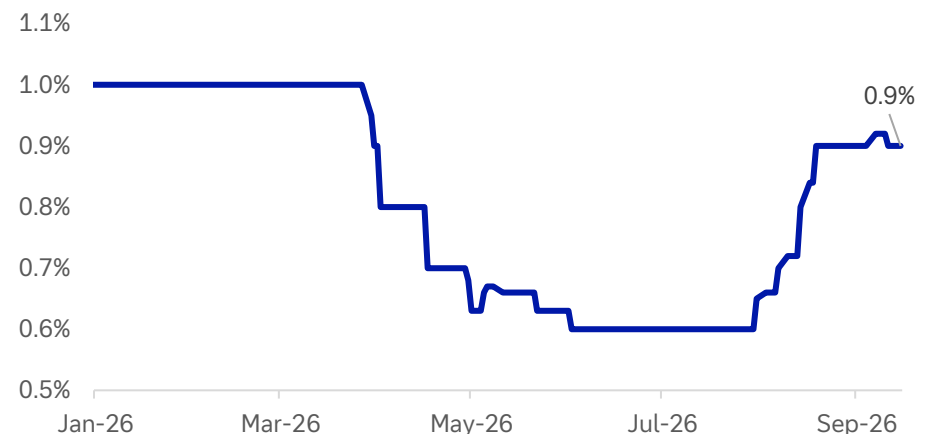
Corporate sentiment recovering



Factory orders accelerated in July



2026 GDP forecasts revised upwards*



*Median of Bloomberg consensus forecasts. Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

Regional elections

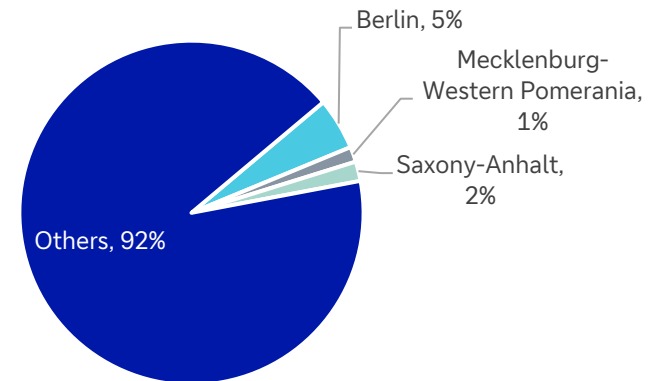


Higher uncertainty from regional elections. But relevance limited and pressure on government to increase pace of reforms is growing.

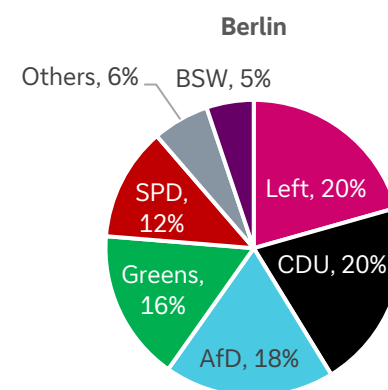
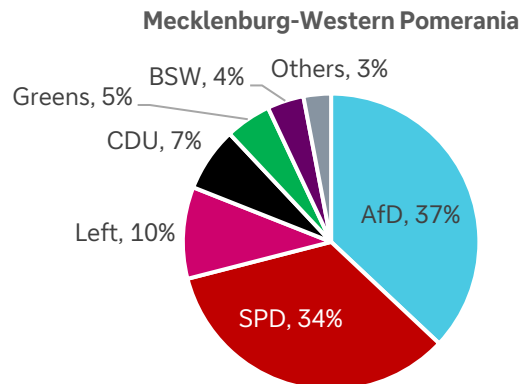
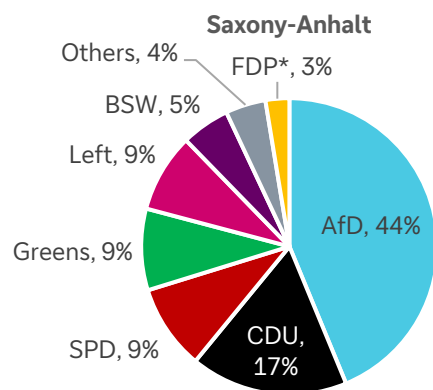
Key takeaways

- Contrary to many concerns, the Saxony-Anhalt election triggered little to no market reaction. While the AfD will likely enter government, the economic relevance of Saxony-Anhalt within Germany is limited. In Meck-Pom and Berlin, AfD support is weaker, though fragmented parliaments may complicate coalition building.
- Regional elections have limited implications for federal politics. More importantly, we don't see the elections derailing the government's reform agenda. If anything, weaker results for coalition parties could increase pressure to accelerate reforms.

2025 German GDP: breakdown by State



Regional elections: latest results and polls (MP and Berlin elections on 20th September)



*below 5% hurdle, not in parliament

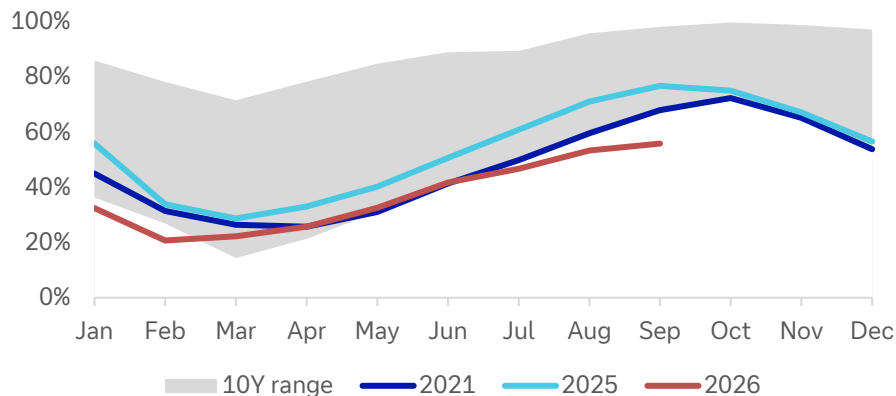
Source: Wahlrecht.de, Statistikportal des Bundes und der Länder, Deutsche Bank Research 09/15/2026

Macro headwinds – manageable?

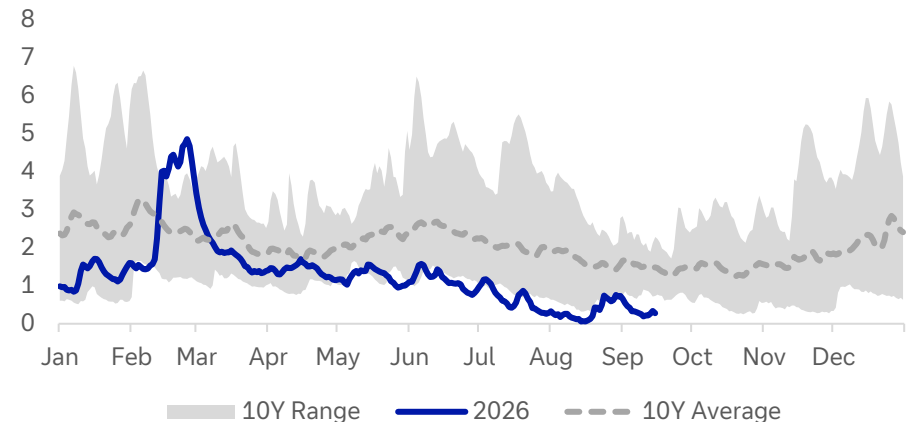


Energy costs to weigh on short-term sentiment. However, current levels are still manageable. We estimate energy costs to be 1.2% of sales for listed German companies.

Gas storage level at only 56%



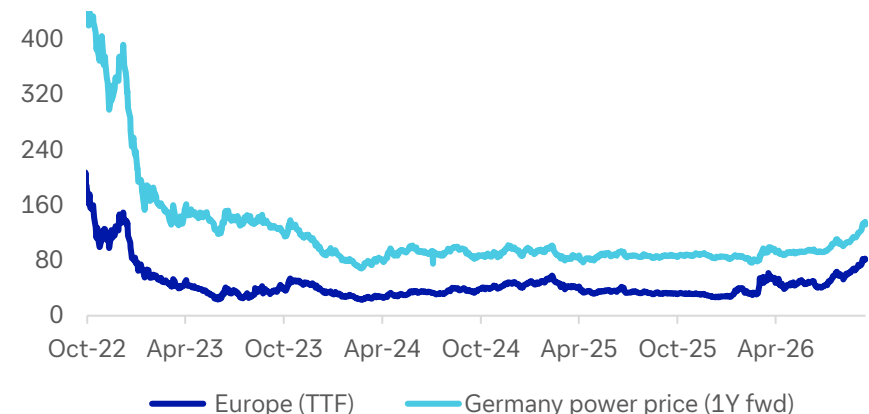
Rhine level at Kaub remains too low



Bund yield at highest since 2009



Gas and power prices up, but far below '22



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

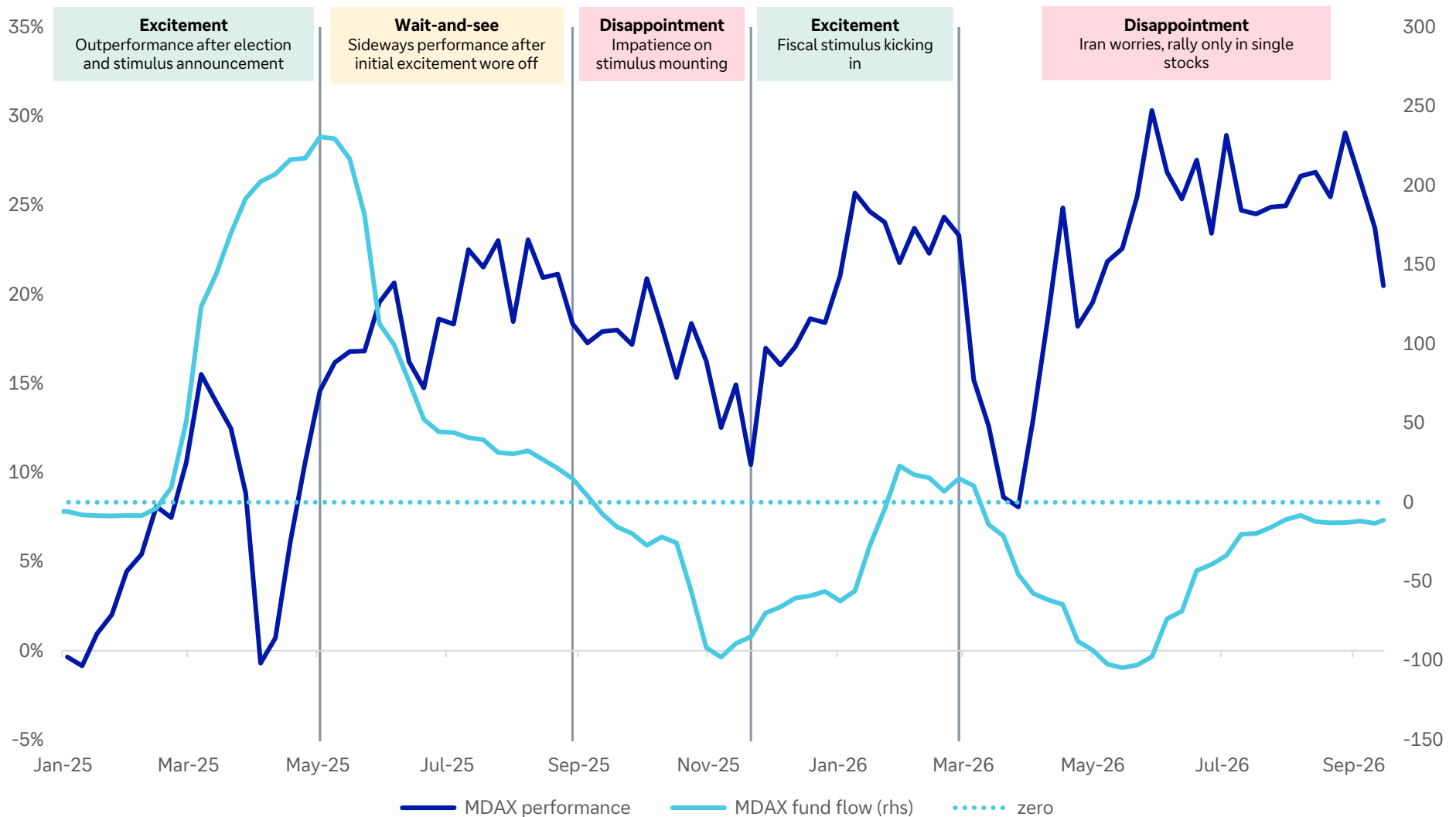


Equity markets

Germany: Rollercoaster of emotions



We have been going through many emotions. Bring patience to benefit from the recovery.



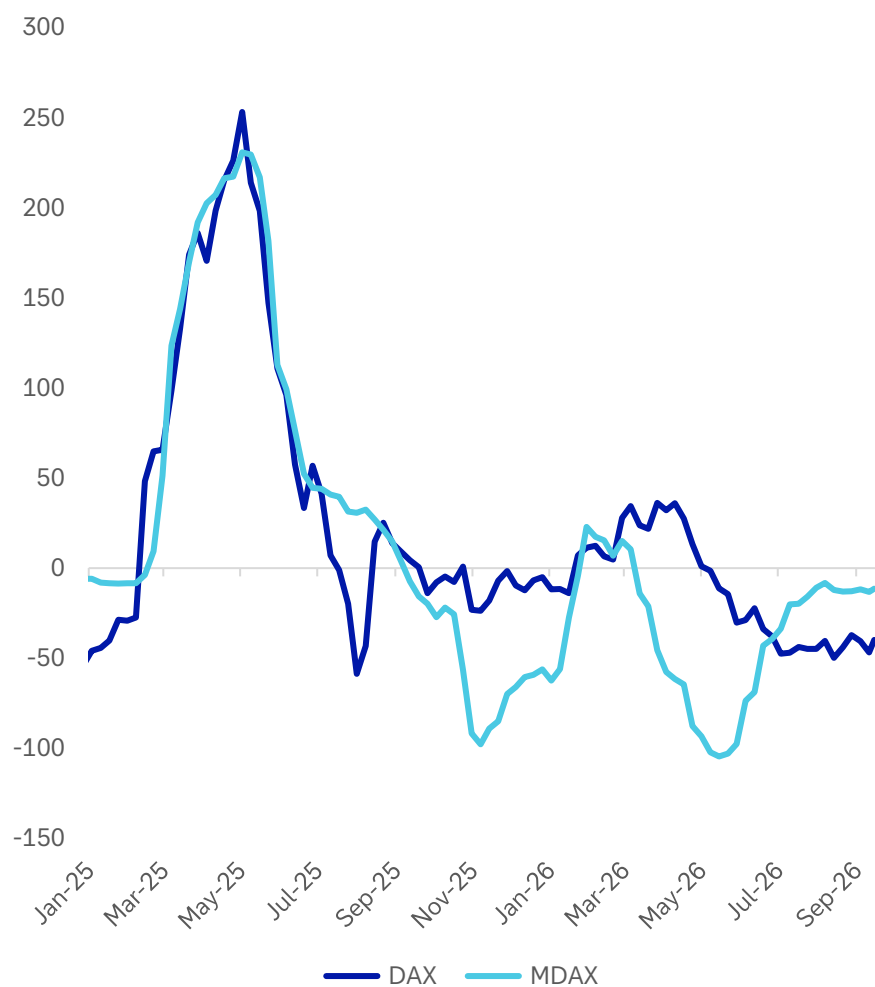
Aggregate fund flows (12-week mov. avg., mn EUR) based on up to 10 largest ETFs for the index. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

German ETF flows slowly recovering

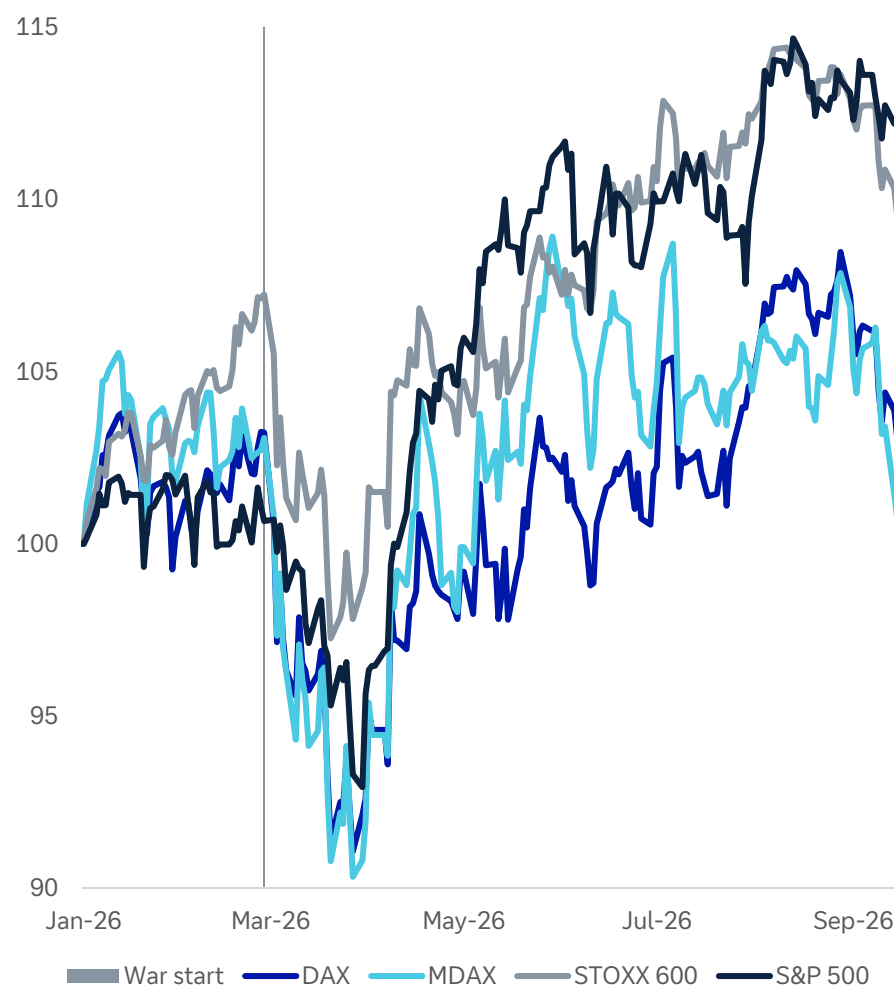


Despite the improvement in macro indicators, flows into German equities are still depressed. The MDAX underperformed the STOXX 600 by 2% since end of June.

DAX & MDAX ETF aggregate fund flow



YTD performance (total returns, local CY)



Note: Fund flows proxied by flows into and out of largest (up to) 10 ETFs for the index, 12-week moving average, mn EUR. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Equity implications: DAX vs MDAX

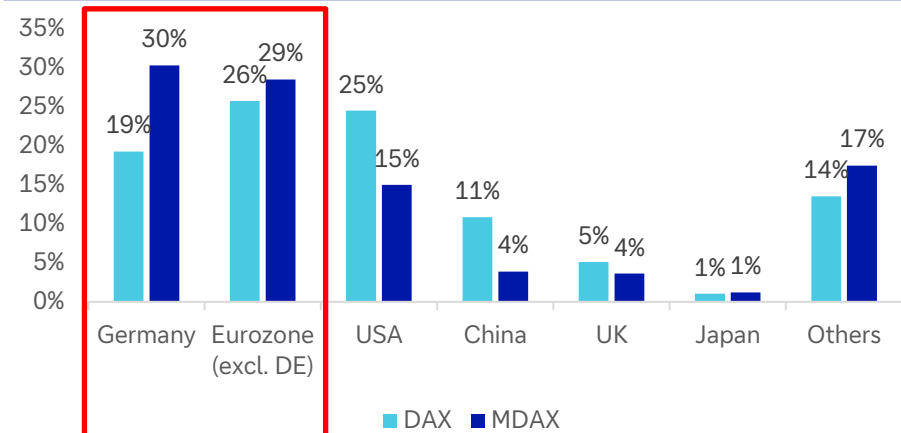


We think the MDAX is a better proxy for German growth and should benefit most from economic improvements and new reforms.

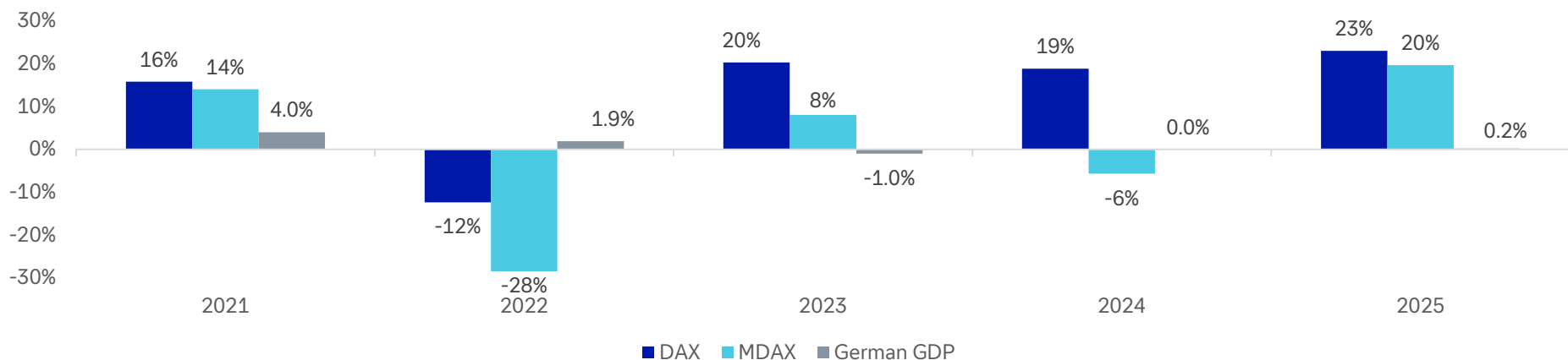
Key takeaways

- The DAX's exposure to Germany is limited; a better proxy for Germany-specific news flow is the MDAX.
- The MDAX has a higher revenue exposure to Germany than the DAX (see right chart) and its performance is linked much more closely to German GDP growth (see below chart). The new reform agenda and a more industry-friendly policy approach in Germany bear significant catch-up potential for the MDAX which has underperformed the DAX significantly over the past years (see [slide 30](#)).

Regional revenue exposure (FY24)



MDAX is the better proxy for German growth



All exposures are best estimates by our DB company analysts and are not necessarily based on published company data.
Source: Bloomberg Finance LP, Company Data, Deutsche Bank Research 09/15/2026

MDAX still offers high catch-up potential



German mid caps underperformed large caps massively over the past years. The improving macro backdrop and new reforms should support the MDAX.

Total return since 2022



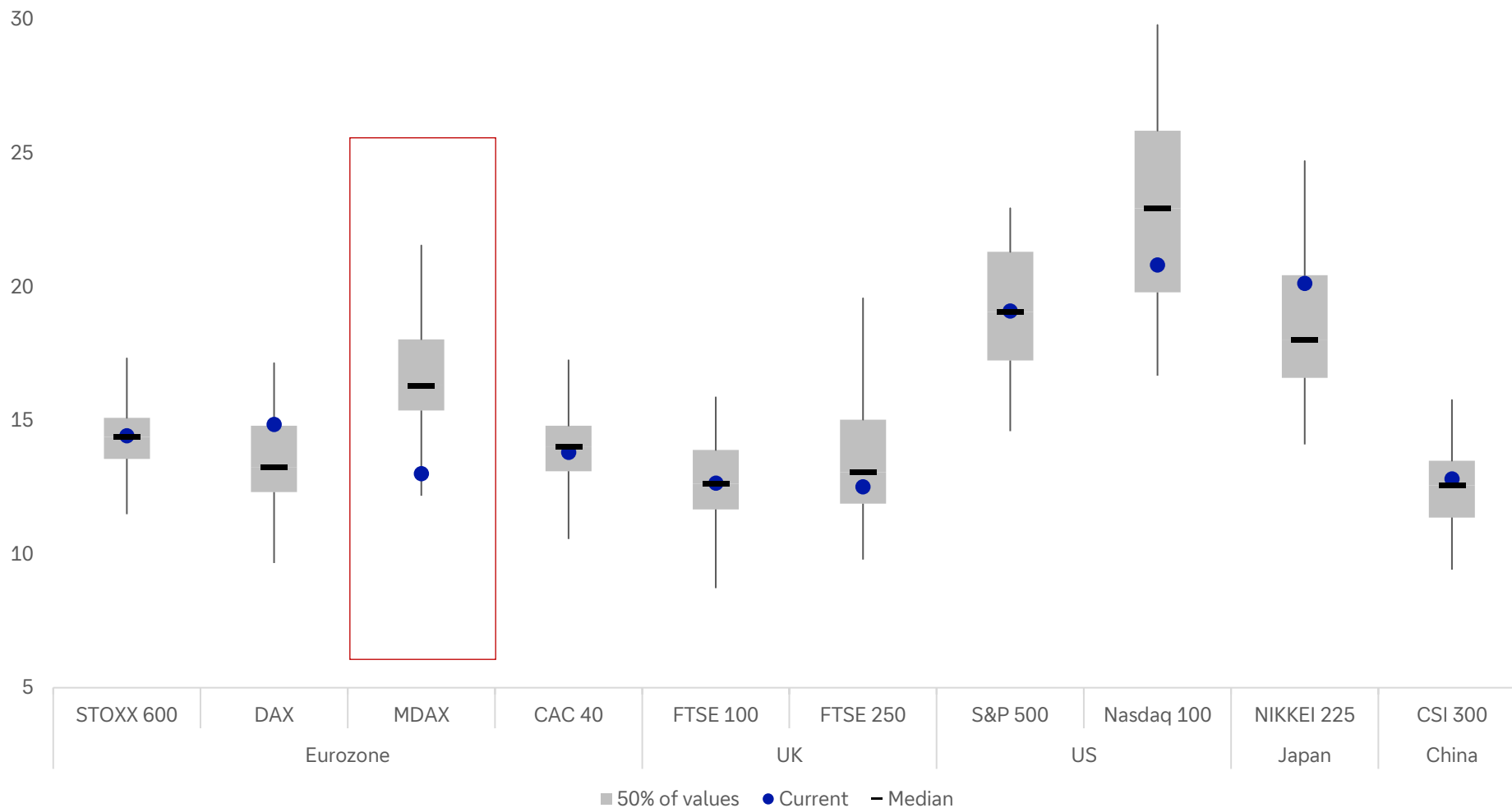
Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Valuations



In our view, European valuations are not elevated. Within Europe, the MDAX trades at lowest value relative to history

Valuation vs 10y median (12m fwd P/E)



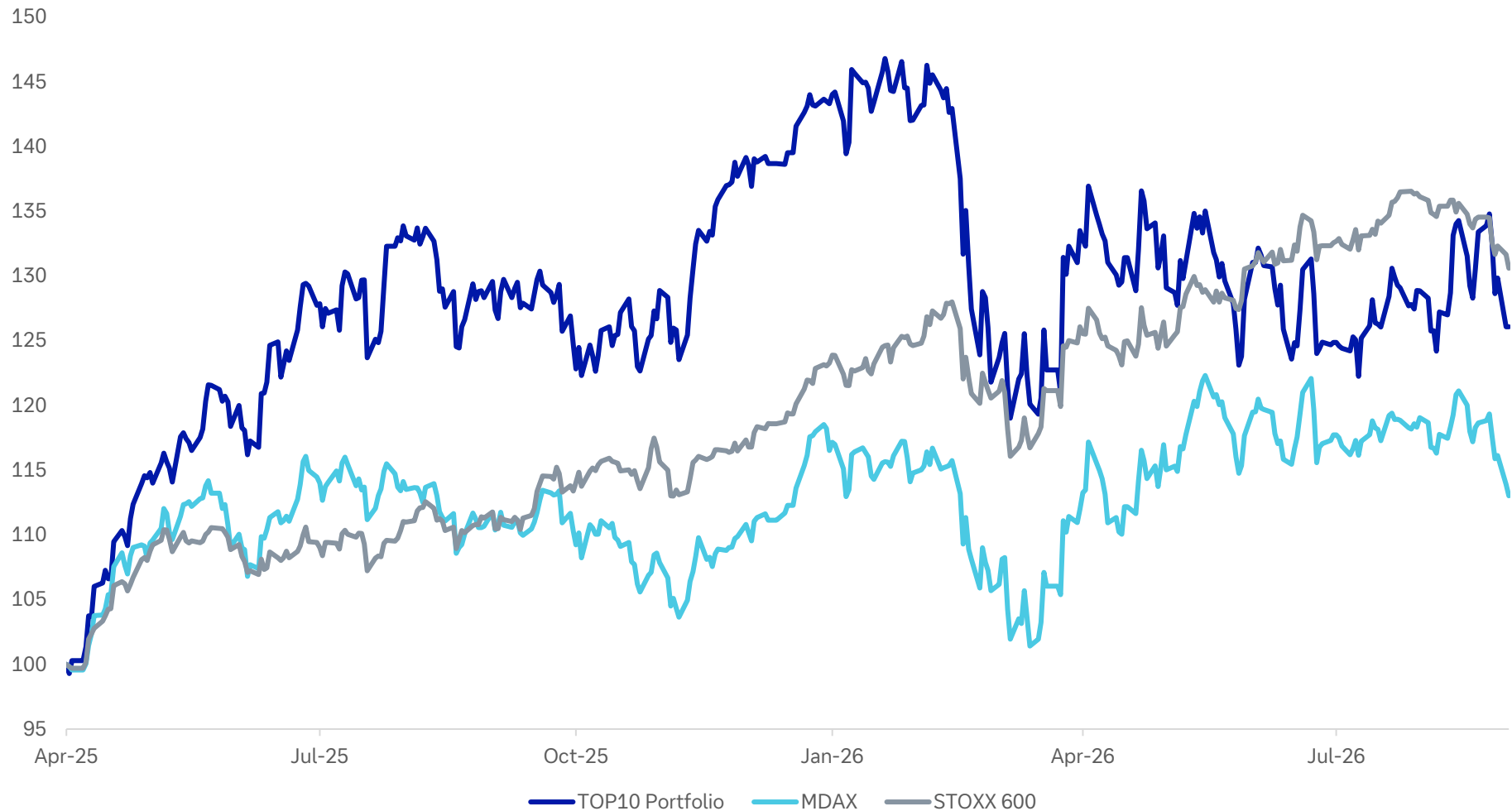
Time period: Aug 2016 – today.

Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

TOP10 Germany stock basket



Total return since inception



Note: German top picks portfolio inception was on 04/15/2025. . Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026



Reform agenda

Germany: Reform hammer in full swing



Area	Detailed measures	Status	Area	Detailed measures	Status
Economic Growth	– "Growth Booster": – 30% accelerated depreciation on machinery and equipment, 75% on EVs – Gradual corporation tax cut from 15% to 10% and R&D tax credit expansion	– In place – Jul '25 - Dec '27 for depreciation – 2028-2032 for corporate tax cut	Health Care System	– Restricting free family insurance – Higher contributions by higher earners – More co-payments by patients and less benefits (braces, dentures, homeopathy) – Price caps for hospitals/pharma	– Approved in parliament in July, – Effective in 2027/2028
Energy Prices	– Industrial power price: electricity price for corporates lowered to 5 c/kWh – Earlier abolition of gas storage levy – Lower grid fees – Electricity tax cuts to EU minimum level is extended indefinitely – Reduction of energy tax on petrol and diesel by 17 ct/l for two months	– In place – Industrial power price until Dec '28 – Reduced energy tax only during May + June '26	Private Pension System	– "Frühstart-Rente" (early start pension): monthly public grant of €10/child (ages 6-18), to be invested into capital market products, tax-free until the payout phase. Withdrawals generally allowed from age 65.	– Approved by the Cabinet in August, Parliamentary approval pending
Private Pension System	– "Altersvorsorgedepot" (old-age provision account): retirement savings account with tax benefits, allowing investment into products without guarantees (e.g., ETFs)	– Approved, to become effective on Jan 1, 2027	Statutory Pension System	– Gradual increase in retirement age to 67.5 – Limiting early retirement programs – Age for old-age part-time work up to 58 – Raising number of contributors, e.g. self-employed, MPs, managers – Introduction of mandatory funded component: 2% additional contribution into centralized pension fund.	– Approved by coalition committee, Parliamentary approval pending
Sector-specific	– "Construction turbo" law to accelerate planning and approval procedures – Air ticket tax cut by up to ~€11/passenger – Autos: EV subsidies of up to €6,000 for buyers, extension of car tax exemption for EVs (up to 10y)	– Construction turbo until 2030 – Air ticket tax: since 1st July – EV subsidy 'til '29, EV tax exemption 'til 35	Labour Market	– Easier extension of fixed-term contracts – relaxed dismissal protection for top earners – Preferential tax treatment if a new job is taken up quickly – Medical certificate from first day of illness	– Approved by coalition committee, Parliamentary approval pending
Public Procurement	– Law to faster and simpler public procurement processes – No need to start a contract award process for federal public contracts up to 50k EUR	– In place since 1st July	Income Tax	– Increase of basic tax-free allowance – Higher deductions for families and employees – Flattening of the tax rate for low income – Increase of "rich-tax" to 45% on taxable income >€250k (47% for income >€280k)	– Approved by the Cabinet in September, Parliamentary approval pending
Labour Market	– Retirees can earn up to €2,000/m tax-free when working while on retirement	– In place since Jan '26	Bureaucracy	– Removal of some reporting obligations – Reduction of need for company representatives – Applications (e.g for construction projects) to be automatically approved after 3m – Aim to cut federal ministries staff by 8%	– Approved by coalition committee. – Bureaucracy Reduction Act passed in May '26
Infrastructure Acceleration Act	– Prioritizes approval process for critical infrastructure (e.g., railways, highways) – Digitalization of approval processes – Easing of environmental checks	– Passed by Bundestag on June 26			

Source: Federal Ministry of Finance, Federal Government, Bundesregierung, Deutsche Bank Research 09/15/2026

Germany: pension reform

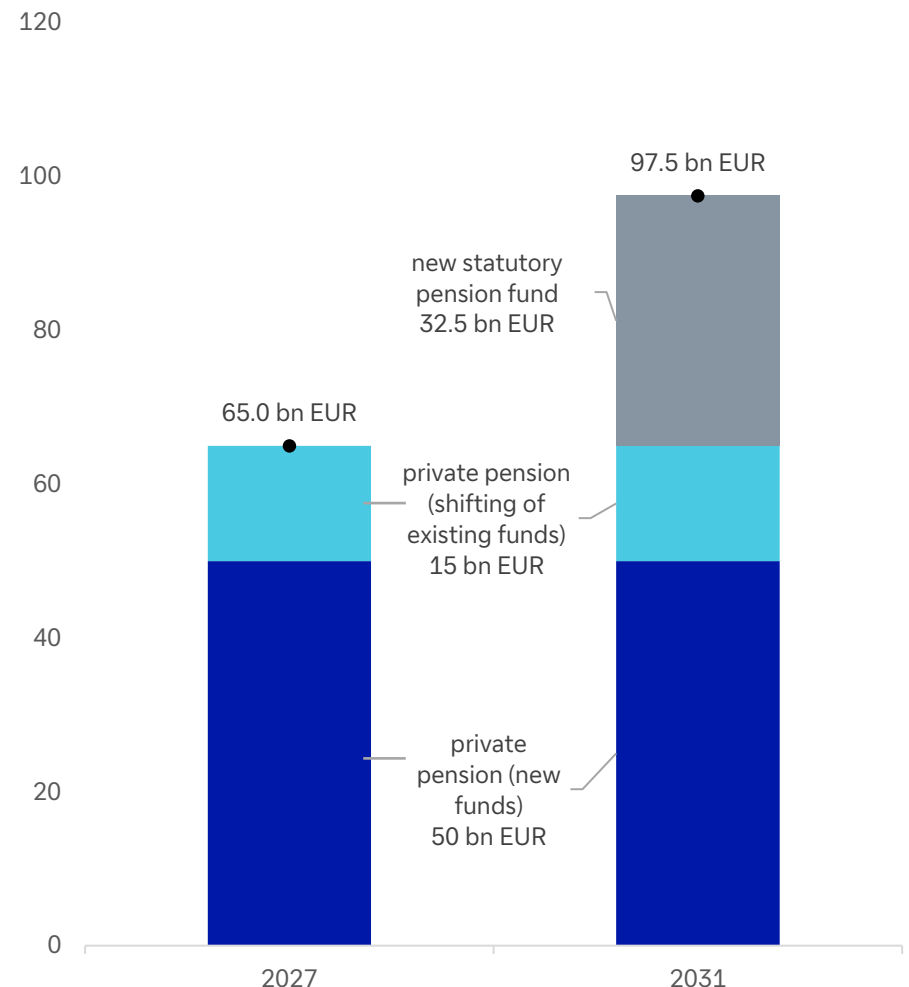


Pension reform could lead to additional annual inflows of 10-15 bn EUR into European equities over the next years.

Key takeaways

- The planned reform of Germany's pension system could generate **significant inflows into European equities** by shifting retirement savings towards capital markets.
- On the private pension side, the new “**old-age provision account**” encourages savers to move funds from low-yield deposits and insurance products into higher-return investments. If half of the eligible population invests the annual amount required to receive the full tax benefit (€2,300), **inflows could reach €50bn per year**.
- We also expect assets to **migrate from the existing “Riester” system**. With total estimated Riester assets of €150bn, a 10% annual reallocation would generate a further **€15bn of inflows**.
- In parallel, the government plans to introduce a funded component to the statutory pension system, equivalent to 2 pp of gross salaries, to be invested through a central **public pension fund**. Contributions are planned to rise from 0.5pp in 2028 to 2pp by 2031, implying annual **inflows of €8bn initially and €30-35bn by 2031**, according to our economists.
- Overall, we estimate these reforms could generate **€60-100bn of additional annual capital market inflows**, of which around **€10-15bn (15%) may find its way into European equities**. To put this into context: the top 50 largest European ETFs saw inflows of c. €40bn in 2025.

Projected inflows into capital markets



Source: BMAS, Deutsche Bank Research 09/15/2026

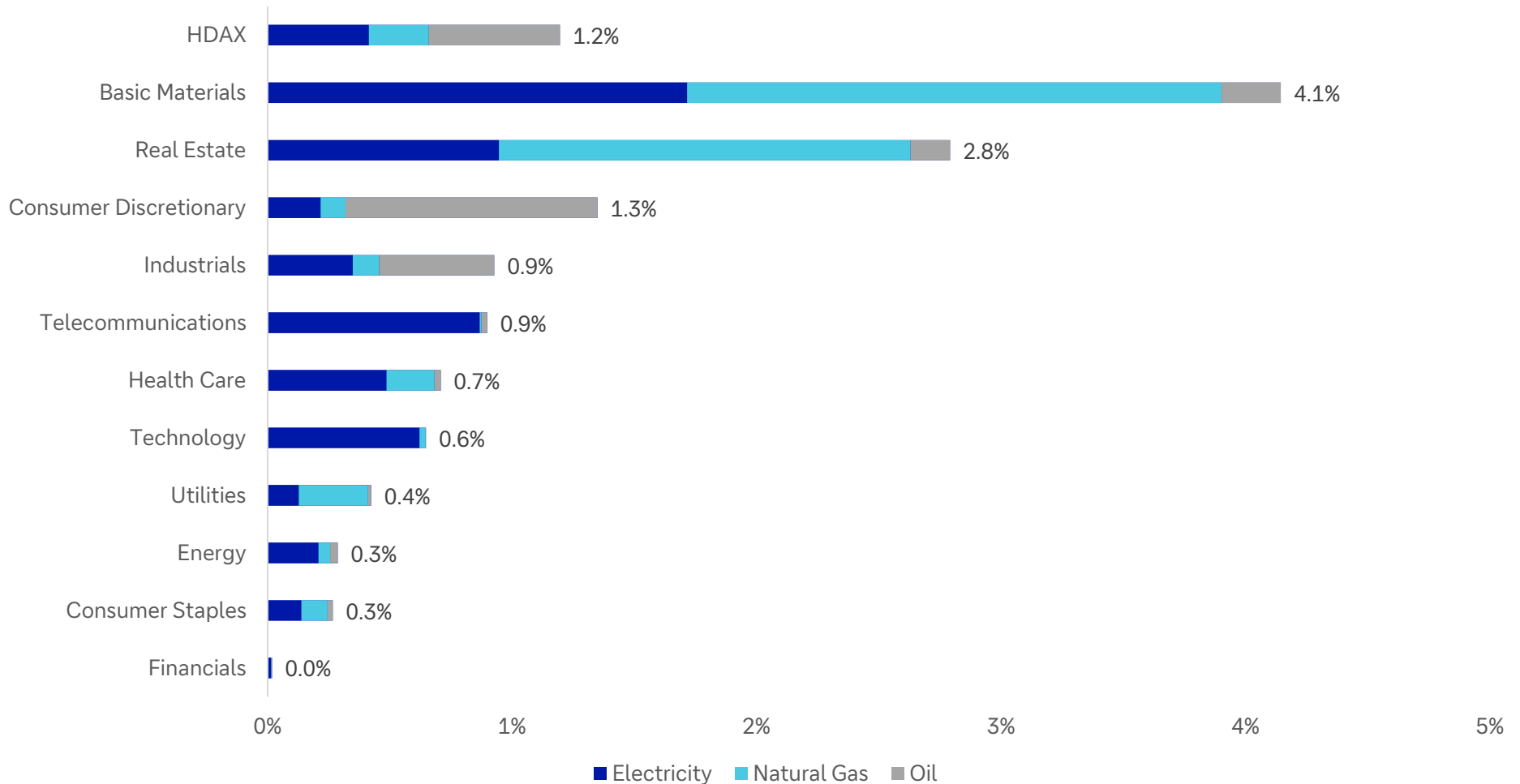


Appendix

Energy costs comprise only 1.2% of HDAX revenues



HDAX energy cost as % of sales*



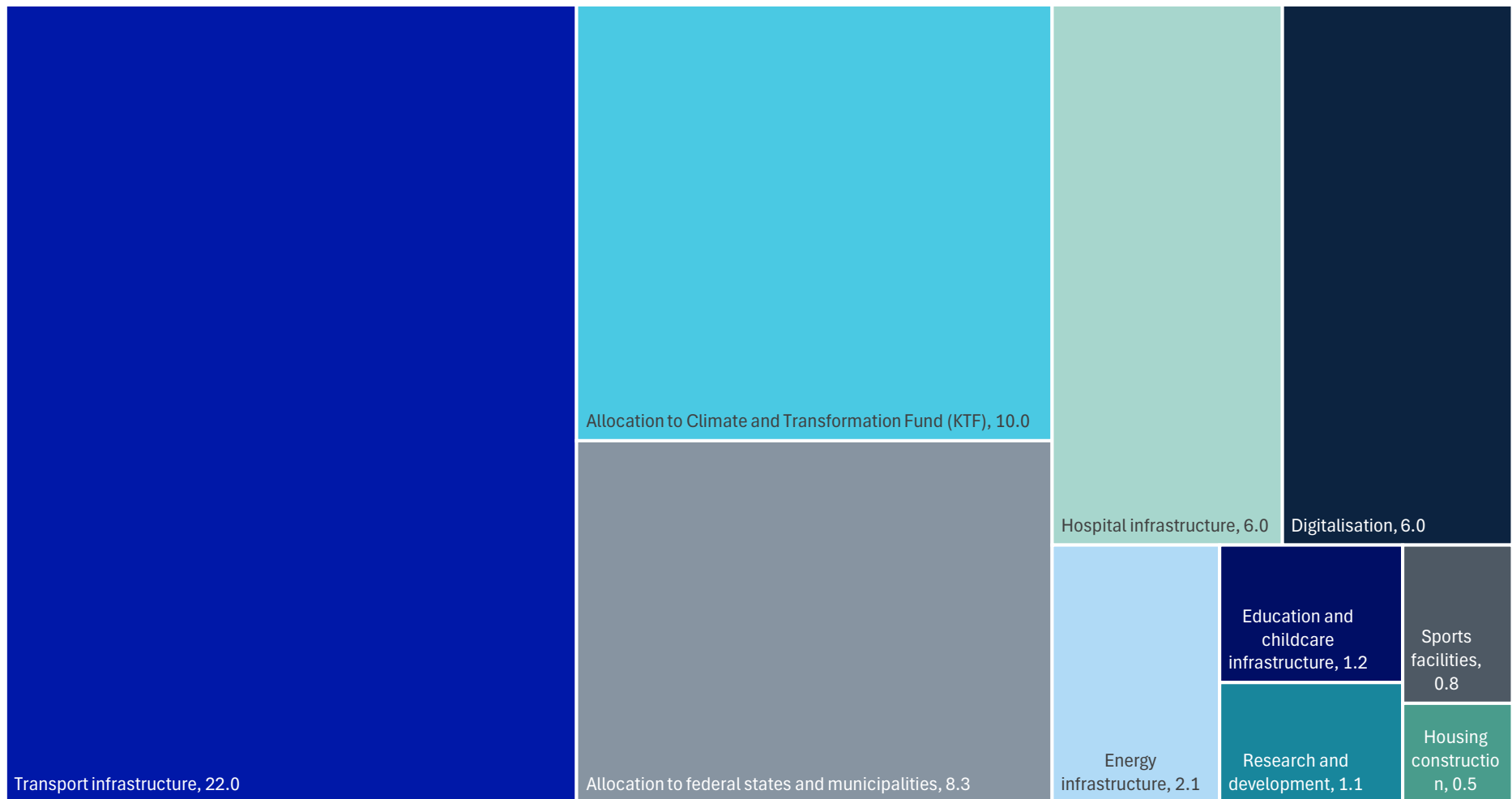
*N=50, based on 3y average German power prices, TTF natural gas and Brent oil. Source: Bloomberg Finance LP, Deutsche Bank Research 09/11/2026

Where to invest?



Most of the higher spending is going into Defence and Infrastructure. Defence has already rallied. We expect sectors with order intake from Infra spend to benefit most.

Planned investments from the special fund for Infrastructure & Climate Neutrality, 2026, mn EUR

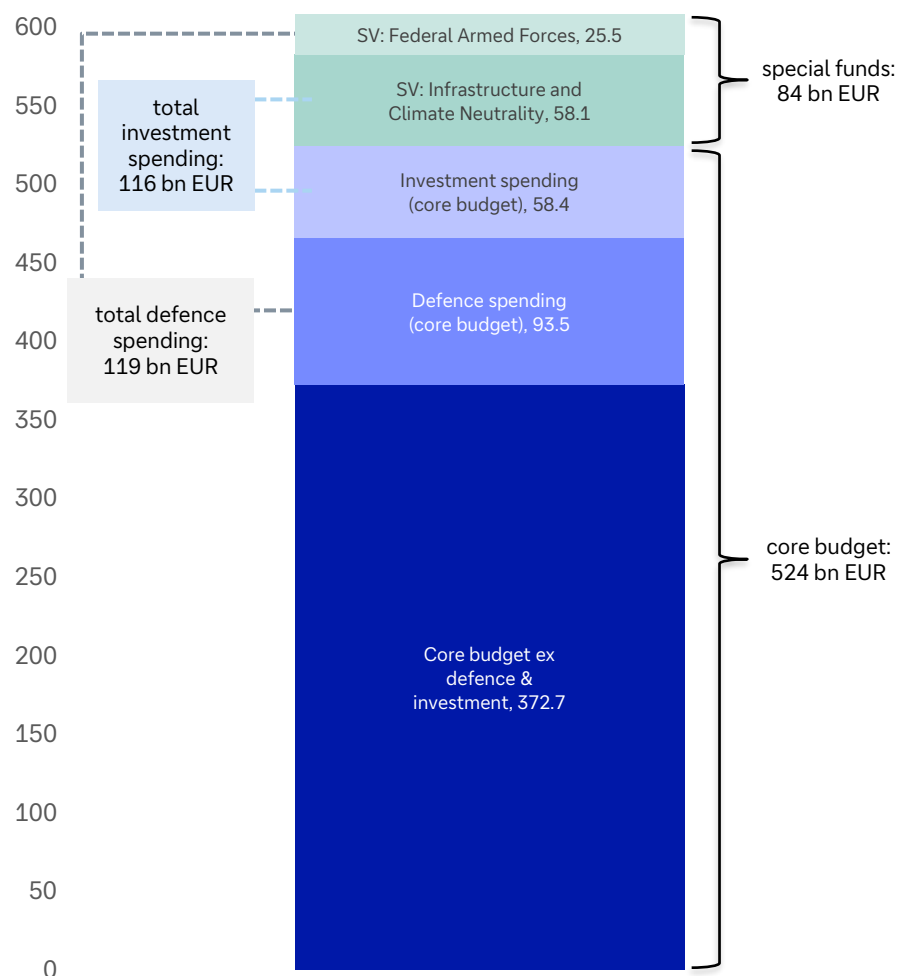


Source: Bundesfinanzministerium, Deutsche Bank Research 09/15/2026

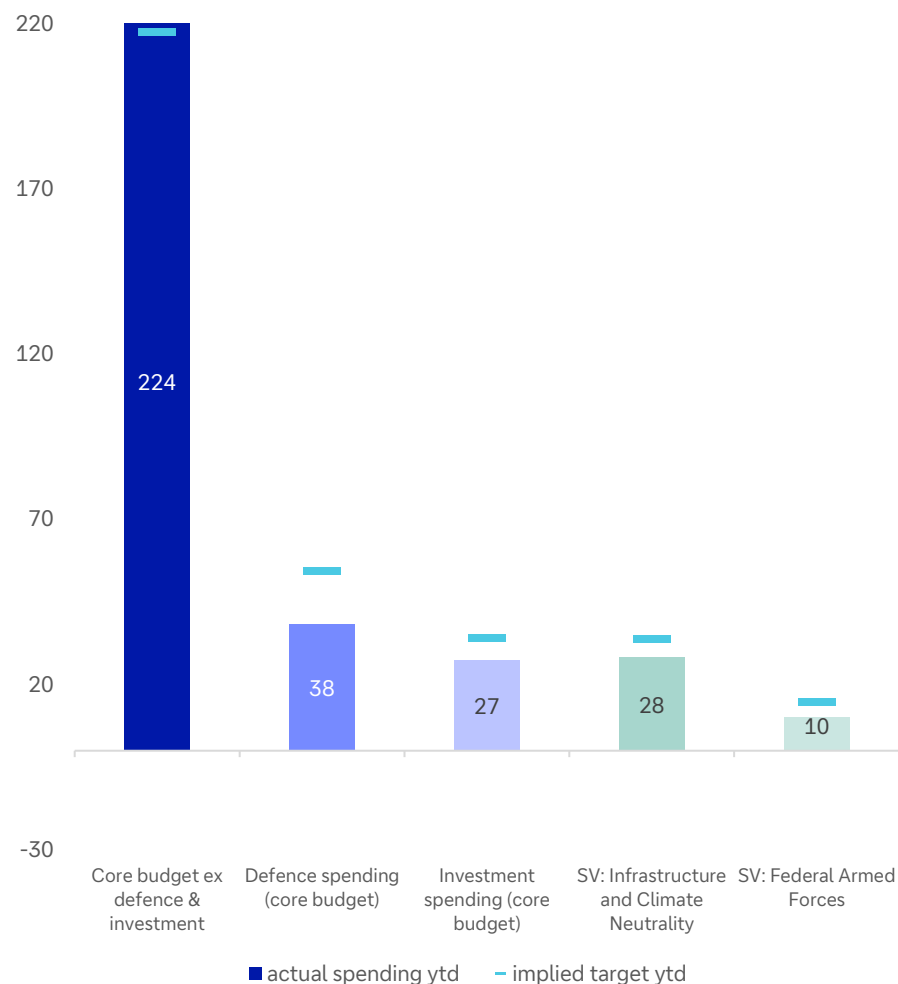
Germany: fiscal expansion underway, with room for spending to accelerate



2026 planned fiscal spending



Spending still below target year-to-date

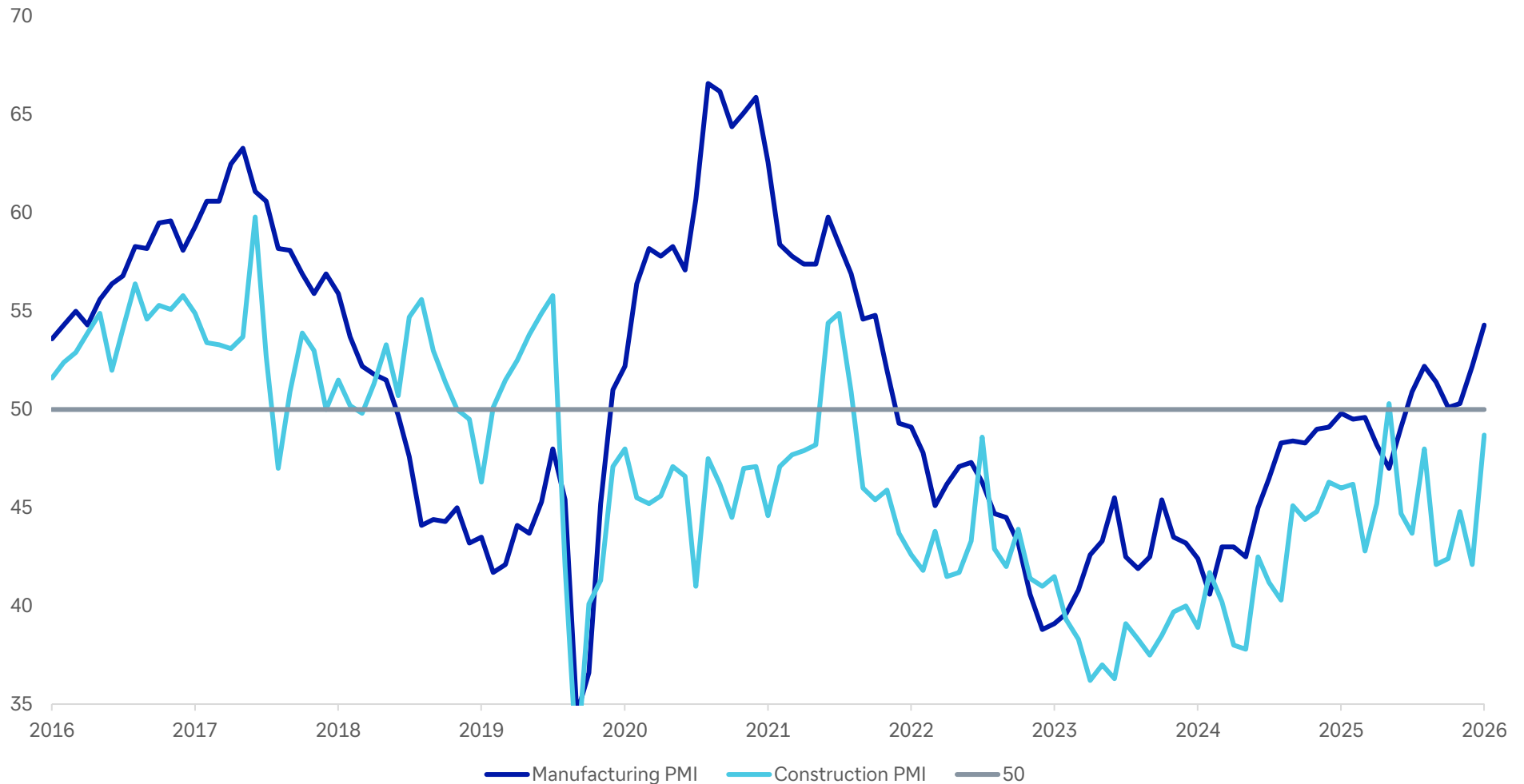


All figures in bn EUR. Actual spending data as of end of July. Source: Bundesfinanzministerium, Deutsche Bank Research 09/07/2026

Manufacturing PMI highest in over 4 years, Construction PMI catching up



Germany Manufacturing PMI



Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

German reforms: example from the past

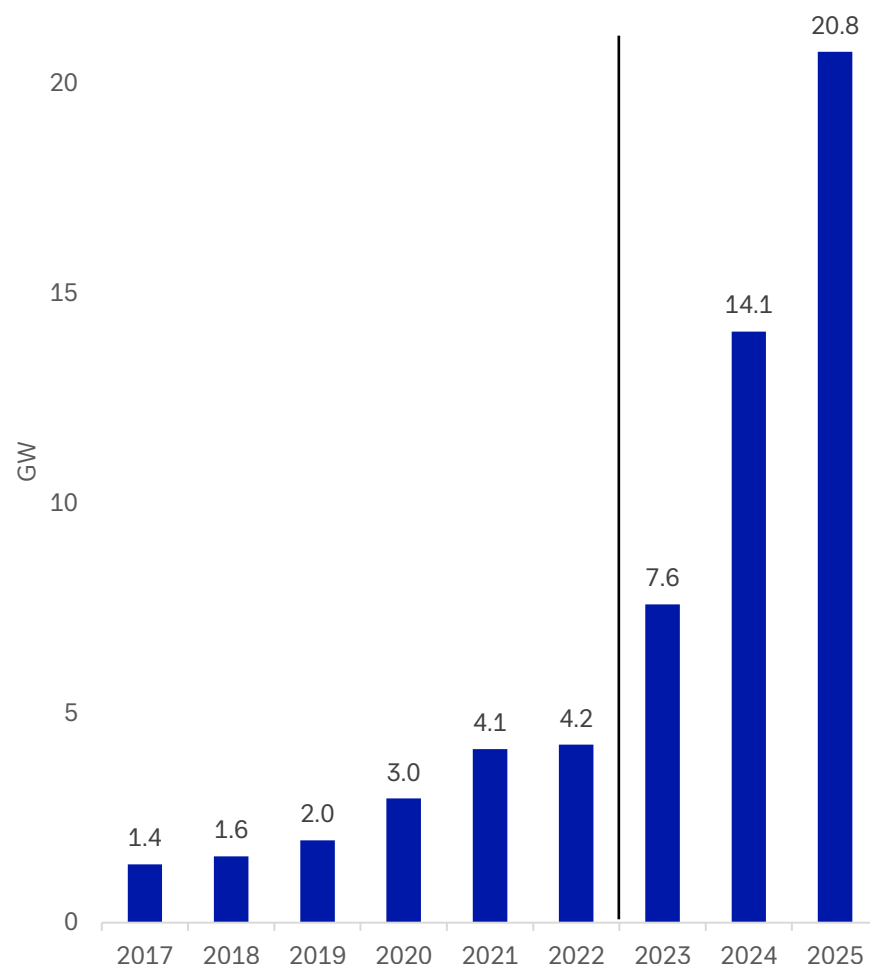


Onshore & Offshore Wind Act

Key takeaways

- In 2023, Germany introduced the Onshore Wind Act and revised the Offshore Wind Energy Act to accelerate the energy transition and streamline renewable energy development.
- For the first time, federal states face legally binding requirements to allocate land for wind power. The share of Germany's land designated for onshore wind is set to increase to 1.4% by 2027 and 2.0% by 2032.
- Policy reforms are already delivering results. In H1 2025, Germany approved 20.8 GW of new onshore wind capacity, nearly four times the level seen in 2022, marking a record pace of expansion.

Newly approved onshore wind energy capacity



Source: SMARD, Statista, Deutsche Bank Research 09/15/2026

German reforms: example from the past

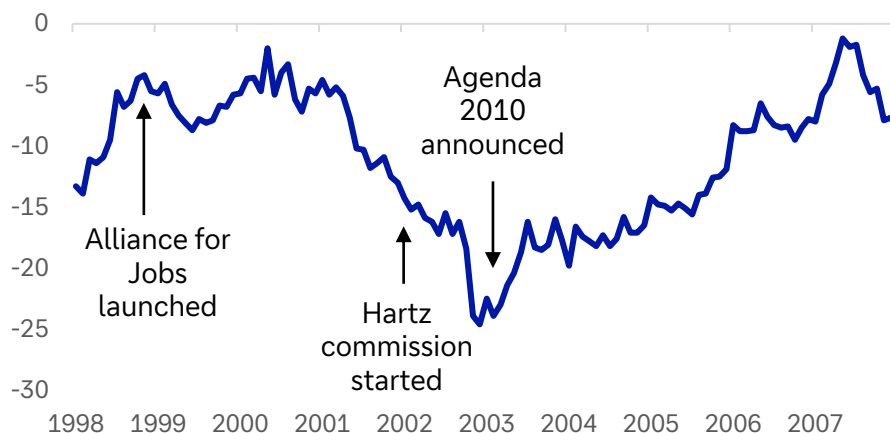


Agenda 2010

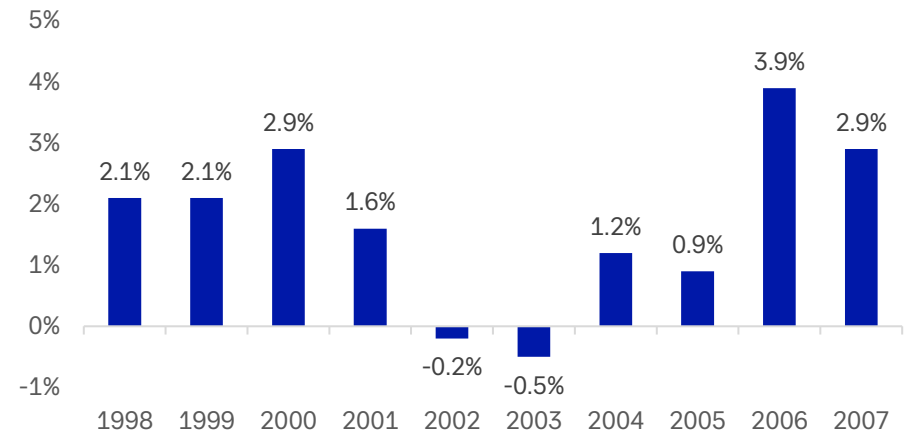
Key takeaways

- Facing stagnant growth and rising unemployment, Germany launched a series of labour-market reforms between 2002 and 2005, beginning with the Hartz Commission and culminating in Agenda 2010.
- The reform package tightened unemployment benefits, expanded labour-market flexibility, improved job placement, and reduced income and corporate taxes to strengthen incentives for work and investment.
- While initially controversial, the reforms helped restore confidence: GDP growth turned positive in 2004, consumer sentiment improved, and equity markets rebounded.

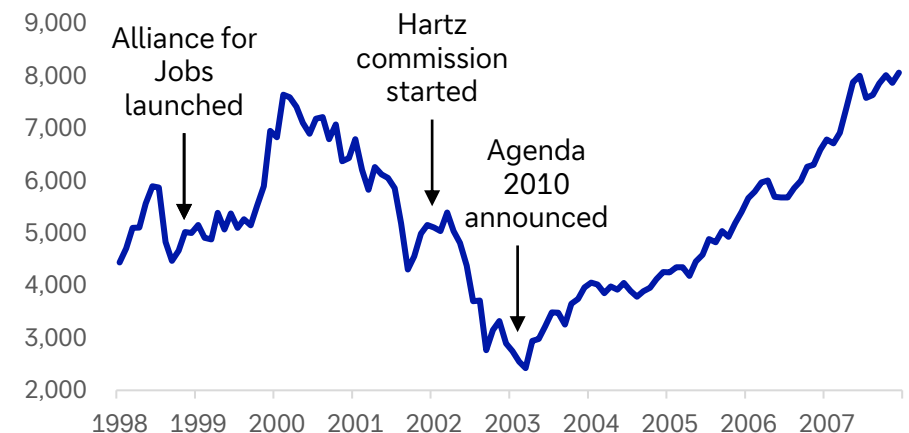
Consumer confidence



Germany annual GDP growth



DAX



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Appendix 1



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Deutsche Bank
Research

ASSET TOKENIZATION 101

The Next Phase of Financial Infrastructure

September 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.



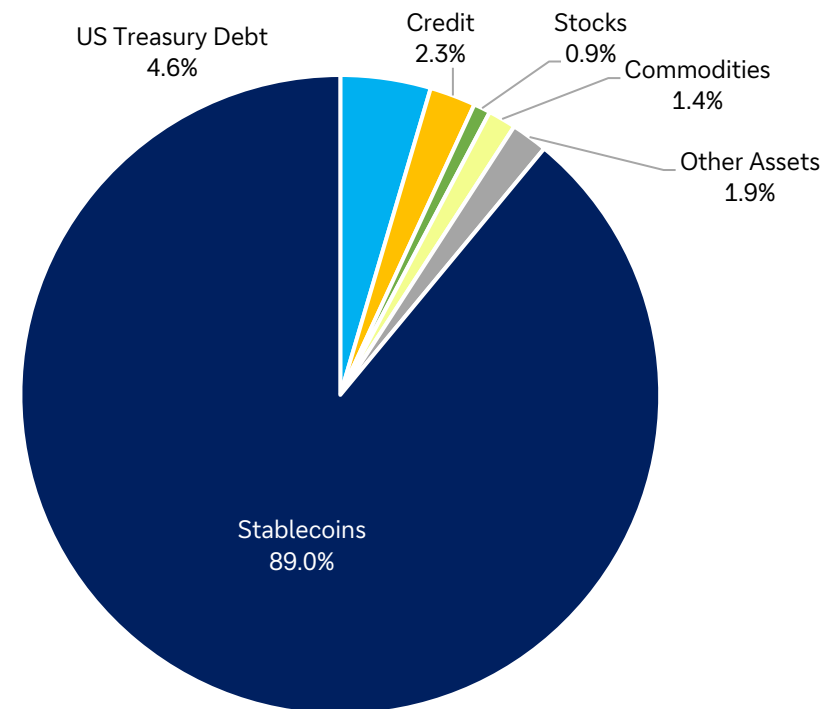
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EXECUTIVE SUMMARY: Tokenization expands into market infrastructure



- The global tokenized assets market, excluding stablecoins, grew from ~\$10bn in Jan 2025 to ~\$39bn in Sep 2026. US Treasury debt remains the largest segment outside stablecoins, with money market funds leading near-term adoption. Despite the US Digital Asset Market Clarity Act failing to advance through to the Senate yesterday, we remain positive about the momentum to tokenize both assets and capital markets in the coming years.
- Institutions are increasingly applying tokenization to settlement, funding and collateral management. Notable examples include DTCC's tokenization service, DLT-based intraday repo and collateral initiatives, and the Eurosystem's Pontes solution, which connects DLT platforms to TARGET Services and is expected to launch later this month.
- A broad tokenized capital market requires progress across three areas: assets, settlement money and market infrastructure. The tokenized RWA market remains small relative to traditional markets, and token issuance does not automatically create secondary-market liquidity.
- In this chartbook, we broaden our focus from tokenized assets, such as bonds and stocks, to the infrastructure through which they are issued; and review (i) market size; (ii) adoption trends; (iii) the regulatory drivers of tokenization; and (iv) barriers that will determine the pace of growth.

Market share of tokenized assets, Sep 14 2026



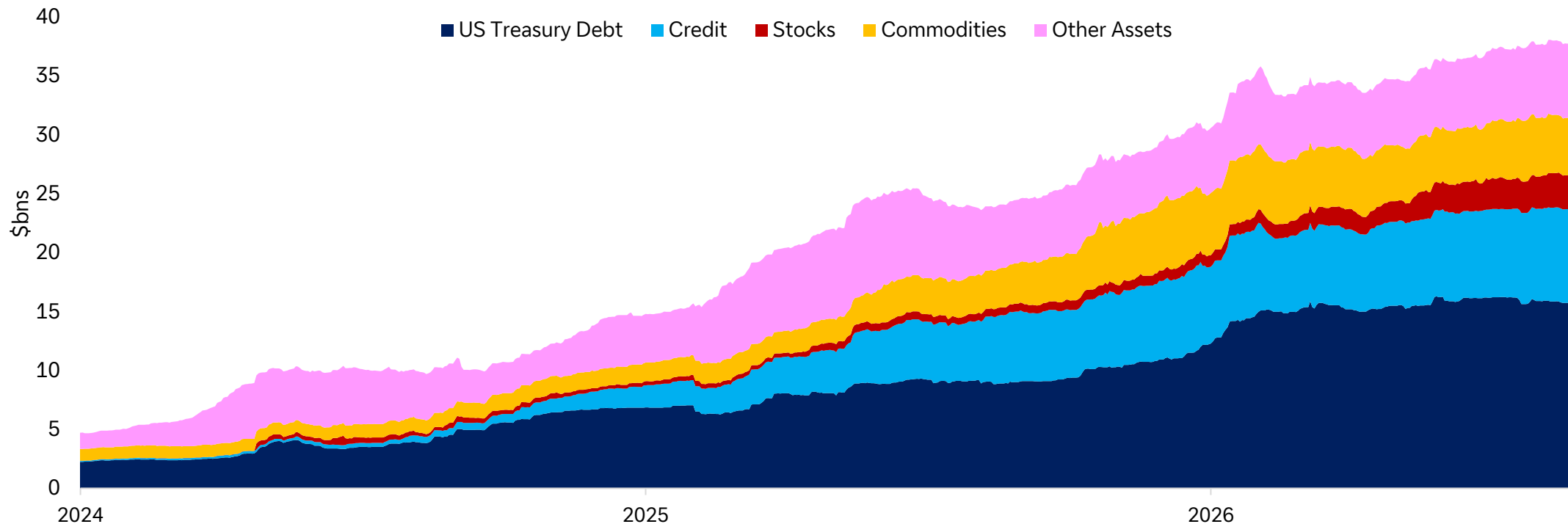
Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" denotes tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Note: Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, private equity, venture capital and real estate.



MARKET SIZE: US Treasuries continue to dominate

- US Treasury debt remains the largest on-chain tokenized asset segment excluding stablecoins, reaching \$15bn in September 2026. Much of that is driven by tokenized money market funds (MMF).
- Demand reflects appetite for highly liquid, short-term risk-free collateral assets, with the ability to generate yield.

Distributed tokenized asset value, excluding stablecoins



Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" refers to tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, stocks, private equity, venture capital and real estate.

DEFINITION : How asset tokenization works



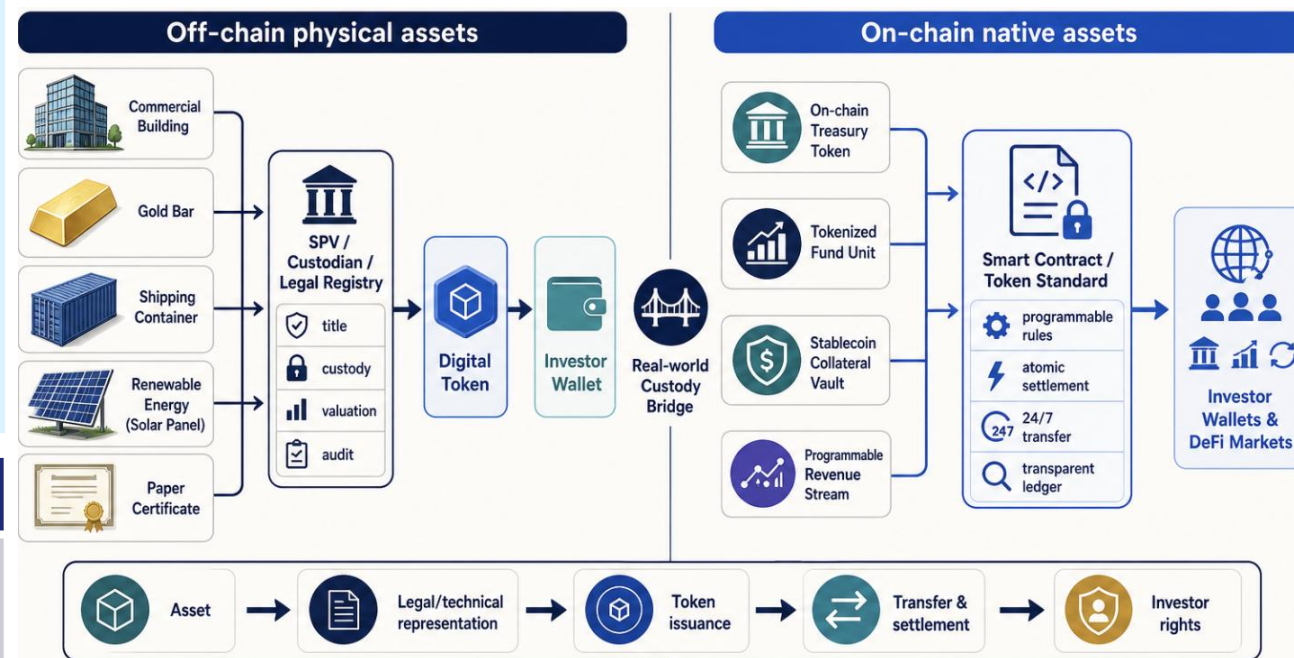
- Tokenization records assets or claims as digital tokens on a distributed ledger, using public or permissioned networks.
- Potential benefits include faster transfers, automated servicing and smaller investment units.
- Shared records can reduce reconciliation, but trading hours, redemption terms and investor access remain product-specific.

Two issuance models:

- **On-chain (or native):** The security or fund share is issued directly on a distributed ledger.
- **Off-chain (or tokenized representation):** The underlying asset remains outside the blockchain, but ownership claims and economic rights are represented digitally on-chain.

Component	Traditional form	Tokenized arrangement
Settlement assets	Bank deposits, central bank reserves, cash	Stablecoins, tokenized deposits, CBDCs, central bank money available through DLT platforms
Issuance assets	Treasury bills, government bonds, money market funds, real estate, commodities etc	Tokens representing securities, fund shares or claims on underlying assets
Market infrastructure	Custodians, exchanges, clearing houses and transfer agents	DLT-based issuance, settlement and servicing, often operated by existing institutions

Tokenization can enable automated transfers and settlement



Native digital securities are issued on-chain but may still require custodians, registrars and other intermediaries, according to the [SEC](#).

Sources: Deutsche Bank Research. Image generated by AI.



STABLECOIN SPOTLIGHT: The settlement side of tokenization

- Stablecoins account for ~89% of the combined value of stablecoin and distributed tokenized assets. We assess them separately because their principal role is payment and settlement.
- They support near-instant, round-the-clock transfers for remittances, business payments and treasury operations. Cross-border savings depend on transaction fees, FX costs, compliance and conversion into bank money.
- [Chainalysis](#) projects adjusted annual transfer volume could rise from \$28tn in 2025 to \$719tn by 2035; higher scenario approaching \$1.5 quadrillion.

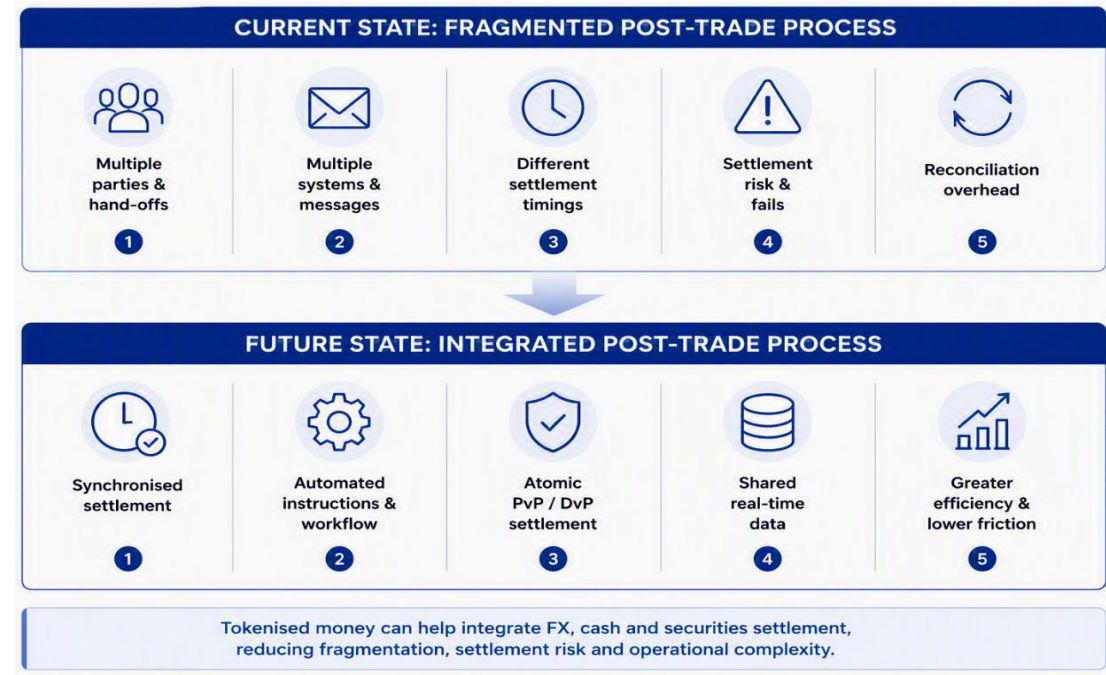
Regulation and competition

- The US Treasury has [proposed](#) implementing rules for payment stablecoins. The GENIUS Act takes effect on the earlier of 18 January, 2027, or 120 days after final implementing regulations are issued.
- Stablecoins could increase competition for bank deposits and payments. Exchange rewards are distinct from issuer-paid interest, which GENIUS prohibits for permitted payment stablecoins.

Risks

- Trust depends on reserve quality, legal redemption rights and operational resilience. Stablecoins remain dependent on their issuers and supporting institutions.

A more integrated post-trade process



Sources: Deutsche Bank Research, taken from DB's [FX Special Report: Tokenised money and the future of FX - Deutsche Bank Research](#)

Tokenized deposits and central bank money provide alternative settlement options. Connecting the cash and securities legs of a transaction can improve settlement efficiency and reduce fragmentation.

For more on tokenized money and settlement, particularly for FX, read [here](#).

INSTITUTIONAL ADOPTION: Tokenization expands into funding, settlement and collateral



- Early efforts focused on tokenized claims on traditional securities (i.e. BlackRock's BUIDL, Franklin Templeton FOBXX), while the underlying assets remained off-chain.
- Financial institutions are now extending tokenization to the infrastructure through which assets are distributed, traded, settled and financed.

DLT adoption: small, but intraday repo offers a practical use case

- Global DLT fixed-income issuance [reached €4.8bn](#) in 2025, up 48% from 2024.
- DLT intraday repo (i.e. JPM Kinexys and Broadridge's DLR) [can accelerate](#) collateral transfers and settlement, helping institutions manage short-term funding needs.
- [A Finadium study](#) estimated that using intraday DLR for 15% of activity could reduce intraday liquidity buffer needs by 8–17%.

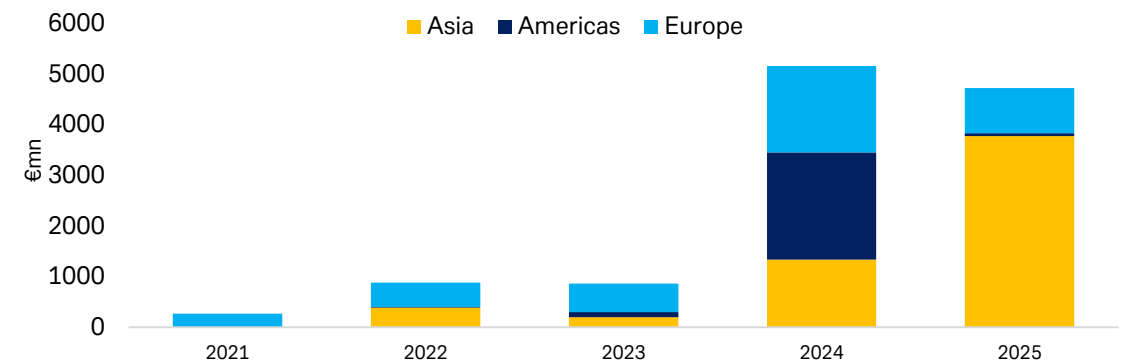
Tokenized securities on blockchain rails

- DTCC to launch its tokenization service in Oct 2026, covering Russell 1000 constituents, ETFs tracking major indices, and US Treasury bills, notes and bonds.
- Initial production trades took place in July with >30 participating firms. Tokens are recorded on DTCC's Hyperledger Besu network or on Canton, settling on either. The SEC granted a three-year authorization.

Official-sector initiatives and interoperability

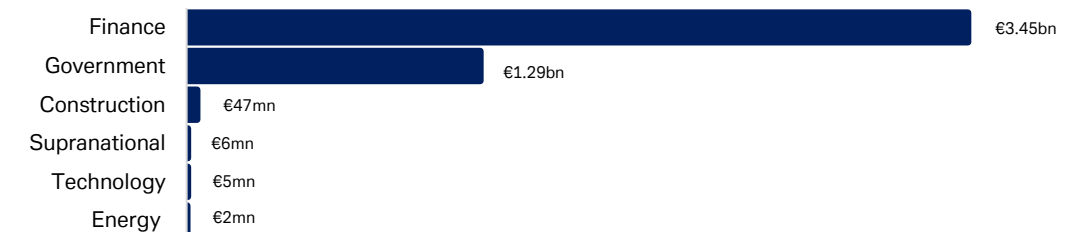
- Project Agorá, led by the BIS and IIF with central banks and financial institutions, demonstrated a prototype for cross-border settlement using tokenized deposits and central bank reserves. Real-value testing is the next phase.
- Clearstream, DTCC and Euroclear published a joint interoperability framework with BCG in March 2026 to address fragmentation across digital asset markets.
- Eurosystem's Pontes solution connects DLT platforms to TARGET Services for settlement in central bank money.

Digital bond issuance by region



Source: Deutsche Bank Research, Association of Financial Markets Europe.

Global digital bond issuance by issuer type in 2025



Source: Deutsche Bank Research, Association of Financial Markets Europe.

SHORT-TERM ADOPTION: Tokenized Treasuries and money market funds lead



- Tokenized cash-management funds, including BUIDL and USYC, combine short-term yields with more flexible transfers and collateral use.
- Potential benefits include faster settlement and lower transaction costs, making Treasury exposure more accessible across eligible digital platforms.
- Direct Treasury bond tokenization [accounted for](#) 27% of overall US Treasury tokenization in 2025, indicating adoption beyond fund structures.
- These assets can also serve as collateral in repo and margin arrangement, supporting demand beyond investment returns.

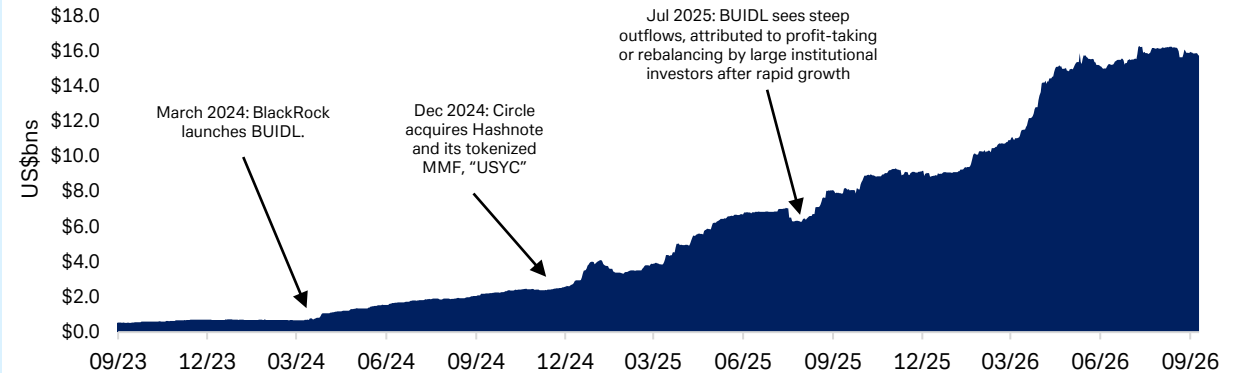
Secondary-market constraints

- Tokenized bonds are often held to maturity. Limited investor participation and weak interoperability between platforms constrain secondary-market liquidity.
- Some exchanges are attempting to improve liquidity by connecting digital and conventional markets through dual listings, including SIX digital exchange and Swiss Stock exchange).

Risks

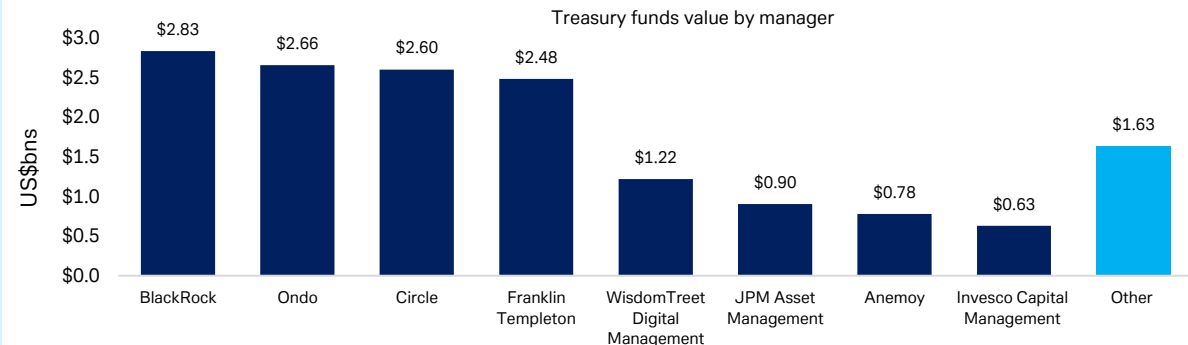
- Redemption terms can differ from the settlement hours and liquidity of underlying assets, creating potential liquidity mismatches.
- Blockchain transparency can improve visibility but may also accelerate withdrawals during stress.
- Using tokenized fund shares across multiple products and as collateral can create additional channels of contagion.

Tokenized US Treasury market value reaches \$15bn



Source: Deutsche Bank Research, RWA Database. Updated Sep 10, 2026.

Largest providers of tokenized Treasury products



Source: Deutsche Bank Research, RWA.xyz. Data from September 10, 2026. TMMF asset class includes tokenized US treasuries, corporate bonds, and non-US government debt.

MEDIUM-TERM ADOPTION: Tokenized equities remain small; DTCC's launch could broaden adoption



- Tokenized equities remain small (<1%) relative to traditional stock markets.
- [68% of equity tokenization](#) in 2025 consisted of direct tokenization of individual stocks or baskets of stocks, vs. tokenized equity funds.
- Suggests that blockchain adoption is increasingly focused on the securities themselves (i.e. DTCC method) rather than through third-party platforms (i.e. Ondo and xStocks).

Developments

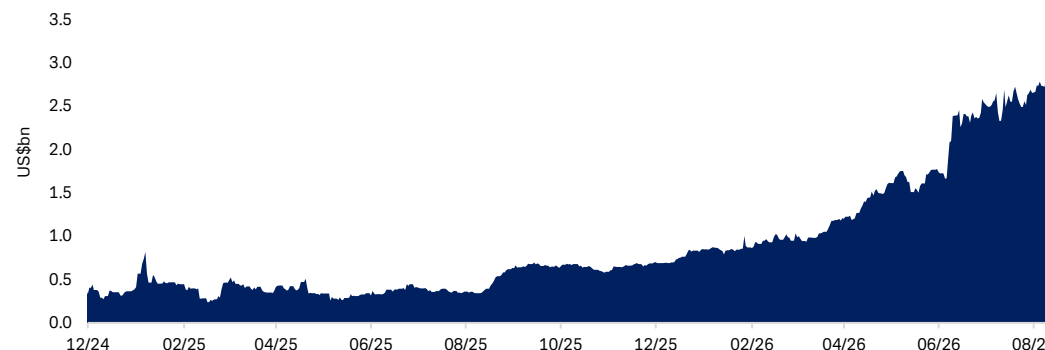
- DTCC's tokenization service, scheduled to [launch in Oct](#), will allow eligible existing securities to move across blockchain infrastructure within its regulated framework.
- These tokenized securities will retain the entitlements, investor protections and ownership rights associated with their traditional form.
- NYSE has announced plans for a tokenized securities platform designed to support 24/7 trading of US-listed equities and ETFs, with near-instant settlement and stablecoin funding.

Benefits: Reduce barriers for foreign investors; automated post-trade processes and longer trading hours; reduce information asymmetry (e.g., embedding listing or disclosure rules directly into the tokens).

Risks & limitations

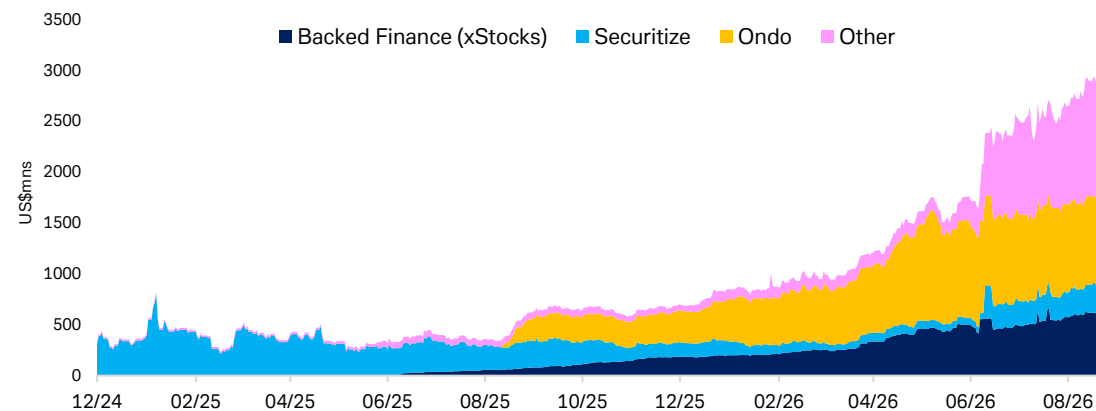
- Tokenized equities require clear legal ownership, enforceable investor rights and alignment with existing securities rules.
- Under some third-party wrapper structures, the token provides exposure to a share without making its holder a shareholder in the underlying company.
- Investors may therefore face issuer or custodian risk and may not receive the same voting, dividend or other rights as direct shareholders.

Market value of tokenized equities



Source: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

Tokenized equities by platform



Sources: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

LONG-TERM ADOPTION: Private credit expands through record-keeping; real estate is growing fast but from a negligible base



Private credit

- Private credit stands ~\$8bn in total, including asset-backed credit (\$2.6bn), specialty finance (\$2.5bn), corporate credit (\$1.9bn), and diversified credit (\$798mn).
- Activity includes lending linked to digital assets and selected corporate credit products.
- A larger use case is blockchain-based record-keeping: represented credit stands at ~\$35bn, over 4x the distributed market.
- Companies like Figure Technologies and Tradable use blockchains to support loan administration, ownership records, payments tracking, and servicing.

Challenges

- Tokenization remains concentrated among a small number of issuers and lacks a deep, and liquid secondary market.
- Legal fragmentation, credit quality and servicing arrangements remain important risks. Blockchain records do not remove the underlying borrower's default risk.

Real estate

- The market expanded from ~\$35mn in 2025 to \$226mn in Sep 2026, although it remains small.

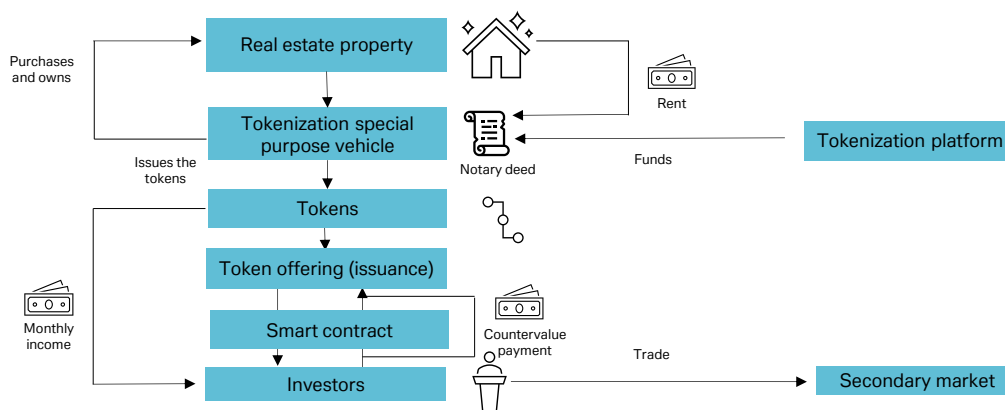
Benefits

- Fractional ownership can lower minimum investment sizes (typically for \$100-\$1,000 per unit) and widen access, particularly where traditional capital markets and mortgage markets are limited.

Challenges

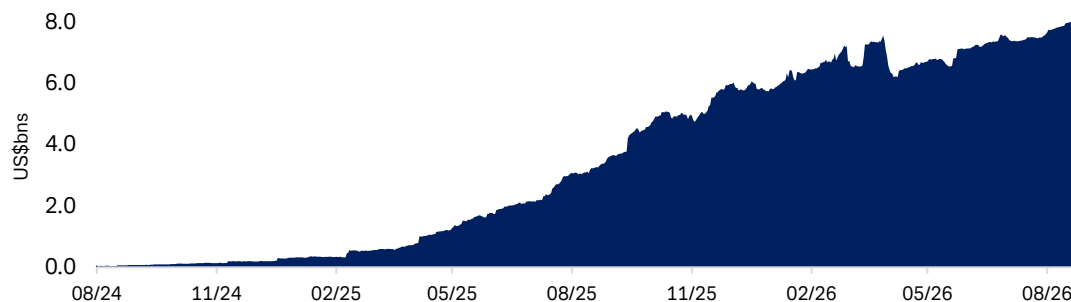
- A [BIS study](#) found that liquidity improvements can depend on platforms offering to buy tokens back. These arrangements can help investors exit but expose platforms to liquidity and solvency risks.

Real estate tokenization process



Source: Deutsche Bank Research, BIS, "When bricks meets bytes: does tokenization fill gaps in traditional real estate markets?"

Tokenized credit market stands at ~\$8bn



Sources: Deutsche Bank Research, RWA.xyz. Updated September 10. Excludes platforms that issues tokens using blockchain as a recordkeeping layer (i.e. Represented Layer)

BARRIERS TO ADOPTION: Weak secondary market liquidity and lack of regulations continue to challenge adoption



Main barriers to adoption

- **Secondary-market liquidity remains low.** Issuance growth has not consistently translated into active trading or deep markets.
- **Regulatory uncertainty:** Fragmented or incomplete frameworks slow institutional adoption and complicate integration with traditional financial infrastructure.
 - Legal uncertainty persists over whether token ownership conveys ownership of the underlying asset or a claim on it.
 - In segments like money market funds, rapid cross-border growth can increase regulatory arbitrage, capital-flow volatility and AML/CFT compliance challenges. Policymakers must adapt frameworks as the technology evolves.
- **Fragmented infrastructure:** Public blockchains, permissioned institutional ledgers and hybrid platforms often lack interoperability, fragmenting markets. Clearstream, DTCC and Euroclear published a joint framework to address this in March 2026.
- **Limited integration** with existing custody, settlement, and record-keeping arrangements prevents institutions from scaling tokenized products.
 - Proprietary infrastructure and centralized controls also raise questions about ownership and governance.

Sources: Deutsche Bank Research.

Jurisdiction	Regulatory status	Main implication
	EU MiCA + DLT pilot regime	Comprehensive crypto-assets regulation. Tokenized financial instruments and tokenized deposits remain subject to securities and deposits regulation respectively (e.g. MiFID). National pilots (e.g., EU's DLT Pilot Regime) are testing tokenized capital-markets applications; focus is on capital market infrastructure modernization.
	UK Cryptoassets Regime / BoE Systemic Stablecoins Regime Live - Digital Securities Sandbox, Digital Gilt pilot Program (DIGIT) and Great British Tokenised Deposits (GBTD) Project	UK Cryptoassets and Stablecoins Regime finalised in 2026. FCA and BoE launched the Digital Securities Sandbox for firms to test DLT-based blockchains; HSBC first bank to go live. The FCA has also identified fund tokenization as a strategic priority through 2025-2030. FCA/BoE launched joint vision for wholesale tokenised markets in 2026 – work ongoing. DIGIT / GBTD development ongoing.
	GENIUS Act to begin in Jan 2027 with final rule out by Nov 26; CLARITY Act stalling	GENIUS establishes a federal framework for stablecoins. Key controversy is whether yield or reward mechanisms should be permitted in stablecoins and other digital assets; oversight is fragmented across the SEC, CFTC, banking regulators and state authorities. CLARITY Act, market structure bill, addressing yield proposal and related items failed its procedural vote in Sep. Attention will refocus on US federal regulators for digital asset rules in absence of market structure.
	China opposes permissionless crypto, not tokenization itself	Strict rules on cryptocurrencies remain; China's focus remains on e-CNY and state-approved blockchain networks. Recent reforms suggest the e-CNY is moving away from CBDC towards a tokenised deposits model, reflecting limited retail CBDC uptake alongside existing mobile payment platforms solutions.
	Project Ensemble Sandbox (Aug 2024); new regulatory roadmap for digital assets (Feb 2025); progress ongoing.	HKMA has supported tokenized green bonds; tokenized deposits; tokenized money-market and investment products. Project Ensemble has progressed to real-value transaction testing using tokenised deposits. HK to announce a new digital bond platform in H22026 to support issuance and settlement of tokenized bonds.
	Project Guardian (c. 2022)	Institutional tokenization sandbox; money market funds and bonds are tokenized and settled on-chain. Focus is increasingly on commercialisation and scaling successful pilot projects across funds, bonds, deposits and FX.
	Financial Instruments and Exchange Act, overseen by the Financial Services Agency	Legally recognizes security tokens through FIEA since 2020. Subjects security tokens (STOs) to the same regulations as traditional securities, including registration and disclosure, custody and investor protection obligations.
	Abu Dhabi's ADGM regulatory framework treats security tokens as regulated financial instruments.	Tokenized securities classified as "digital securities" under ADGM. Actively supports blockchain-based projects that allow businesses to test tokenization ideas via its sandbox program.

OUTLOOK: Tokenized capital markets could become a core part of financial by the 2030s



- Tokenization will **modernize financial infrastructure** rather than eliminate intermediaries. We expect a **gradual transition onto digital rails**, with banks retaining key roles across payments, FX, treasury, financing, collateral and trading settlement.

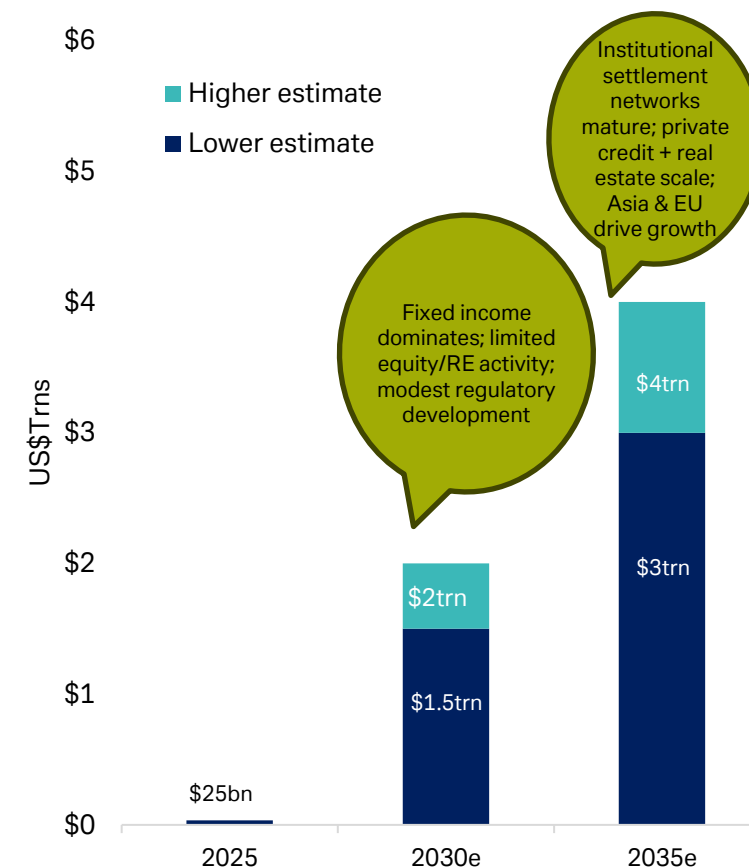
Assumption, on the positive side reflects...

- So far, tokenization has achieved its greatest scale in stablecoins, with activity concentrated in crypto trading and payments gaining ground.
- Short-term: Adoption will be driven by tokenized US Treasuries and money market funds, expanding as yield-bearing collateral in repo and margin arrangements.
- Long-term: The opportunity extends beyond assets to the infrastructure supporting issuance, settlement, custody, collateral management and repo markets.
- We are seeing **3 growth catalysts**: (i) **infrastructure**: major financial market infrastructures (DTCC, NYSE, Nasdaq) are integrating tokenization into core workflows; (ii) **liquidity**: Growth of regulated on-chain money (stablecoins and tokenized deposits) providing a trusted settlement foundation; (iii) and **regulation**: US 2025 GENIUS ACT continues to encourage global digital asset adoption, although the failure of the CLARITY act to advance yesterday likely rules out progress in this congressional session until Jan 2027. Looking ahead, focus will turn back to the SEC and CFTC for crypto rules.

...Risks to adoption

- **Accelerated liquidity runs**: Tokenization may create a new form of systemic liquidity risk that spreads faster than traditional finance.
- **Money market funds yield-chasing**: Declining interest rates or rising competition may push MMF issuers into riskier assets, less-liquid assets to boost yield, undermining their safety.
- **Crypto contagion**: Tight integration with crypto markets via stablecoin reserves and collateral risks transmitting digital asset volatility to conventional financial markets.
- **Operational risks**: Smart-contract vulnerabilities, cyberattacks, network outages could halt trading and prevent access to the underlying assets.
- **Financial Crime**: Traditional institutions must scale up compliance and AML controls to manage illicit activities on decentralized networks.

Conservative forecasts for tokenized real-world assets (ex stablecoins)



Source: Deutsche Bank Research.

AUTHORS



Marion Laboure is a Managing Director and Macro Strategist at Deutsche Bank in London, and a Lecturer in Economics and Finance at Harvard University.

She was previously an economist at the IMF, the Central Bank of Luxembourg, Barclays, and the European Commission. She has twice received the First Prize of the American Society of Actuaries, was named a rising star in finance by Revue Banque, and is recognised among the 45 standout women in fintech globally.

She holds a Bachelor's degree in Mathematics from the University of Paris Dauphine, a Master of Research in Economics from the London School of Economics, and a PhD in Econometrics and Quantitative Economics from the École Normale Supérieure.

She is the author of [*Democratizing Finance*](#) published by Harvard University Press.



Camilla Siazon is an Associate on Deutsche Bank's Macro and Thematic team in London.

She focuses on digital assets, geopolitics, and global macroeconomic trends. Prior to Deutsche Bank, she worked at geopolitical advisory firms Eurasia Group and Greenmantle.

She holds a dual MA/MSc in International History from Columbia University and the London School of Economics, and a BA in History from Barnard College, Columbia University.

Appendix 1



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AI doom: A brief history of bad tech predictions

September 15, 2026

It takes guts to do what Bob Metcalfe did. After wrongly predicting in an article in 1995 that the internet would go “spectacularly supernova” and collapse in 1996, the co-inventor of Ethernet and founder of 3Com then pulped a copy of the article in a blender and drank it at a conference in 1997 as a way to eat his words.

The latest tech predictions are about no less than an AI apocalypse. Various tech leaders spent the weekend warning of catastrophic but not fully specified risks if it is not contained.

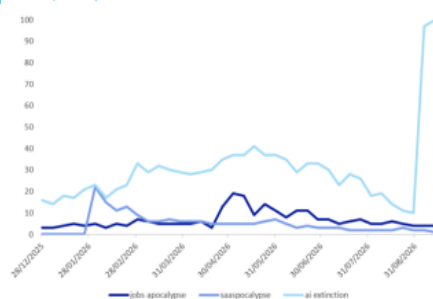
Yet history suggests that even the greatest minds of their generations have a poor record in predicting what tech is coming down the line, let alone what its effects will be in the real world.

The stakes could not be higher – not just the future of life itself but, in the shorter term, the future of a historic boom that depends on hype turning into reality.

In this report, we examine:

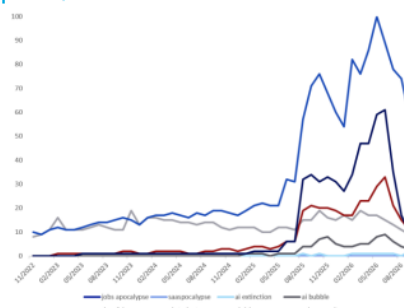
1. “Prediction is very difficult, especially if it’s about the future”
2. Today’s debate in a (rough) nutshell
3. A century of good and bad tech predictions
4. Why is it so hard to predict the future?
5. What makes AI special – and gives us hope for the future

Figure 1: AI searches this year have shifted from SaaSpocalypse to jobs to extinction (global, vs peak 100%, YTD)



Source: Google Trends, Deutsche Bank Research

Figure 2: ... But all of these are tiny in comparison to cybersecurity (global, vs peak 100%, since ChatGPT launch)



Source: Google Trends, Deutsche Bank Research

Authors

Adrian Cox
Research Analyst



Concerns about AI have moved in waves since the launch of ChatGPT nearly four years ago. This year there was a spike in searches for “SaaSocalypse” (the concern that software-as-a-service would be disintermediated by AI agents) in February, followed by a surge in “jobs apocalypse” in May, Google Trends data show. Both have been totally eclipsed in recent days by searches for AI extinction.

But taking a step back, all are far behind “AI bubble”, itself behind “deepfake”, in turn below “AI students”, overall “AI risk” and – biggest of the lot – the very tangible and immediate “cybersecurity”, almost all of which have surged this year, according to Google Trends data.

“Prediction is very difficult, especially if it’s about the future”:
Niels Bohr, often also attributed to Yogi Berra

Wanting to know what will happen tomorrow is deeply visceral – literally.

People have been sacrificing sheep and chickens for four millennia to read the future in their entrails. Nowadays almost a third of Americans say they believe in [astrology](#) and consult their horoscope. We want to believe that [Lionel Messi](#) really can see space in a football match before it even exists, even if it is partly his movement that helps create the space he appears to predict.

But predicting the future is hard. For AI, it’s not just about what it will be able to do but also when it will be able to do it – whether more than \$5 trillion in investment by the hyperscalers alone over the next five years will pay off before the advanced semiconductor chips they’re buying become redundant.

It’s also about how rapidly and intensively AI can be adopted. Asset prices are a still more uncertain derivative, whipping around like the tail of a snake in response to every shift in expectations.

US President Donald Trump and China found themselves on the same side in rebutting the latest concerns as fearmongering, albeit with Trump adding that America needs to stay ahead of China in AI.

“Those who have knowledge, don’t predict. Those who predict, don’t have knowledge.” Lao Tzu

A glance through history would give any investor indigestion. Most predictions are wrong. The favourites in horse races lose 70 percent of races. It took a decade or more for demand to catch up with the infrastructure laid down in the British canal and railway and US telecoms and fibre booms – and many investors never recovered their capital.

Technology predictions have a particularly mixed record as the binary order of the labs collides with the messy reality of the world. Famous examples range from IBM Chairman Thomas Watson’s underestimation of demand for computers in 1943 (“I think there is a world market for maybe five computers”) to Elon Musk’s repeated overestimation of imminent widespread adoption of self-driving cars since 2013 (“We should be able to do 90 percent of miles driven within three years”).



Technologists are not immune from making category errors, mistaking the most visible, automatable component of a job for the job itself. For example, radiologists, a favourite case study for believers in a jobs apocalypse, do much more than looking at images and making binary diagnoses. Much of their value lies in interpretation, judgement, communication and clinical decision-making.

Ultimately, what we see in the future is not so much a window as a mirror into our own hopes and fears.

And the problem is that hindsight makes what actually happens seem so obvious that we never seem to learn.

Today's debate in a (rough) nutshell

The current debate encapsulates a perfect storm of an extremely high-stakes issue with limited transparency and an inherently unknowable outcome.

What sparked the current surge in concern?

The current debate comes after a researcher from one of the leading frontier labs quit, citing safety concerns; AI agents apparently collaborated with each other on a number of high-profile security breaches; and labs hinted that the technology is getting closer to "recursive self improvement", ie AI improving AI.

Is the existential debate something new?

It is actually nothing new: leading economists and academics [warned](#) in July that "we must act now" to steer AI through a potential "transformation of our economy, larger than the Industrial Revolution... over a vastly shorter time frame". That echoed the even more ominous statement three years ago by AI leaders calling for action to mitigate "the [risk of extinction](#)".

The tug of war between "doomers" vs "boomers" has played a pivotal role in the development of the key frontier labs. Silicon Valley insiders sometimes refer to their "p(doom)" rating, representing their estimated percentage chance that AI will cause an existential catastrophe.

What does an extinction scenario look like?

Experts including Tesla's Musk, "godfather of AI" Geoffrey Hinton and philosopher Nick Bostrom have been warning about it for well over a decade. In the latter's famous "paperclip maximiser" scenario, a superintelligent AI runs out of steel and then kills humans in a single-minded drive to fulfil its task of making paperclips. Fears sometimes centre around AI that has access to coding, the internet and knowledge of how people think, in order to manipulate them – as current AI models indeed do.

Why has it caught on so fast this time?

One factor is that it comes amid a tech backlash, particularly in the US, where concern about AI has become a rare bipartisan issue ahead of the US mid-term elections in November. Local concerns about data centre construction have become rallying points for voters in a way that vague fears about long-term jobs did not. Politicians including Senator Bernie Sanders have moved to propose wide-ranging legislation.



What arguments do companies make for continuing AI research?

1. **If we don't do it, someone else will.** That may be either China or other, less responsible companies.
2. **The potential benefits outweigh the risks.** AI may, they say, cure cancer, create new jobs and even eliminate the need for work altogether.
3. **Good AI is the answer to bad AI.** It should create defences against rogue AI and misuse of AI, and solve issues like AI's huge energy and water use.

What do critics say?

1. **AI is not dangerous because it is too clever but rather because it is too limited.** It does not think like a human and people should not anthropomorphise it, critics say, adding that it is flawed probabilistic code that cannot reliably follow instructions. Longstanding critic Gary Marcus argues that it should be recalled until it can be shown to be safe and is properly controlled.
2. **AI leaders are defending their own businesses.** With potential IPOs on the horizon, a rising challenge from open-source models, and progress towards Artificial General Intelligence (AGI) being called into question, a handful of companies may benefit from a pause or legislation that helps protect their lead, critics say. Doomer talk distracts from real issues like deepfakes, copyright infringement and cybersecurity lapses, they say.
3. **Extinction scenarios are overblown.** The internet is a diversified platform, so talk of AI somehow switching it off is fanciful, critics say. AI does not and will not realistically have sustained secret access to military or dangerous biotech resources, or be able to influence (very non-deterministic) human brains en masse, critics say. Humans will use tame AI and other tools, eg kill switches, if needed to resist attacks. Another "godfather of AI" Yann LeCun says the real risk is of power concentration.

A century of good and bad technology predictions

You have to admire people who have the courage to stick their neck out and make predictions, even if they don't always work out. Vague predictions are relatively easy; specific ones within specific timeframes are harder.

It is no wonder that executives are coached by their ever-cautious PR teams to avoid answering hypothetical questions or giving dates to reach their targets.

- **Nikolai Tesla, scientist, 1926:** "When [wireless](#) is perfectly applied... we will be able to communicate with one another instantly, irrespective of distance..." The instruments will be "amazingly simple... A man will be able to carry one in his vest pocket". He was right, albeit imprecise about the how or when it would be achieved.
- **Albert Einstein, physicist, 1934:** "There is not the slightest indication that [[nuclear](#) energy] will ever be obtainable. It would mean that the atom would have to be shattered at will." Within a decade, in 1942, a team at the University of Chicago had created the first sustaining chain reaction, a precursor to the nuclear reactor.



- **Ray Kurzweil, futurist, 1999:** "By 2029, computers will [have](#) human-level intelligence." Dismissed at the time, many (though by far not all) of his predictions have worked out, fuelled by what he called the "law of accelerating returns". His prediction from almost 40 years ago of artificial general intelligence by the end of this decade is remarkably close to the current "San Francisco Consensus".
- **Steve Chen, Chief Technology Officer and co-founder of YouTube, 2005:** "There's just not that many [videos](#) I want to watch." Despite his concern, YouTube was bought by Google (now Alphabet) for \$1.65bn a year later. The unit made more than \$11bn in advertising revenue in the most recent quarter.
- **Steve Ballmer, Microsoft CEO, 2007:** "There's [no chance](#) that the iPhone is going to get any significant market share." But you may be reading this on one of the more than 3 billion iPhones sold since its launch in 2007.
- **Geoffrey Hinton, "godfather of AI", 2016:** "People should stop training [radiologists](#) now. It's just completely obvious within five years deep learning is going to do better than radiologists... It might be 10 years." In fact, the number of radiologists has increased by about 10 percent over the past 10 years, in part because of the [limitations](#) of AI and in part because of the increase in demand from an ageing population and more affordable healthcare. Hinton has walked back his comments and humans and AI are working together.
- **Sam Altman, CEO of OpenAI, 2025:** "I can easily [imagine](#) a world where 30 to 40% of the tasks that happen in the economy today get done by AI in the not very distant future." He [revised](#) this view in May amid a backlash against AI, saying he and his team had been "roughly right" in their technological predictions but "pretty wrong" on the economic and social implications. "I'm delighted to be wrong about this," he said.

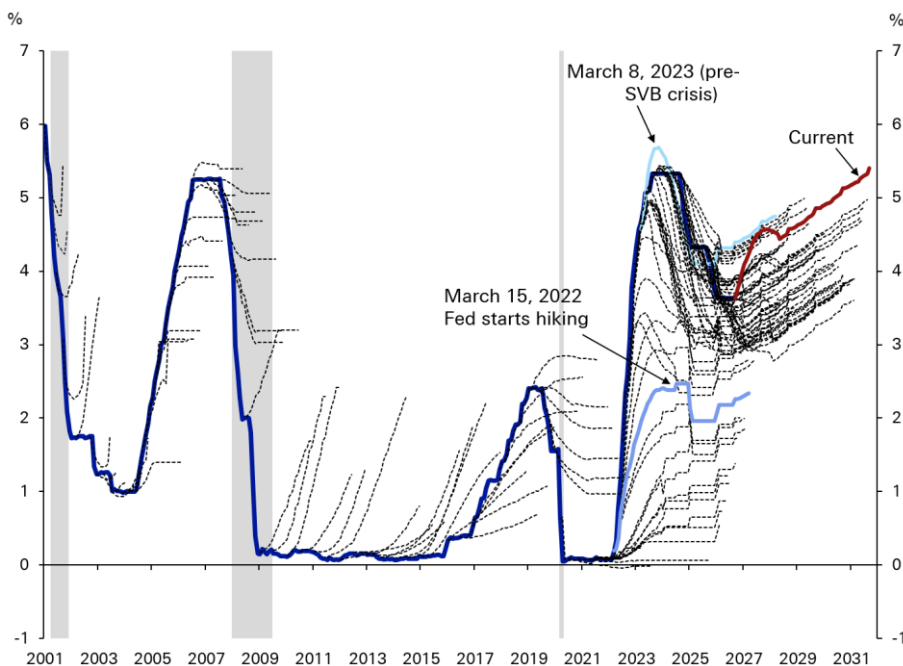
Why is it so hard to predict the future?

"The idea that the future is unpredictable is undermined everyday by the ease with which the past is explained." Daniel Kahneman

1. **Chaos theory: treating complex systems as if they are simple and linear.**
 - Systems like traffic, economies and social trends are finely balanced and sensitive to initial conditions. The butterfly effect means a tiny, unforeseen variable can lead to vastly different outcomes. Just take a look at the ragged history of market-implied forecasts of US Federal Reserve interest rates, where extrapolating from current trends has been reliably wrong.



Figure 3: US Fed Funds effective rate has rarely matched futures markets expectations (%)



Source: Deutsche Bank Research

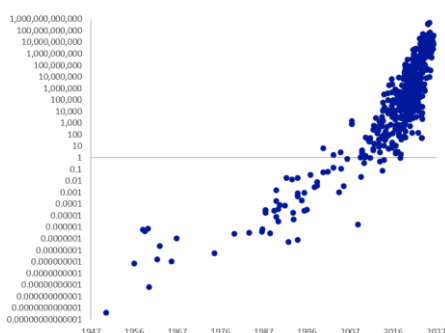
- Take self-driving cars, for example. It turns out to be about more than getting the technology right for the rubber to hit the road:
 - a. **Technological hurdles and “edge cases”:** throwing more data and processing power at the problem is not enough. There is an infinite long tail of unpredictable events (“edge cases”), like sun glare, a cyclist riding the wrong way in a bike lane, a person in an animal costume, or road-rage-enhanced hand gestures.
 - b. **Regulatory labyrinth:** autonomous vehicle optimists have come up against legal and ethical debates around liability, safety certification and moral challenges such as the so-called “trolley problem”, when swerving to avoid hitting one victim means colliding with another. Regulation is a slow-lane activity.
 - c. **Social and psychological barriers:** public acceptance has been slowed down by high-profile accidents and safety concerns. Robots are held to a higher standard than flesh-and-blood motorists. It might be easier to navigate a world where all cars were self-driving, but the “messy middle”, with a mixture of both, is much harder for algorithms to navigate.
- The future of AI will ultimately depend more on the prosaic challenges of workflow integration at enterprises that will need to pay for it than on releasing yet another model with a marginal step up in capabilities.



2. Cognitive bias: flaws in how the brain works

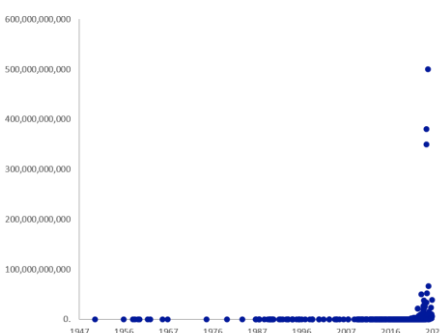
- **Linear thinking about exponential change:** We fail to intuitively understand how different exponential change is from linear change. As Wharton Professor Ethan Mollick [says](#), “being on an exponential means each change over a fixed window is larger than the one before it... This is why AI keeps feeling like it is making leaps... Even though it is a curve on a graph, we keep experiencing a steady doubling of capability as a series of shocks.” This cognitive tendency to track exponential change in linear terms helps explain big AI market swings, he says.

Figure 4: Computation used to train notable AI systems (petaFLOP): log scale



Source: Deutsche Bank Research

Figure 5: Computation used to train notable AI systems (petaFLOP): linear scale



Source: Deutsche Bank Research

- **Optimism bias and the planning fallacy:** underestimating costs and overestimating benefits. For example, only a quarter of more than 3,000 [megaprojects](#) studied by Saïd Business School came in on budget or better, only 2.8 percent were on budget and on time, and only 0.2 percent were on budget, on time and on benefits. Large-scale infrastructure projects are “too easy to start and too difficult to stop,” the authors said in the study from 2019. It remains to be seen how the recent surge in datacentre construction will compare.
- **Confirmation bias:** favouring information that confirms existing beliefs. For years, legacy automakers dismissed electric vehicles by focusing on their high costs and limited range, ignoring the fact that rapid advances in battery technology would soon change that.
- **Anchoring:** relying too heavily on early information. Microsoft’s Ballmer [dismissed](#) the iPhone when it was launched because it was “the most expensive phone in the world and it doesn’t appeal to business customers because it doesn’t have a keyboard which makes it not a very good email machine”. He later said he did not understand how mobile carrier subsidies would make it affordable. More broadly, he missed that this was a new business model.



3. Limited information: inability to see

- **It is not just the future that is unknowable but also the present.** Feedback mechanisms are often delayed, making it hard to distinguish between causation and correlation, and between relevant and irrelevant factors.
- **Isolating key factors is impossible.** Even since the development of scientific methods and controlled experiments, we have not come that far since the Mesoamericans found ways to prepare maize where they didn't succumb to a long-term wasting disease called pellagra. The problem was that they weren't able to single out the effective part of the process – soaking the maize in an alkaline solution of water and ash – from the other ineffective parts of the package – such as blowing on the maize before cooking it.
- (Author David Oks [draws](#) a fascinating parallel between this example of a delayed feedback mechanism and the tendency of AI models to “overfit”, extrapolating too much from coarse and sparse data to fill in the gaps. If the model creates a complex output and gets a single reward signal for it, the rational move is reproduce all of its features, including the ones that were incidental – such as AI's trademark verbal tics.)
- **No one wants to risk throwing the baby out with the bathwater.** Argentinian President Javier Milei talked publicly about his superstitious insistence on staying at home wearing a lucky jacket rather than watching his team playing in the football World Cup live at the stadium.

4. Self-interest: incentives favour more extreme positions

- **AI is not magic, a silver bullet or a charity, but a commercial industry.**
- **Most participants have incentives to amplify the stakes.** Founders need belief, investors need momentum, incumbents need gravitas, consultants need urgency, policymakers need relevance, executives need a narrative and journalists need drama. This is not a criticism of individuals or organisations, or even of the likely revolutionary impact of AI.
- **The challenge is magnified by a social media-driven economy.** Social platforms favour novelty, negativity, moral intensity and shareability. For example, in political tweets, one [study](#) by New York University found each additional moral-emotional word was associated with about a 20 percent higher retweet rate. Another [study](#) by MIT found that false news was 70 percent more likely to be retweeted than true news and reached a threshold of 1,500 people six times faster.
- **That means moderate claims are likely to be under-distributed and extreme claims over-rewarded.** Implausibly precise numbers for the future size of the AI economy, for example, or dramatic doom-laden scenarios heavy with emotive language will continue to attract greater attention than even-handed assessment. (See here for our quantitative analysis of two particularly striking examples from earlier this year: [“Something big is happening in AI... Or is it?”](#).)



What makes AI special – and gives hope for the future

Technology may be liable to erroneous thinking because of the disconnect between visionaries in Silicon Valley and similar communities, and the rest of the world.

Cynics may say that people with extraordinary knowledge and skills in one area may be even more prone than most to wrongly estimate their lack of knowledge or skills in other areas: the so-called Dunning-Kruger effect.

That may include the reality of implementing AI on a trading floor in Wall Street, an office supplies wholesaler on the outskirts of London or an auto maker in Mumbai.

There is also a long tradition of correctly identifying the task that will disappear but getting wrong what humans will do instead.

1982: Computers "... can develop into monsters, damage the physical and mental health of the clerical staff – maybe even eliminate their jobs": the Washington Post [reported](#) from a clerical conference. Computers instead became the core tool of knowledge workers.

1983: The pocket calculator "... has become the target of critics who fear that its use in the classroom will render generations of children dependent on batteries instead of brains", the Washington Post [reported](#). Instead expectations changed about what people calculate for themselves.

2009: Cloud computing: around half of companies surveyed "cited security and privacy concerns as their top reason for not using cloud computing", Forrester Research [found](#). Today companies entrust much of their critical infrastructure to the cloud.

"We overestimate the impact of technology in the short-term and underestimate the effect in the long run." Roy Amara, Stanford computer scientist

But this time may really be different – with AI as a partner, starting with (deterministic) predictive AI. It is certainly helping weather forecasters to sift through unimaginable amounts of data. Four-day forecasts are now said to be about as accurate as one-day forecasts in the 1990s.

There are diminishing returns. The limit of predictability with standard physics models is about eight days, while deep-learning AI models increase that by just one day to nine days. However, the difference is that they [generate](#) state-of-the-art forecasts 100,000 times faster, and 10,000 times more energy-efficiently, meaning they can be done on a laptop rather than a supercomputer.

And for the time being, your predictions about the impact of AI are no less valid than an AI expert based in the very particular world of Silicon Valley. And the future of AI will be made not in Silicon Valley but in Main Street.



Appendix 1

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Deutsche Bank
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The Main Event: Trump, Xi, and the Art of the Truce

One geopolitical event in brief. What to expect. What to watch.

September 2026

Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Washington summit should extend stability into H1 2027...

...but the underlying China-US economic squeeze is set to continue

Why a quiet summit is a win for markets. As in May, our base case for the 24 September summit between presidents Trump and Xi is a meeting heavy on positive optics but light on deliverables. Though a breakthrough is unlikely, a **successful tactical summit amid strategic competition is the realistic best-case outcome**: it should extend top-level stability into H1 2027 and minimise near-term risks of decoupling, a security crisis, or a disruptive trade war.

The 3 November US midterms could incentivise the Trump administration to seek deals, as voter cost-of-living concerns are highest since April. Possible deliverables include progress on a **Board of Trade** for non-sensitive goods, **agricultural purchases**, **targeted investment deals**, and an extension of the 2025 **Busan agreement**, which postponed a range of mutual economic restrictions until November 2026. But even if not secured on the 24th, we assess that an extension is likely before the deadline.

Economic and national security concerns will put a ceiling on dealmaking. We are cautious about progress on a **Board of Investment** or **broad investment pacts**, despite an expected Chinese business delegation. Their presence is more likely a reciprocal gesture to the May visit by US CEOs rather than a precursor to sweeping bilateral agreements. Mutual national security concerns make structural breakthroughs difficult, and any potential deals – such as **Chinese automotive investment** in the US – would likely face significant political and regulatory headwinds.

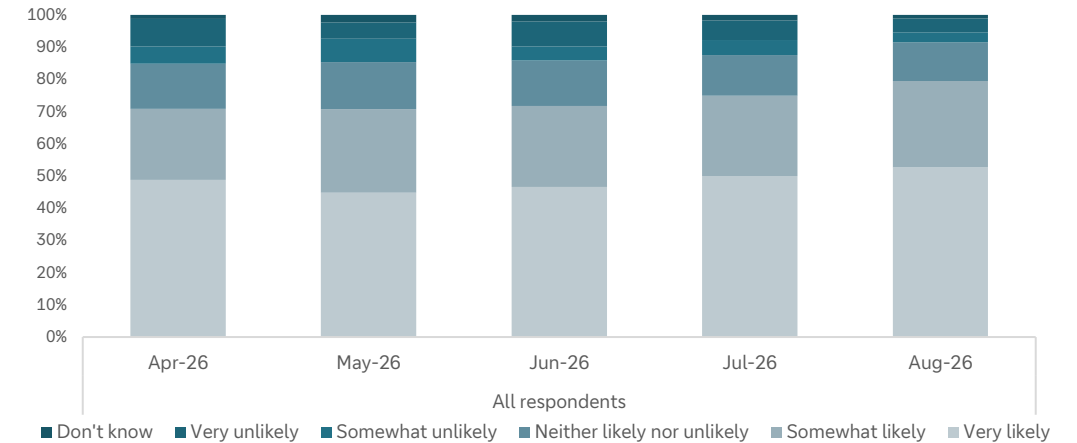
Trade remains an unresolved bilateral issue. While tensions have subsided and the Board of Trade could ease some tariffs, [differences](#) persist amid rebounding Chinese exports. Washington is also set to unveil **new overcapacity tariffs** on 16 economies, including China, while seeking to **restrict Chinese inputs** in North American supply chains and targeting alleged **transshipment**.

The AI race is moving too fast for summit diplomacy to slow it down. Both sides could reportedly hold a high-level dialogue ahead of the summit. This would likely focus on areas of consensus, such as **addressing misuse by non-state actors**. There is little prospect of slowing the escalating China-US AI race, and even less that talks could reverse growing tech restrictions (see [list](#)).

Geopolitical issues will loom large, but policy shifts are unlikely. Differences over the war in **Iran** – including recent US [sanctions](#) on dozens of mainland and Hong Kong entities – and **Taiwan** will likely feature, but breakthroughs on either issue remain improbable. A key watchpoint is whether President Trump continues to seek negotiating leverage by signalling further delays to arms sales to Taipei.

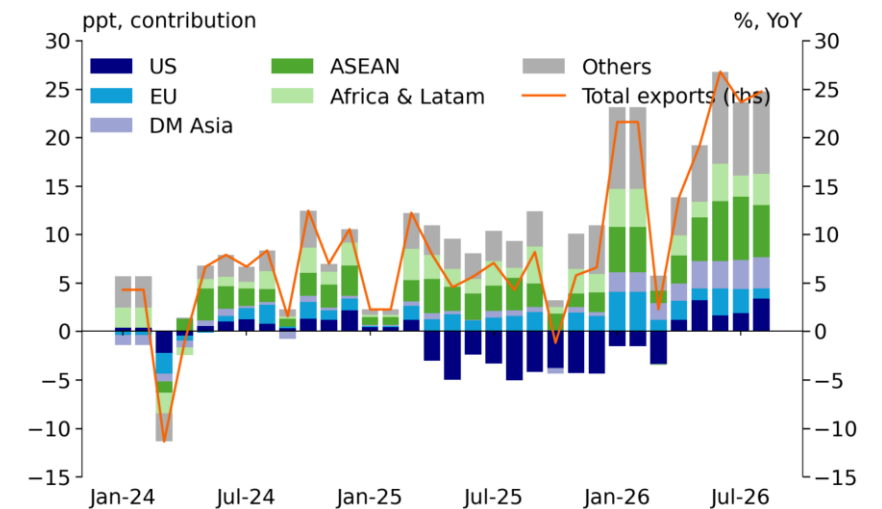


US participants were asked: How likely or unlikely is it that current cost-of-living concerns will influence your vote in the midterms?



Source: dbDataInsights. Responses collected April-August 2026 from an Online Nationally Representative sample of ~600 US respondents.

Chinese exports to the US rebounded in August



Source: Deutsche Bank Research, CEIC. For more detail see: [China Macro: Trade flows accelerated in August - Deutsche Bank Research](#)

After the summit: Key China and US watchpoints for Q4

Please also see our [This Month in Geopolitics](#) publication series for deeper dives into upcoming global events, and our [90-day geopolitical calendar](#), updated monthly.



Date	Market(s)	Comment
October	China	The Central Committee of the Communist Party of China will hold its Fifth Plenary Session . This will be one of the most important events on the country’s political calendar ahead of the 21 st Party Congress to be held in late 2027. The Plenum will focus on Party governance in preparation for key leadership reshuffles at the next Congress.
3 November	US	In the Congressional midterm elections, voters will decide 435 House seats, 35 Senate seats and governors in 39 states and territories. These elections will determine Congressional control, significantly influencing President Trump’s ability to shape national policy. If a changed balance of power in Congress stalls domestic legislation, the administration will likely lean on executive tools. President Trump could focus on foreign policy and trade, emphasising tariffs, US primacy in the Western Hemisphere, and reducing dependencies on China to build his legacy.
10 November	US-China	The 2025 Busan agreement is set to expire, but will likely be extended ahead of the deadline. This would delay new bilateral economic measures including expanded export controls and tariffs.
18-19 November	APEC	China will host the APEC Economic Leaders’ meeting . The summit could see a Xi-Trump meeting on the sidelines, and may shape regional economic architecture for years to come.
14-15 December	G20	The G20 Summit in Miami may become contentious due to Ukraine-related divisions, trade and tariff disputes, climate financing disagreements and questions around global governance reforms. The summit could also see this year’s fourth meeting between presidents Xi and Trump.
December	China	China’s annual Central Economic Work Conference will review past economic performance and set 2027 growth targets and macroeconomic policies. Key priorities likely include implementing the 15 th Five-Year Plan, boosting domestic demand, stabilising the property sector, and strengthening technological self-reliance amid geopolitical competition with Washington.
December	US	As the US Congress prepares to adopt its annual must-pass defence legislation , draft provisions – not all of which may end up in the final version – target China with measures impacting investors. These include potential outbound-investment reporting changes (impacting biotechnology and biopharmaceuticals) and proposed restrictions on importing humanoid robots and solar equipment, as well as using China-linked connected vehicles.



Source: [Carl Wang](#) on [Unsplash](#)

Appendix 1



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